LCD-1

IVY Bridge (rPGA989)

Intel PCH (Panther Point)

DY:No stuff
SWG:SWG SKU

PSL: KBC795 PSL circuit for 10mW solution installed.
10mW: External circuit for 10mW solution installed.

BOM1

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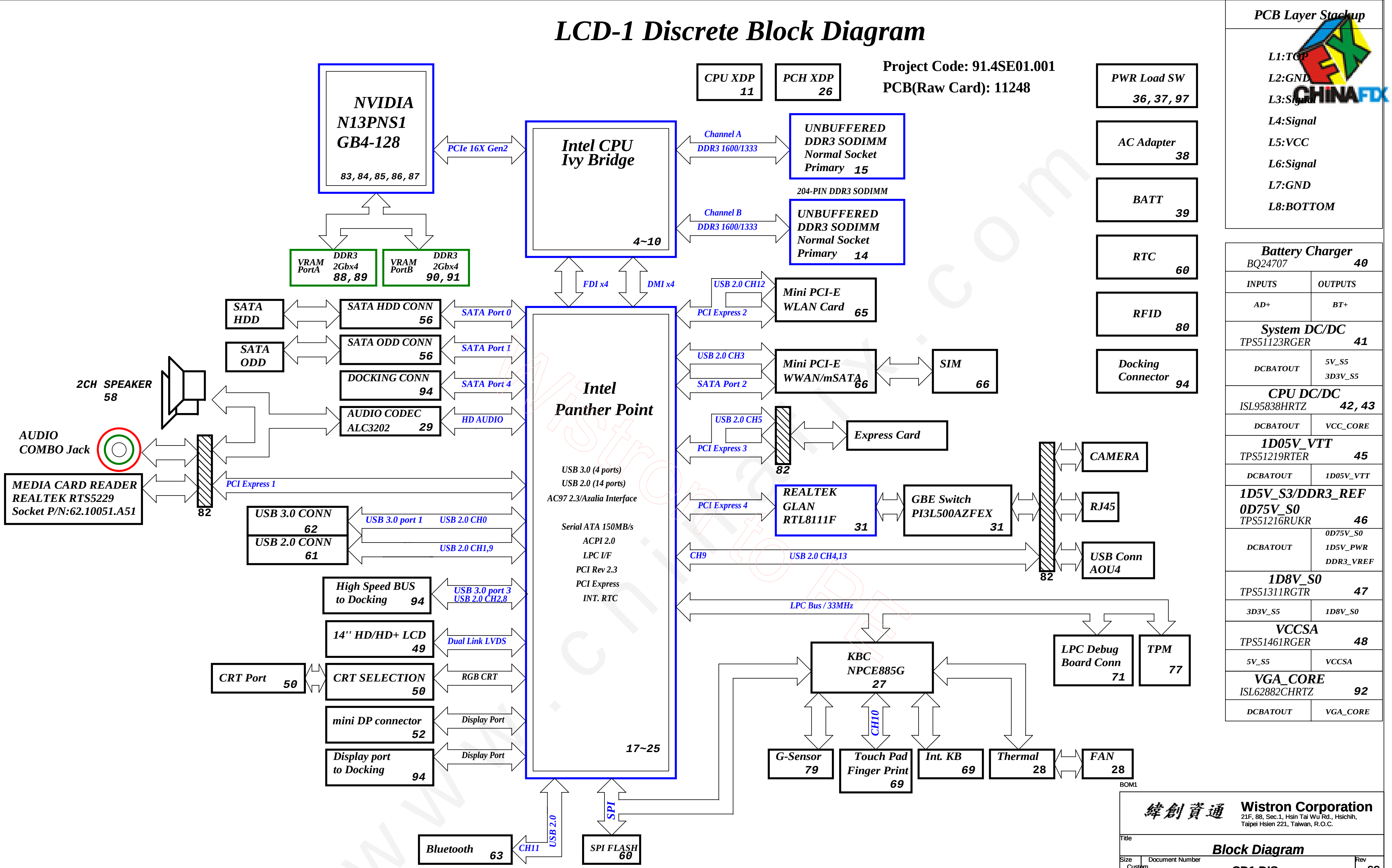
Title	*Cover Page*	
Size A3	Document Number CD1 DIS	Rev SC
Date: Tuesday, December 13, 2011	Sheet 1 of 102	

LCD-1 Discrete Block Diagram

Project Code: 91.4SE01.001
PCB(Raw Card): 11248



BOM1

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Title: Block Diagram

Size: Custom | Document Number: CD1 DIS | Rev: SC

Date: Tuesday, December 13, 2011 | Sheet 2 of 102

PCH Strapping   Chief River Schematic Checklist Rev0.72

Name	Schematics Notes
SPKR	**Reboot option at power-up** **Default Mode:** Internal weak Pull-down. **No Reboot Mode with TCO Disabled:** Connect to Vcc3_3 with 8.2-kΩ - 10-kΩ weak pull-up resistor.
INIT3_3V#	Weak internal pull-up. Leave as "No Connect".
GNT3#/GPIO55 GNT2#/GPIO53 GNT1#/GPIO51	GNT[3:0]# functionality is not available on Mobile. Mobile: Used as GPIO only Pull-up resistors are not required on these signals. If pull-ups are used, they should be tied to the Vcc3_3power rail.
SPI_MOSI	**Enable Danbury:** Connect to Vcc3_3 with 8.2-k? weak pull-up resistor. **Disable Danbury:** Left floating, no pull-down required.
NV_ALE	**Enable Danbury:** Connect to +NVRAM_VCCQ with 8.2-kohm weak pull-up resistor [CRB has it pulled up with 1-kohm no-stuff resistor] **Disable Danbury:** Leave floating (internal pull-down)
NC_CLE	DMI termination voltage. Weak internal pull-up. Do not pull low.
HAD_DOCK_EN# /GPIO[33]	Low (0) - Flash Descriptor Security will be overridden. Also, when this signals is sampled on the rising edge of PWROK then it will also disable Intel ME and its features. High (1) - Security measure defined in the Flash Descriptor will be enabled. Platform design should provide appropriate pull-up or pull-down depending on the desired settings. If a jumper option is used to tie this signal to GND as required by the functional strap, the signal should be pulled low through a weak pull-down in order to avoid asserting HDA_DOCK_EN# inadvertently. Note: CRB recommends 1-kohm pull-down for FD Override. There is an internal pull-up of 20 kohm for DA_DOCK_EN# which is only enabled at boot/reset for strapping functions.
HDA_SDO	Weak internal pull-down. Do not pull high. Sampled at rising edge of RSMRST#.
HDA_SYNC	Weak internal pull-down. Do not pull high. Sampled at rising edge of RSMRST#.
GPIO15	Low(0) - Intel ME Crypto Transport Layer Security (TLS) cipher suite with no confidentiality. High(1) - Intel ME Crypto Transport Layer Security (TLS) cipher suite with confidentiality. Note : This is an un-muxed signal. This signal has a weak internal pull-down of 20 kohm which is enabled when PWROK is low. Sampled at rising edge of RSMRST#. CRB has a 1-kohm pull-up on this signal to +3.3VA rail.
GPIO8	GPIO8 on PCH is the Integrated Clock Enable strap and is required to be pulled-down using a 1k +/- 5% resistor. When this signal is sampled high at the rising edge of RSMRST#, Integrated Clocking is enabled, When sampled low, Buffer Through Mode is enabled.
GPIO27	**Default = Do not connect (floating)** High(1) = Enables the internal VccVRM to have a clean supply for analog rails. No need to use on-board filter circuit. Low (0) = Disables the VccVRM. Need to use on-board filter circuits for analog rails.

Processor Strapping   Chief River Schematic Checklist Rev0.72

Pin Name	Strap Description	Configuration (Default value for each bit is 1 unless specified otherwise)	Default Value
CFG[2]	**PCI-Express Static Lane Reversal**	1: Normal Operation. 0: Lane Numbers Reversed 15 -> 0, 14 -> 1, ...	1
CFG[4]		1: Disabled - No Physical Display Port attached to Embedded DisplayPort. 0: Enabled - An external Display Port device is connectd to the EMBEDDED display Port	0
CFG[6:5]	**PCI-Express Port Bifurcation Straps**	11 : x16 - Device 1 functions 1 and 2 disabled 10 : x8, x8 - Device 1 function 1 enabled ; function 2 disabled 01 : Reserved - (Device 1 function 1 disabled ; function 2 enabled) 00 : x8, x4, x4 - Device 1 functions 1 and 2 enabled	11
CFG[7]	**PEG DEFER TRAINING**	1: PEG Train immediately following xxRESETB de assertion 0: PEG Wait for BIOS for training	1

PCIe Routing

LANE1	Card Reader
LANE2	Mini Card1(WLAN)
LANE3	Express Card
LANE4	GBE LAN
LANE5	X
LANE6	X
LANE7	X
LANE8	X

Voltage Rails

POWER PLANE	VOLTAGE	ACTIVE IN	DESCRIPTION
5V_S0 3D3V_S0 1D8V_S0 1D5V_S0 1D05V_VTT 1D0V_S0 VCCSA 0D75V_S0 VCC_CORE VCC_GFXCORE 1D8V_VGA_S0 3D3V_VGA_S0 1V_VGA_S0	5V 3.3V 1.8V 1.5V 1.05V 1.0V 0.9 - 0.675V 0.75V 0.35V to 1.5V 0.4 to 1.25V 1.8V 3.3V 1V	S0	CPU Core Rail Graphics Core Rail
5V_USBX_S3 1D5V_S3 DDR_VREF_S3	5V 1.5V 0.75V	S3	
BT+ DCBATOUT 5V_S5 5V_AUX_S5 3D3V_S5 3D3V_AUX_S5	6V-14.1V 6V-14.1V 5V 5V 3.3V 3.3V	All S states	AC Brick Mode only
1D05V_LAN	1.05V	S0/M0, SX/M3	ON whenever iAMT is active
3D3V_M 1D05V_M	3.3V 1.05V	S0/M0, SX/M3, WOL_EN	ON for iAMTLegacy WOL
3D3V_AUX_KBC	3.3V	DSW, Sx	ON for supporting Deep Sleep states
3D3V_AUX_S5	3.3V	G3, Sx	Powered by Li Coin Cell in G3 and 3D3V_S5 in Sx

USB Table

Pair	Device
0	USB3.0 port 0
1	USB2.0 port 1
2	USB3.0 Docking
3	WWAN
4	USB2.0 port (AUO4)
5	New Card
6	X
7	X
8	USB2.0 Docking
9	USB2.0 port 2
10	FPR
11	BLUETOOTH
12	WLAN
13	Camera

RESISTOR

Symbol name	Value	Tolerance (J: 5%, F: 1%, D: 0.5%, B: 0.1 %)	Rating 0402=> 1/16W, 25V 0603 => 1/16W, 75V 0805 => 1/10W, 100V	Size 2=>0402, 3=>0603, 5=>0805, 6=>1206, 0=>1210
10KR3	10K Ohm	If no letter, it means J: 5%	1/16W, 75V	0603
33D3R5	33.3 Ohm	If no letter, it means J: 5%	1/10W, 100V	0805
1KR3F	1K Ohm	F: 1%	1/16W, 75V	0603

The naming rule is value + R + size + tolerance
For the value, it can be read by the number before R. (R means resistor)
For the tolerance, it can be read from the last letter.
For the rating, we don't show on the symbol name.
For the size, R2=>0402, R3=>0603, R5=>0805,....

SMBus ADDRESSES

I^2C / SMBus Addresses Device	Ref Des	Chief River CRV Address Hex Bus
EC SMBus 1 Battery CHARGER		BAT_SCL/BAT_SDA BAT_SCL/BAT_SDA BAT_SCL/BAT_SDA
EC SMBus 2 PCH eDP		SML1_CLK/SML1_DATA SML1_CLK/SML1_DATA SML1_CLK/SML1_DATA
PCH SMBus SO-DIMMA (SPD) SO-DIMMB (SPD) Digital Pot G-Sensor MINI		PCH_SMBDATA/PCH_SMBCLK PCH_SMBDATA/PCH_SMBCLK PCH_SMBDATA/PCH_SMBCLK PCH_SMBDATA/PCH_SMBCLK PCH_SMBDATA/PCH_SMBCLK PCH_SMBDATA/PCH_SMBCLK

SATA Table

SATA	
Pair	Device
0	HDD
1	ODD
2	mSATA
3	N/A
4	Docking
5	N/A

CAPACITOR

Symbol name	Value	Tolerance (M: +/-20, K: +/-10, Z: +80/-20)	Rating	Size 2=>0402, 3=>0603, 5=>0805, 6=>1206, 0=>1210
SCD1U10V2MX-1	0.1uF	M/X5R	10V	0402
SC10U6D3V5MX	10uF	M/X5R	6.3V	0805
SC2D2U16V5ZY	2.2uF	Z/Y5V	16V	0805

The naming rule is
Capacitor type + value + rating + size + tolerance + material
SCD1U10V2MX-1
SC=> SMT Ceremic, TC=> POS cap or SP cap
D1U => 0.1uF
10V => the voltage rating is 10V
2=> 0402, 3=>0603, 5=>0805
M=>tolerance M, K, Z
X=> X7R/X5R, Y=> Y5V
-1 => symbol version, nonsense to EE characteristic

BOM1

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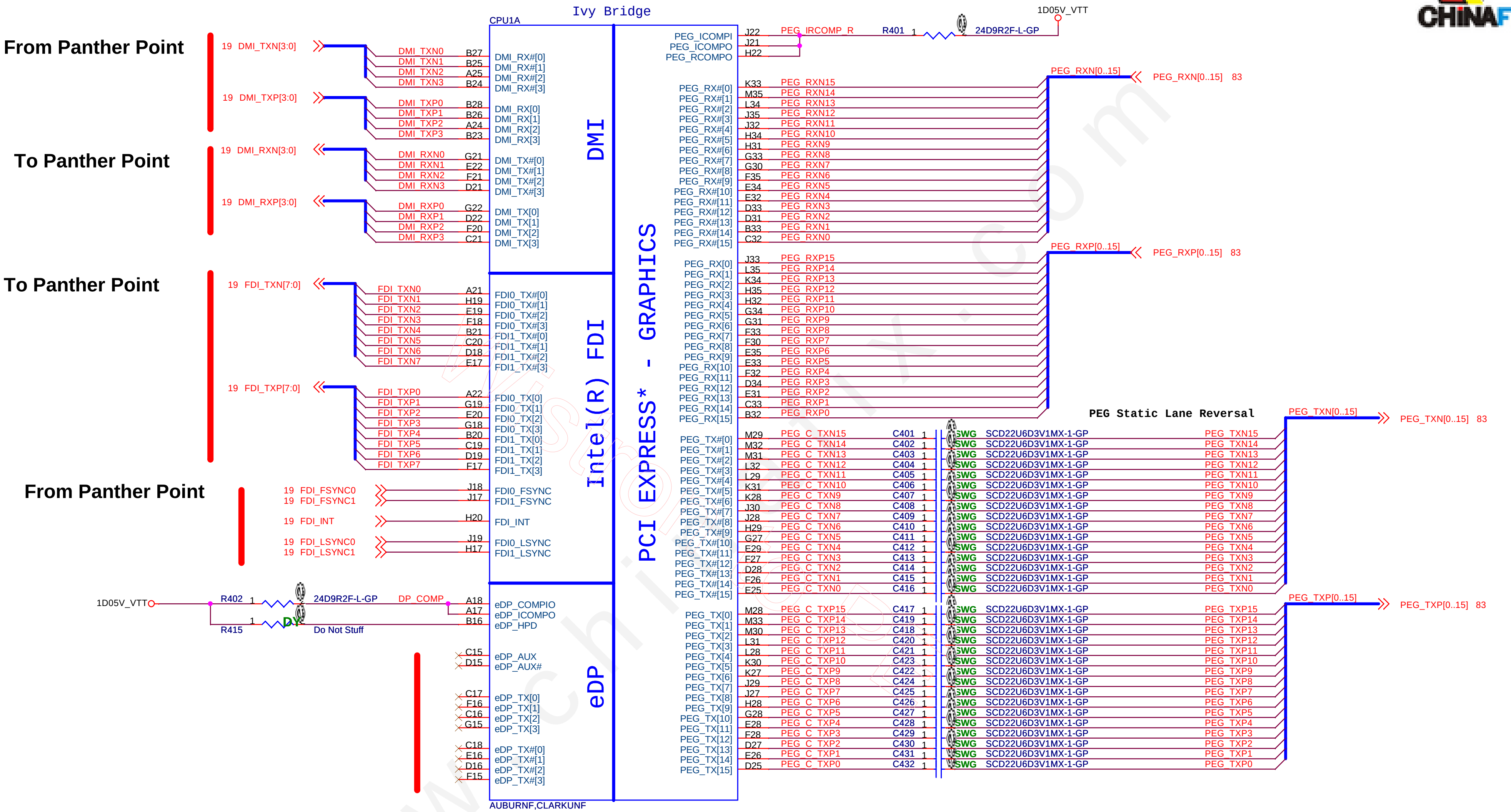
Title: Table of Content

Size: Custom | Document Number: CD1 DIS | Rev: SC

SSID = CPU

Signal Routing Guideline:
PEG_ICOMPO keep W/S=12/15 mils and routing length less than 500 mils.
PEG_ICOMPI & PEG_RCOMPO keep W/S=4/15 mils and routing length less than 500 mils.



BOM1

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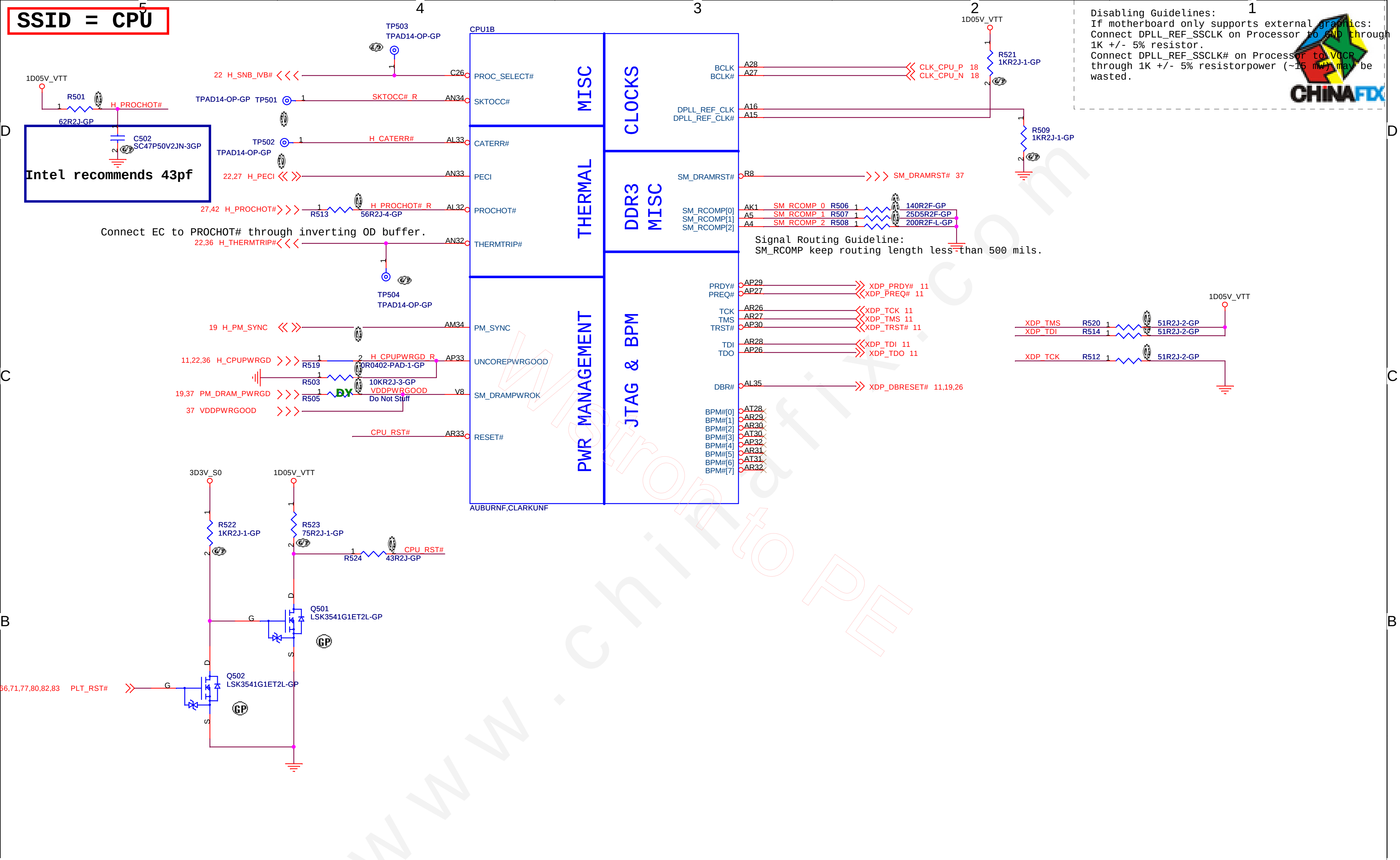
Title	**CPU (PCIE/DMI/FDI)**	
Size A3	Document Number **CD1 DIS**	Rev **SC**
Date: Tuesday, December 13, 2011	Sheet 4 of 102	



BOM1

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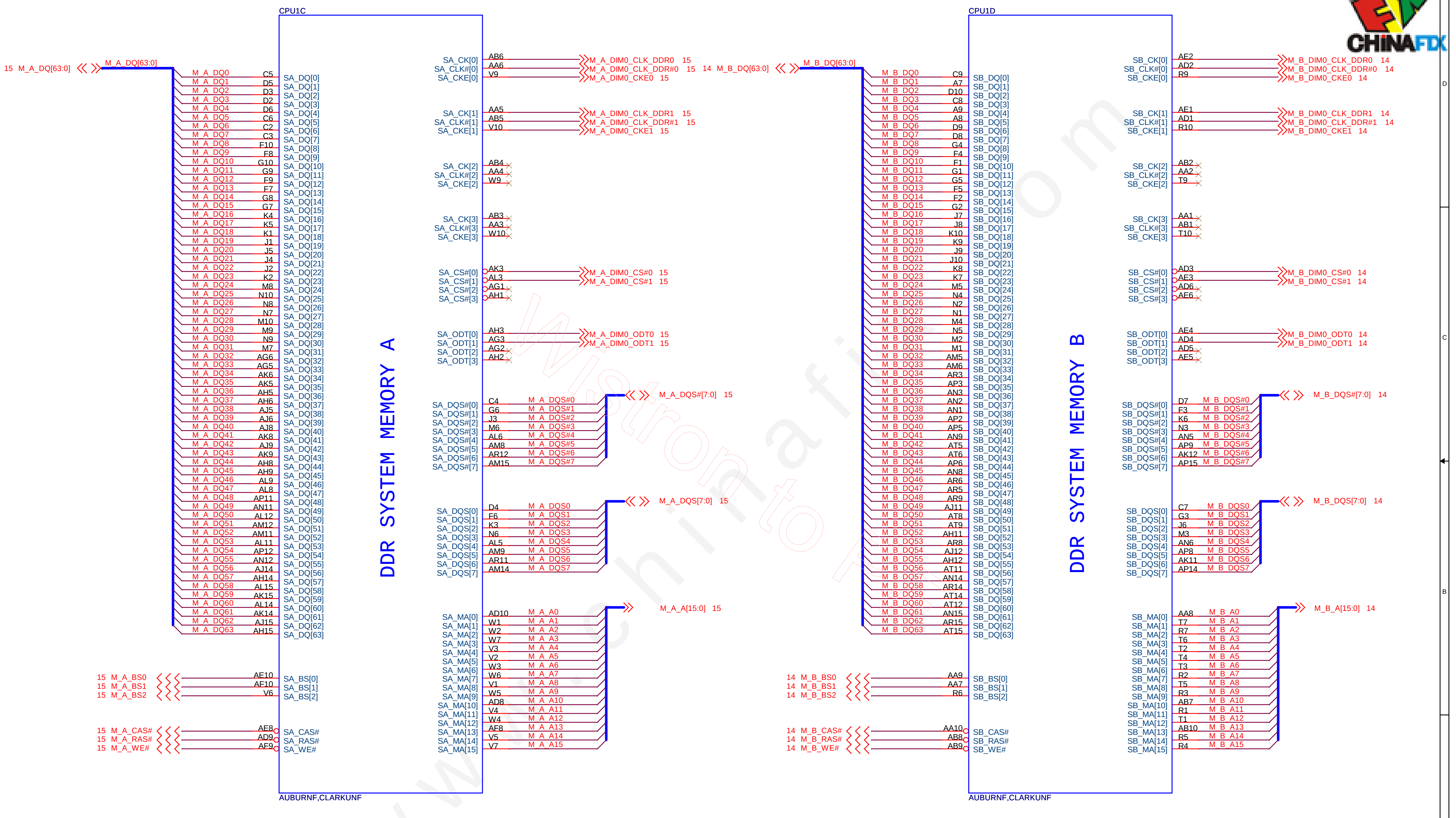
Title	**CPU (THERMAL/CLOCK/PM)**	
Size A3	Document Number **CD1 DIS**	Rev **SC**
Date: Tuesday, December 13, 2011	Sheet 5 of 102	

SSID = CPU



BOM1

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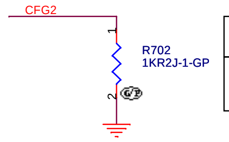
Title	CPU (DDR)	
Size A3	Document Number CD1 DIS	Rev SC
Date: Tuesday, December 13, 2011	Sheet 6 of 102	

SSID = CPU

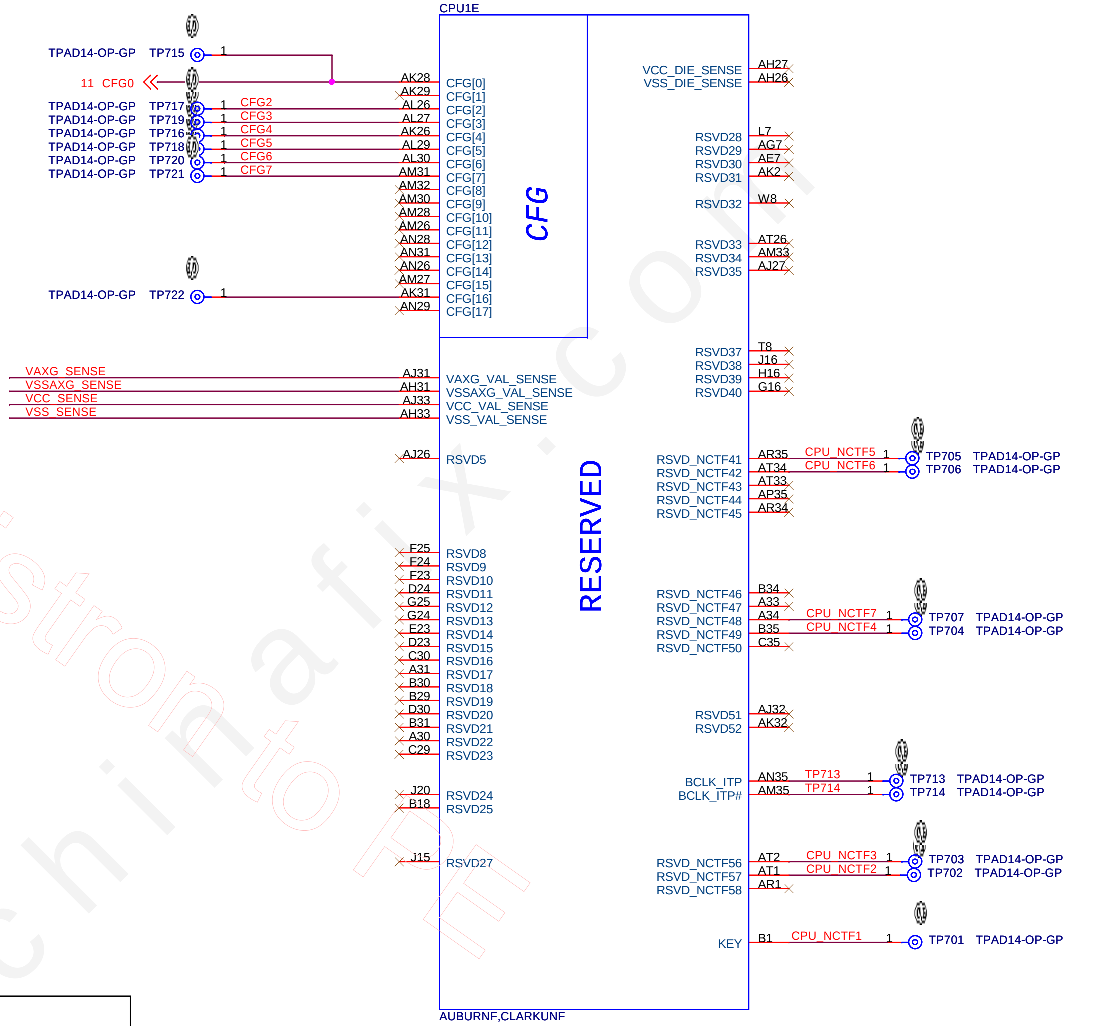


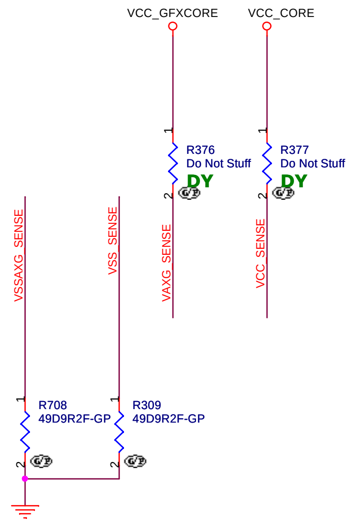
PEG Static Lane Reversal - CFG2 is for the 16x

CFG2	1: Normal Operation; Lane # definition matches socket pin map definition 0:Lane Reversed

Display Port Presence Strap

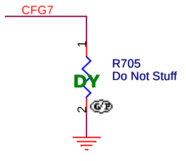
CFG4	1: Disabled; No Physical Display Port attached to Embedded Display Port 0: Enabled; An external Display Port device is connected to the Embedded Display Port





PCIE Port Bifurcation Straps

CFG[6:5]	11: x16 - Device 1 functions 1 and 2 disabled 10: x8, x8 - Device 1 function 1 enabled ; function 2 disabled 01: Reserved - (Device 1 function 1 disabled ; function 2 enabled) 00: x8,x4,x4 - Device 1 functions 1 and 2 enabled



PEG DEFER TRAINING

CFG7	1: PEG Train immediately following xxRESETB de assertion 0: PEG Wait for BIOS for training

BOM1

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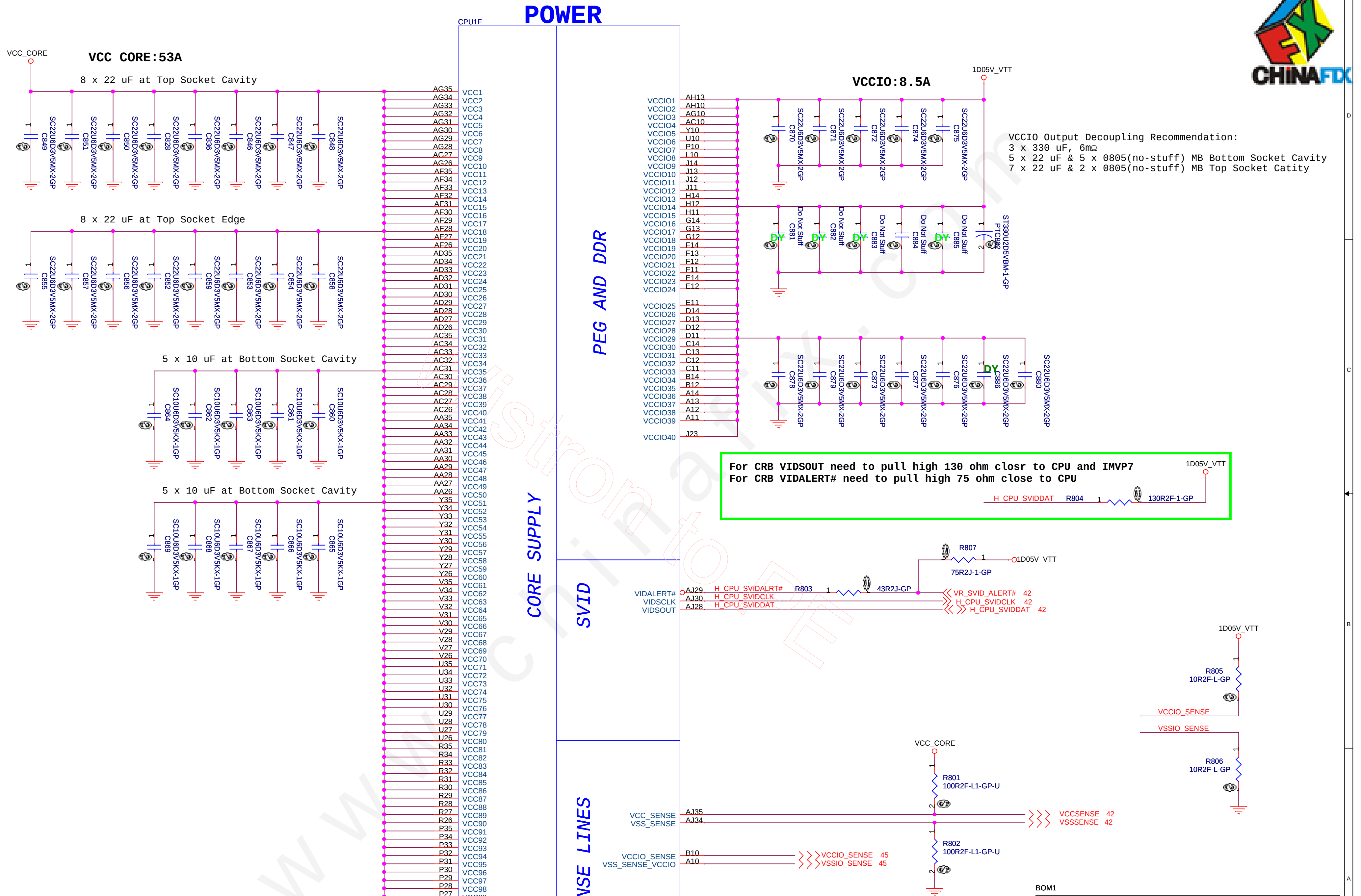
Title: CPU (RESERVED)

Size: A3 | Document Number: CD1 DIS | Rev: SC

Date: Tuesday, December 13, 2011 | Sheet 7 of 102



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Title: **CPU (VCC_CORE)**

Size	Document Number	Rev
Custom	CD1 DIS	SC

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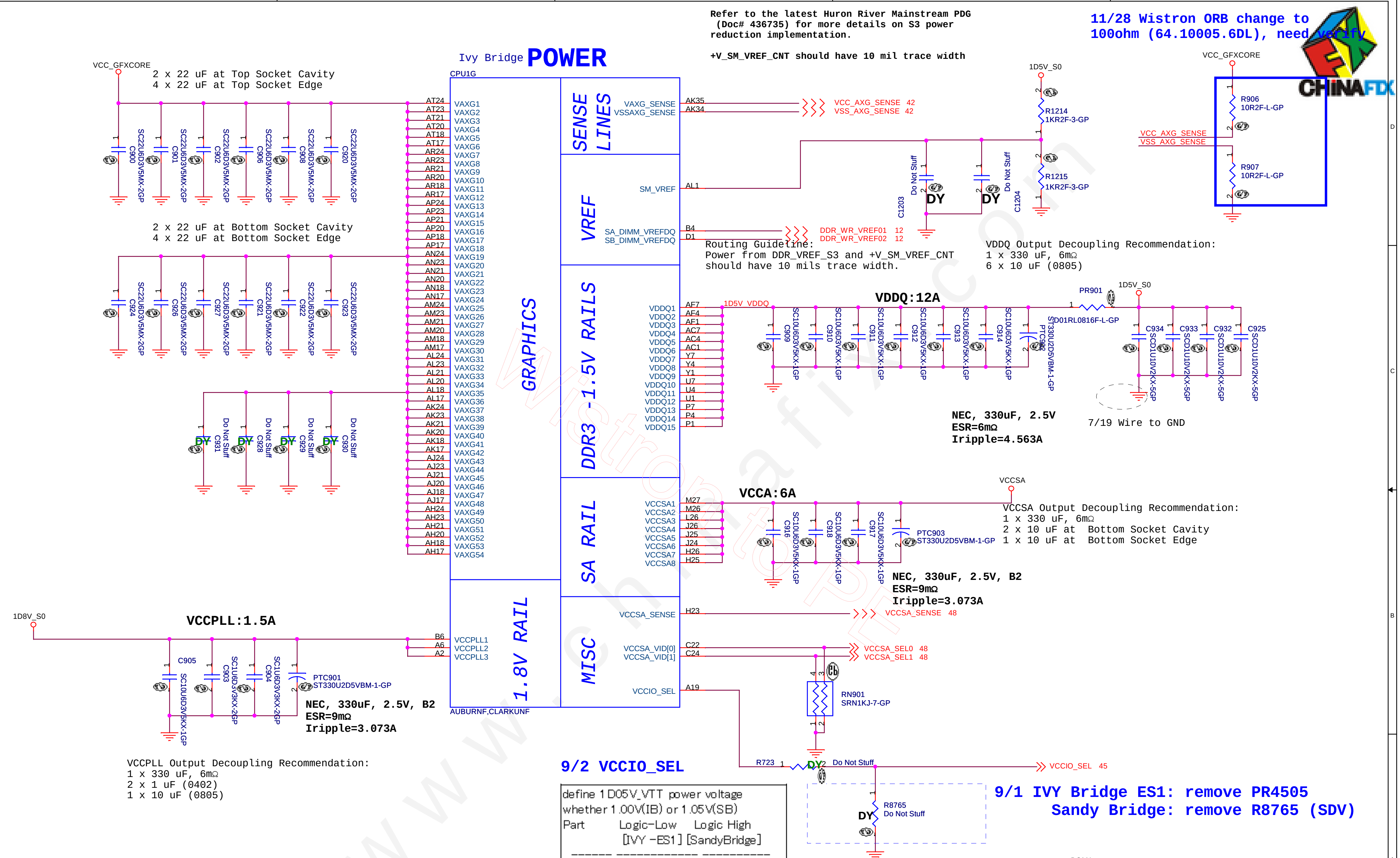

define 1D05V_VTT power voltage
whether 1.00V(IB) or 1.05V(SB)

Part	Logic-Low [IVY-ES1]	Logic High [SandyBridge]
PR4505	DY	ASM
PR4525	(DY)	DY
PR4511	(DY)	DY
PR4518	ASM	ASM
R8765	ASM	DY

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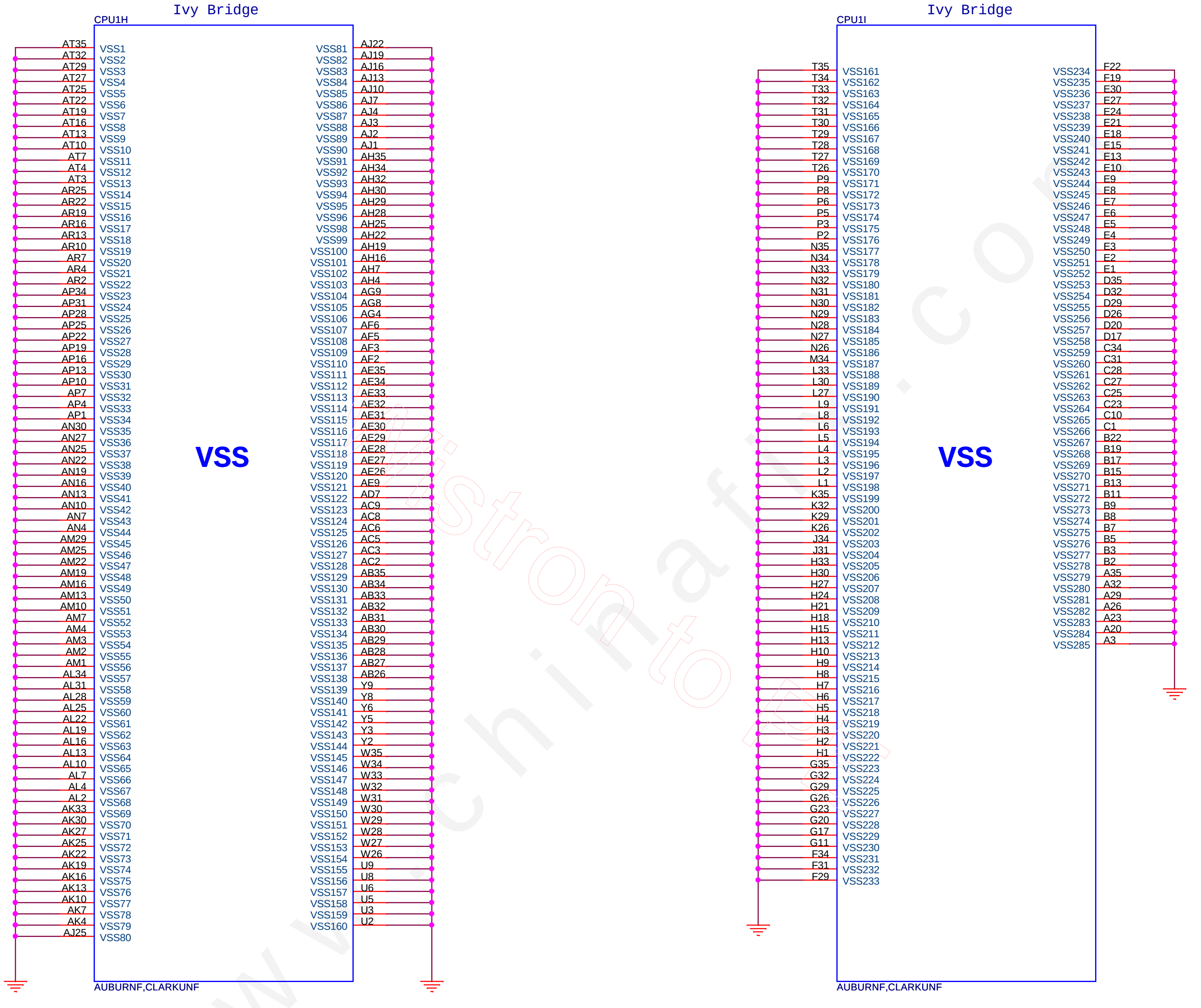
Title: **CPU (VCC_GFXCORE)**

Size	Document Number	Rev
A3	CD1 DIS	SC

Date: Tuesday, December 13, 2011 Sheet 9 of 102

SSID = CPU



BOM1

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Title		
CPU (VSS)		
Size: A3	Document Number: CD1 DIS	Rev: SC
Date: Tuesday, December 13, 2011	Sheet 10 of 102	

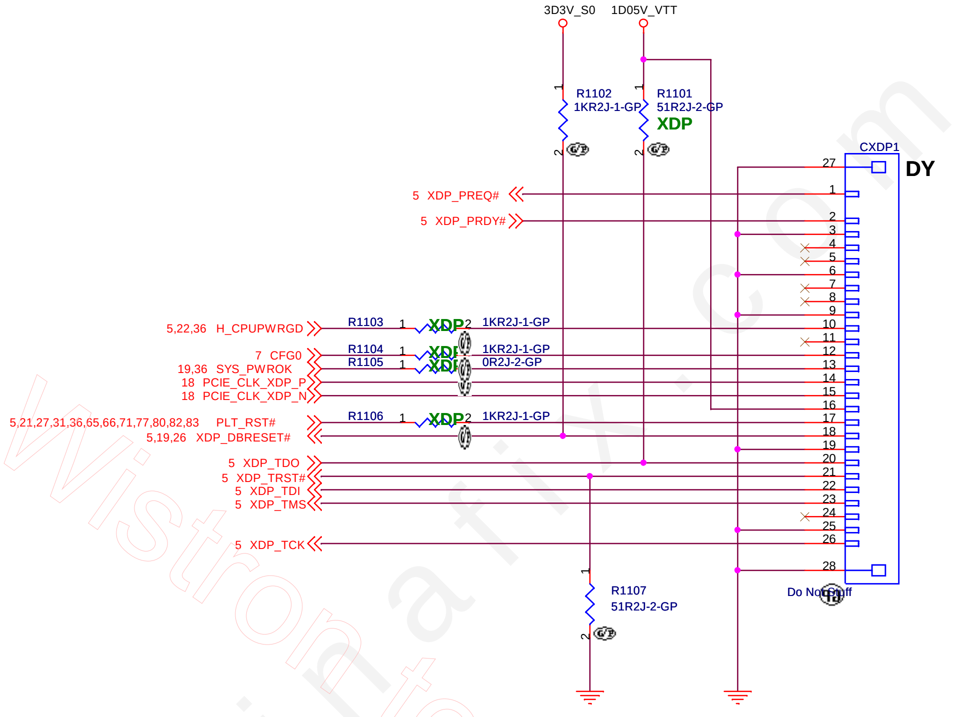
In production, All of parts should be not moounted except of pulldown 51 ohm on TRSTn and Pullup DBR#.

SIGNAL	REF DES	ENABLE	DISABLE
TDO	R1101	ASM	NOASM
TRST#	R1107	ASM	ASM
DBRESET#	R1102	ASM	ASM
PLT_RST#	R1106	ASM	NOASM
CFG0	R1104	ASM	NOASM
CPUPWRGD	R1103	ASM	NOASM
SYS_PWROK	R1105	ASM	NOASM
	CXDP1	ASM	NOASM

LOGIC

9/2 CPU_XDP

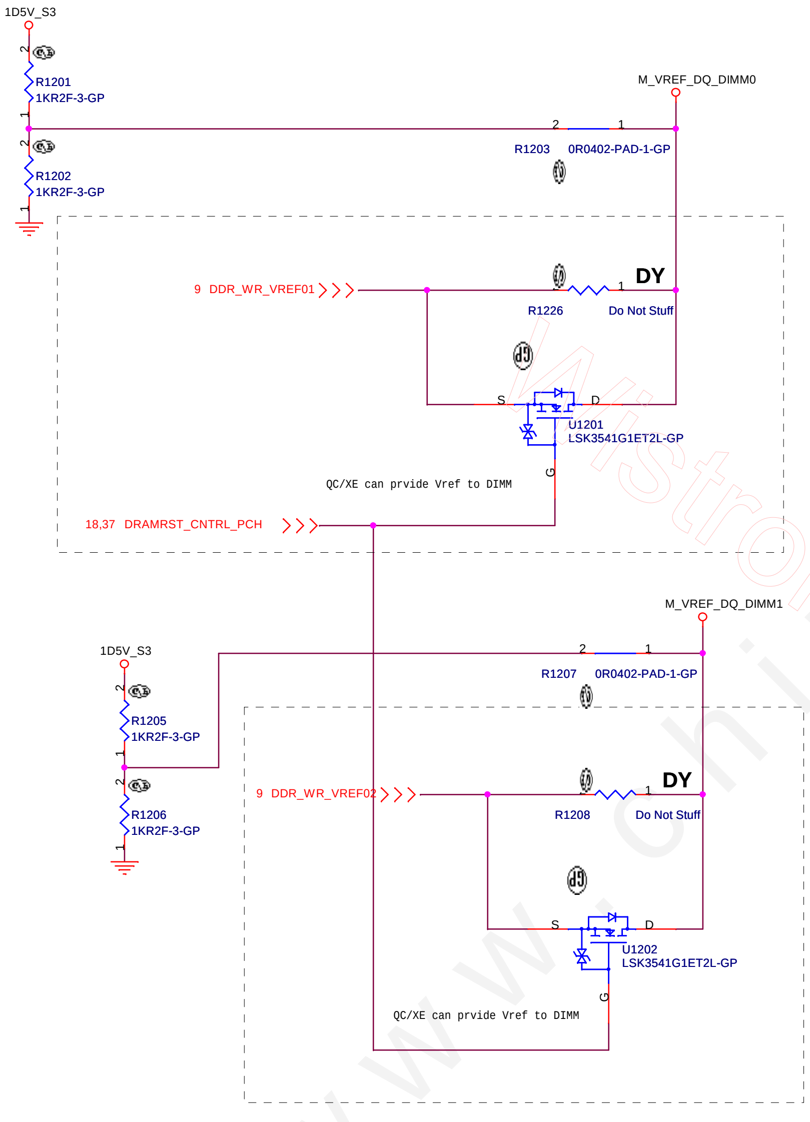
Part	Enable	Disable
R1101	ASM	DY
R1107	ASM	ASM
R1102	ASM	ASM
R1106	ASM	DY
R1104	ASM	DY
R1103	ASM	DY
R1105	ASM	DY
CXDP1	ASM	DY



CPU XDP SFF 26pin IF
Pin 1 OBSFN_A0 (PREQ#, I/O)
Pin 2 OBSFN_A1 (PRDY#, I/O)
Pin 3 GND
Pin 4 OBSDATA_A0 (Open, I/O)
Pin 5 OBSDATA_A1 (Open, I/O)
Pin 6 GND
Pin 7 OBSDATA_A2 (Open, I/O)
Pin 8 OBSDATA_A3 (Open, I/O)
Pin 9 GND
Pin 10 HOOK0 (PWRGD, In)
Pin 11 HOOK1 (BP_PWRGD_RST#, Out)
Pin 12 HOOK2 (CFG0, Out)
Pin 13 HOOK3 (vr_READYSYS_PWROK,Out)
Pin 14 HOOK4 (BCLK, In)
Pin 15 HOOK5 (BCLK#, In)
Pin 16 VCCOBS_AB (VCCP Voltage of CPU, In)
Pin 17 HOOK6 (RESET#, Out)
Pin 18 HOOK7 (DBR#, Out)
Pin 19 GND
Pin 20 TDO, In
Pin 21 TRST#, Out
Pin 22 TDI, Out
Pin 23 TMS, Out
Pin 24 TCK1 (Open)
Pin 25 GND
Pin 26 TCK0 ,Out

BOM1

VREF circuit -M1 (Voltage Driver Network) & M3 (Driven by Processor) Implementation



BOM1

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Title: **M3**		
Size: A3	Document Number: **CD1 DIS**	Rev: SC
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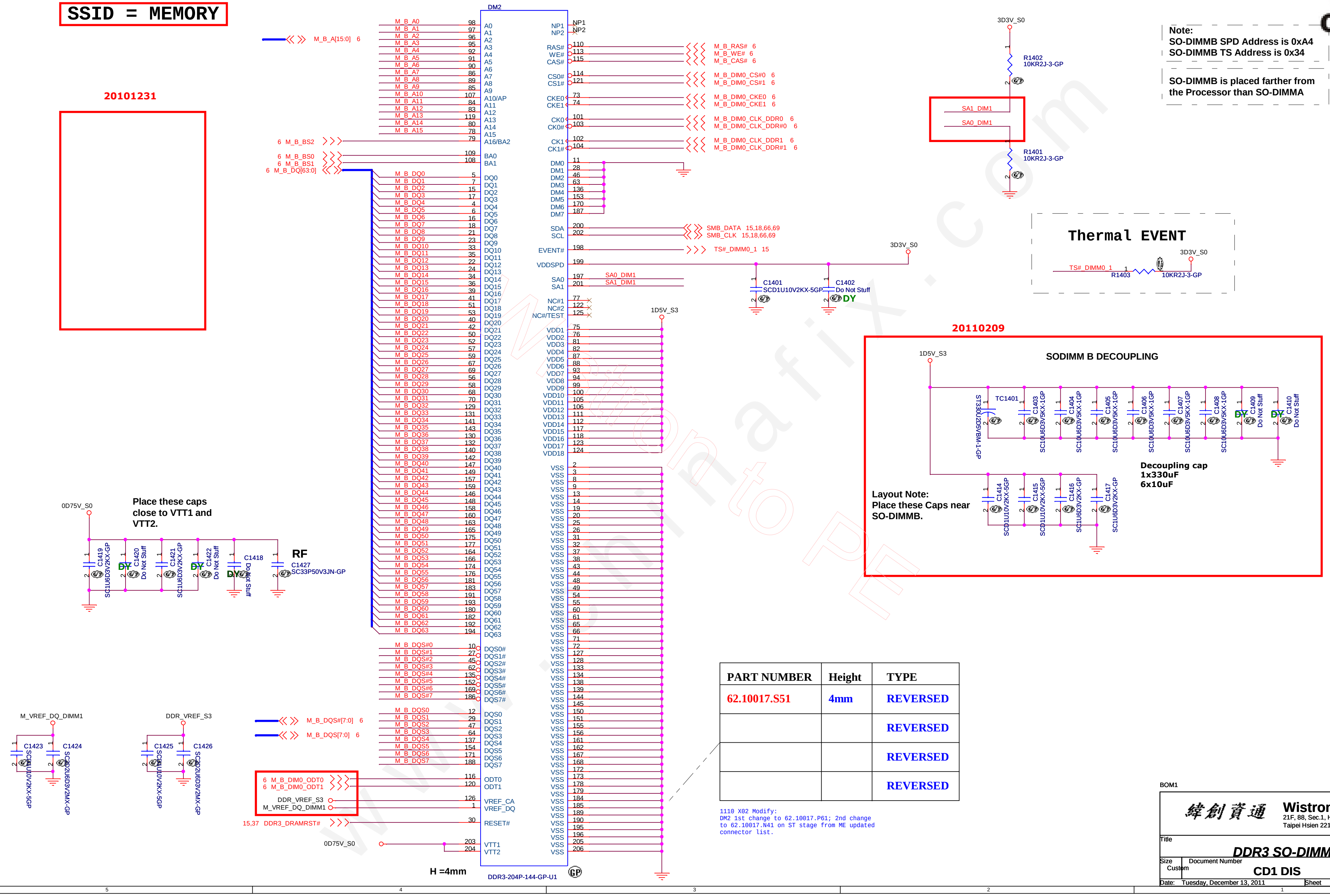
(Blanking)

SSID = MEMORY

20101231



Note:
SO-DIMMB SPD Address is 0xA4
SO-DIMMB TS Address is 0x34

SO-DIMMB is placed farther from the Processor than SO-DIMMA

PART NUMBER	Height	TYPE
62.10017.S51	4mm	REVERSED
		REVERSED
		REVERSED
		REVERSED

1110 X02 Modify:
DM2 1st change to 62.10017.P61; 2nd change to 62.10017.N41 on ST stage from ME updated connector list.

BOM1

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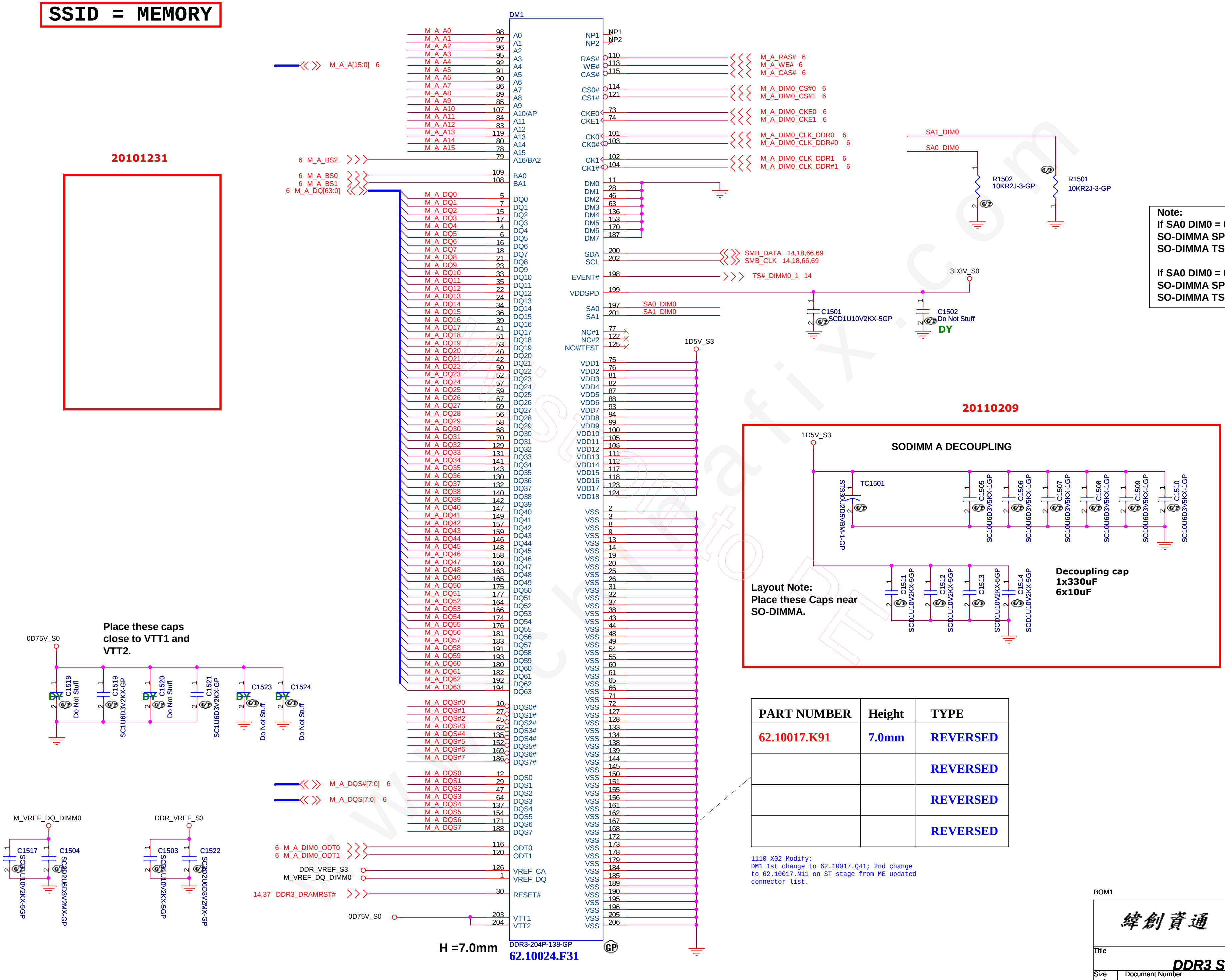
Title: DDR3 SO-DIMM2

Size: Custom | Document Number: CD1 DIS | Rev: SC

SSID = MEMORY



Note:
If SA0 DIM0 = 0, SA1_DIM0 = 0
SO-DIMMA SPD Address is 0xA0
SO-DIMMA TS Address is 0x30

If SA0 DIM0 = 0, SA1_DIM0 = 1
SO-DIMMA SPD Address is 0xA2
SO-DIMMA TS Address is 0x32

PART NUMBER	Height	TYPE
62.10017.K91	7.0mm	REVERSED
		REVERSED
		REVERSED
		REVERSED

1110 X02 Modify:
DM1 1st change to 62.10017.Q41; 2nd change to 62.10017.N11 on ST stage from ME updated connector list.

BOM1

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Title: **DDR3 SO-DIMM1**

Size	Document Number	Rev
Custom	CD1 DIS	SC

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BOM1

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Title	***Reserved***	
Size A3	Document Number **CD1 DIS**	Rev **SC**
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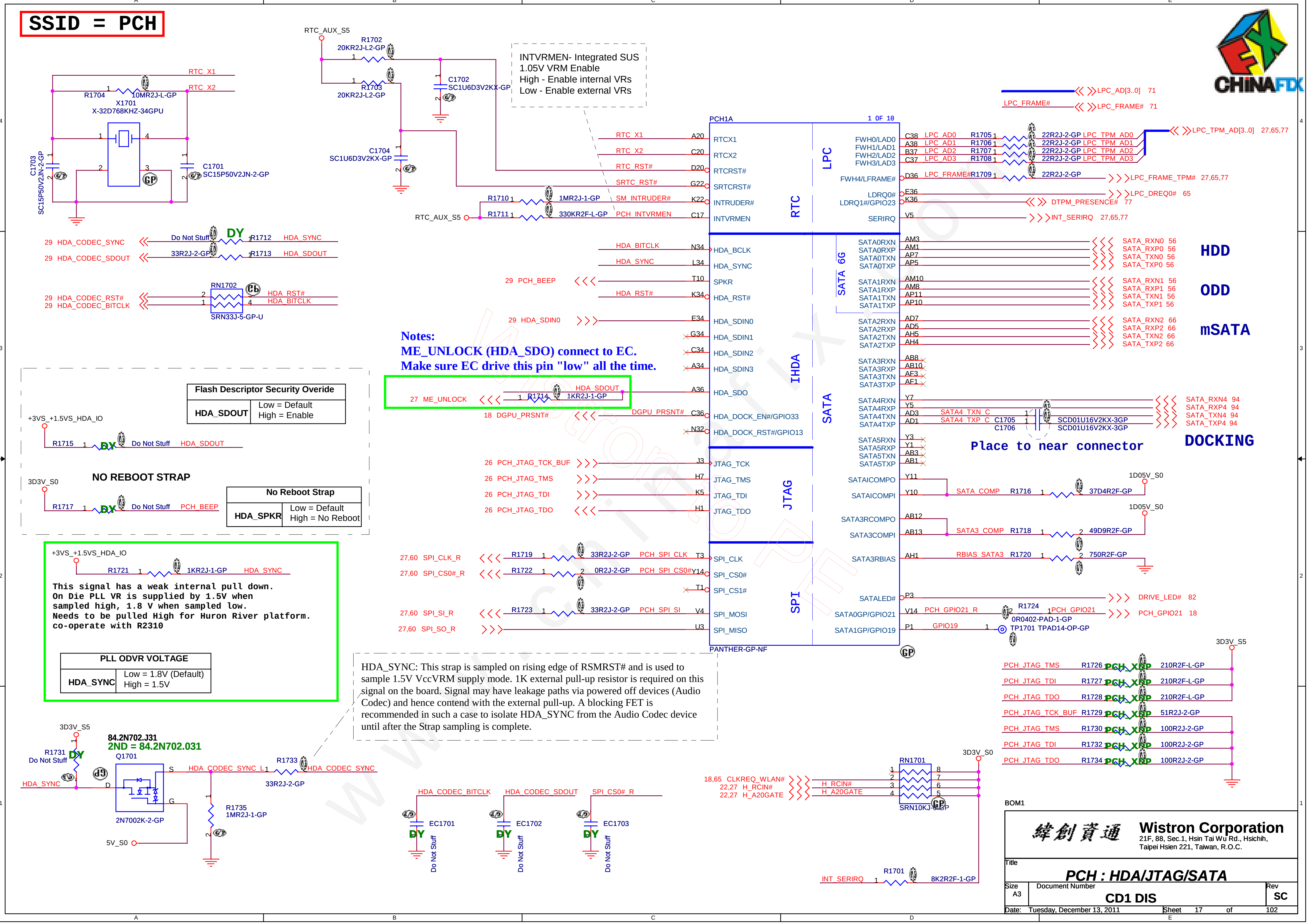
SSID = PCH



Notes:
ME_UNLOCK (HDA_SDO) connect to EC.
Make sure EC drive this pin "low" all the time.

Flash Descriptor Security Overide	
HDA_SDOUT	Low = Default High = Enable

NO REBOOT STRAP

No Reboot Strap	
HDA_SPKR	Low = Default High = No Reboot

This signal has a weak internal pull down.
On Die PLL VR is supplied by 1.5V when
sampled high, 1.8 V when sampled low.
Needs to be pulled High for Huron River platform.
co-operate with R2310

PLL ODVR VOLTAGE	
HDA_SYNC	Low = 1.8V (Default) High = 1.5V

HDA_SYNC: This strap is sampled on rising edge of RSMRST# and is used to sample 1.5V VccVRM supply mode. 1K external pull-up resistor is required on this signal on the board. Signal may have leakage paths via powered off devices (Audio Codec) and hence contend with the external pull-up. A blocking FET is recommended in such a case to isolate HDA_SYNC from the Audio Codec device until after the Strap sampling is complete.

BOM1

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Title	PCH : HDA/JTAG/SATA	
Size A3	Document Number CD1 DIS	Rev SC
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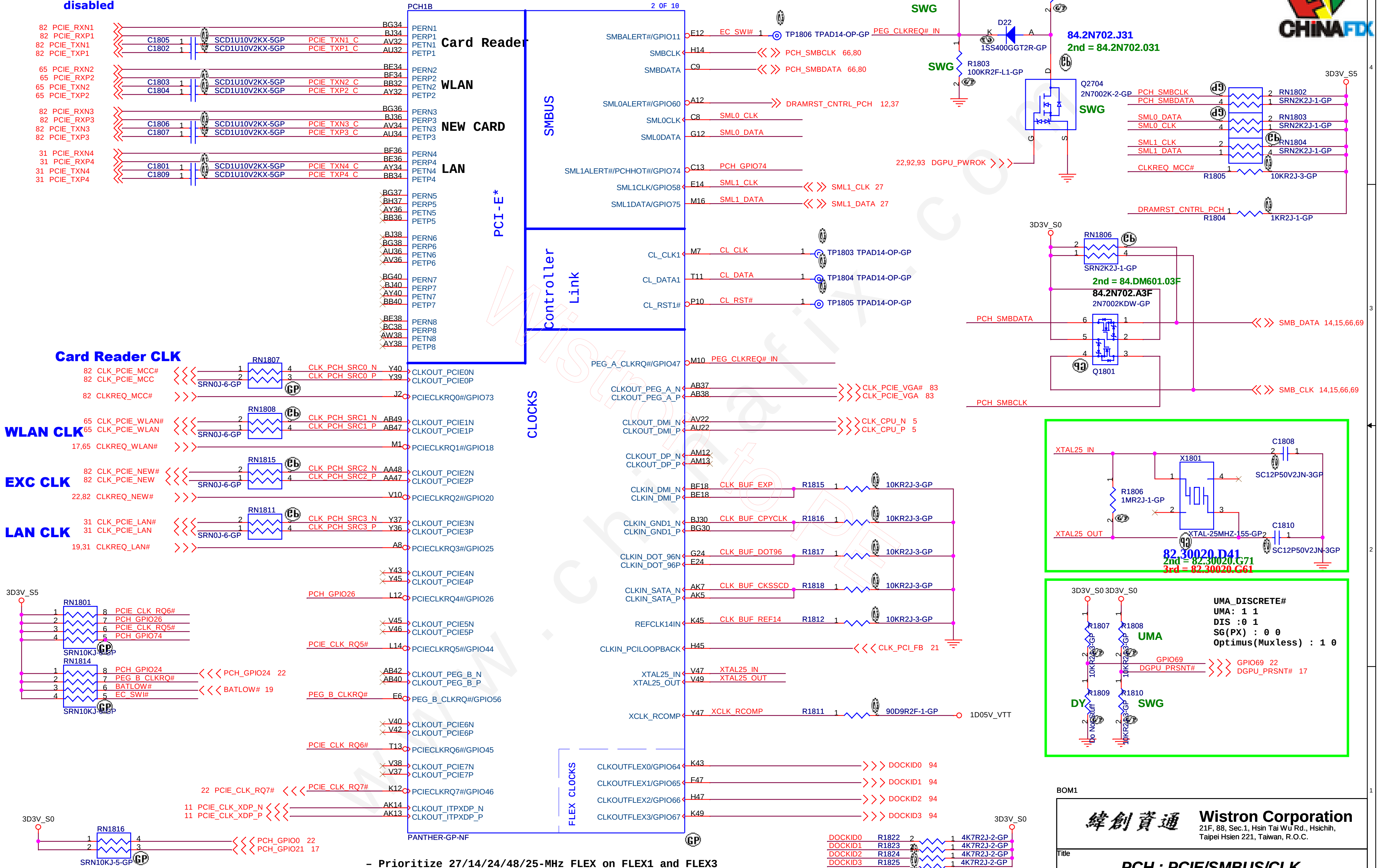
SSID = PCH

If PCIE port 1 is disabled, it will cause all PCIE port disabled

9/25 add



PCIECLKRQ1# and PCIECLKRQ2#
Support S0 power only

- Prioritize 27/14/24/48/25-MHz FLEX on FLEX1 and FLEX3
- Do not configure 27/14/24/48/25-MHz FLEX clock on FLEX0 and FLEX2 if more than 2 PCI clocks + PCI loopback are routed.

BOM1

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Title	PCH : PCIE/SMBUS/CLK	
Size A3	Document Number CD1 DIS	Rev SC
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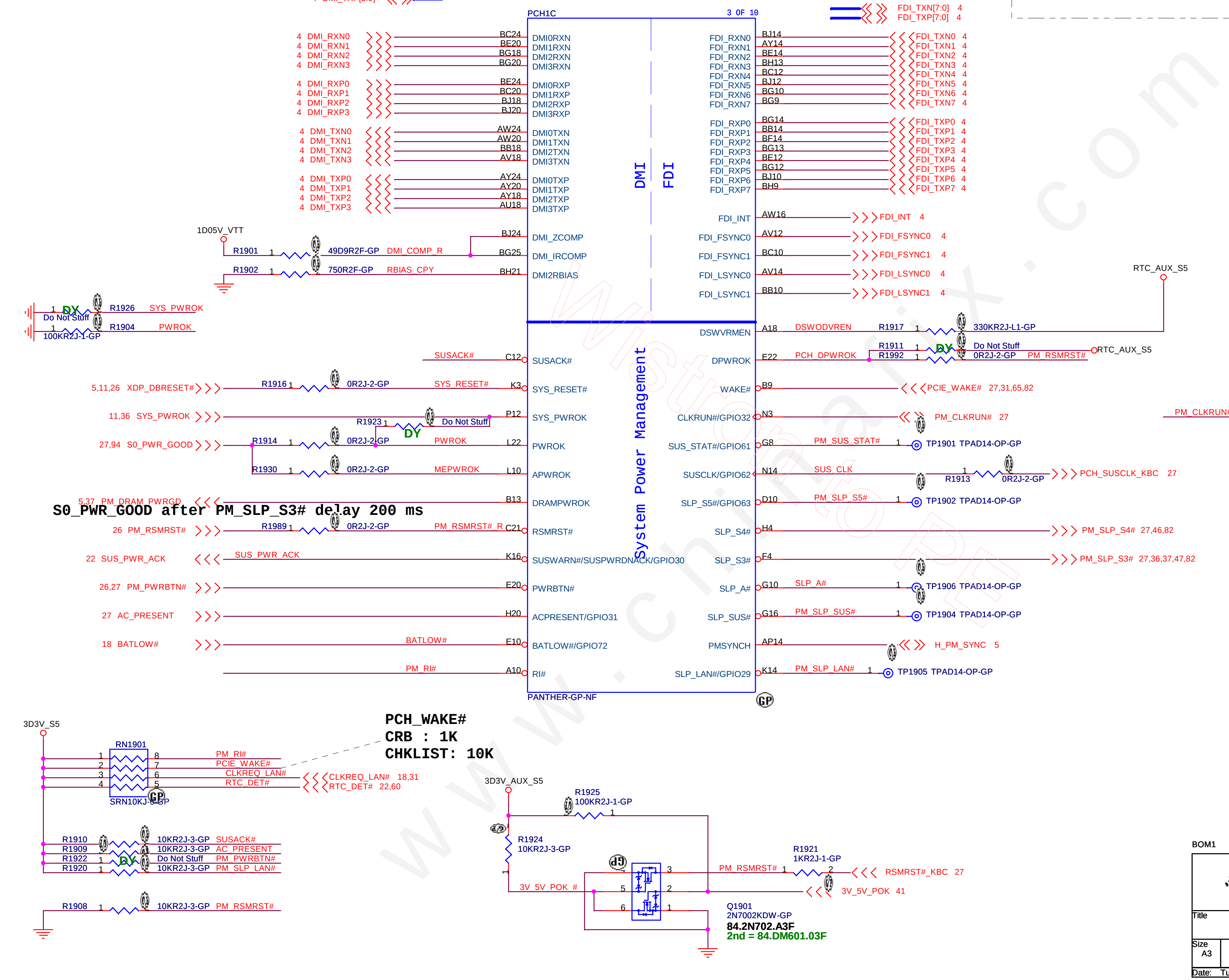
Signal Routing Guideline:
DMI_ZCOMP keep W=4 mils and routing length less than 500 mils.
DMI_IRCOMP keep W=4 mils and routing length less than 500 mils.

SSID = PCH

For platforms not supporting Deep S4/S5
1.VccSUS3_3 and VccDSW3_3 will rise at the same time (connected on board)
2.DPWROK and RSMRST# will rise at the same time (connected on board)
3.SLP_SUS# and SUSACK# are left as ‘no connect’
4.SUSWARN# used as SUSPWRDNACK/GPIO30



BOM1

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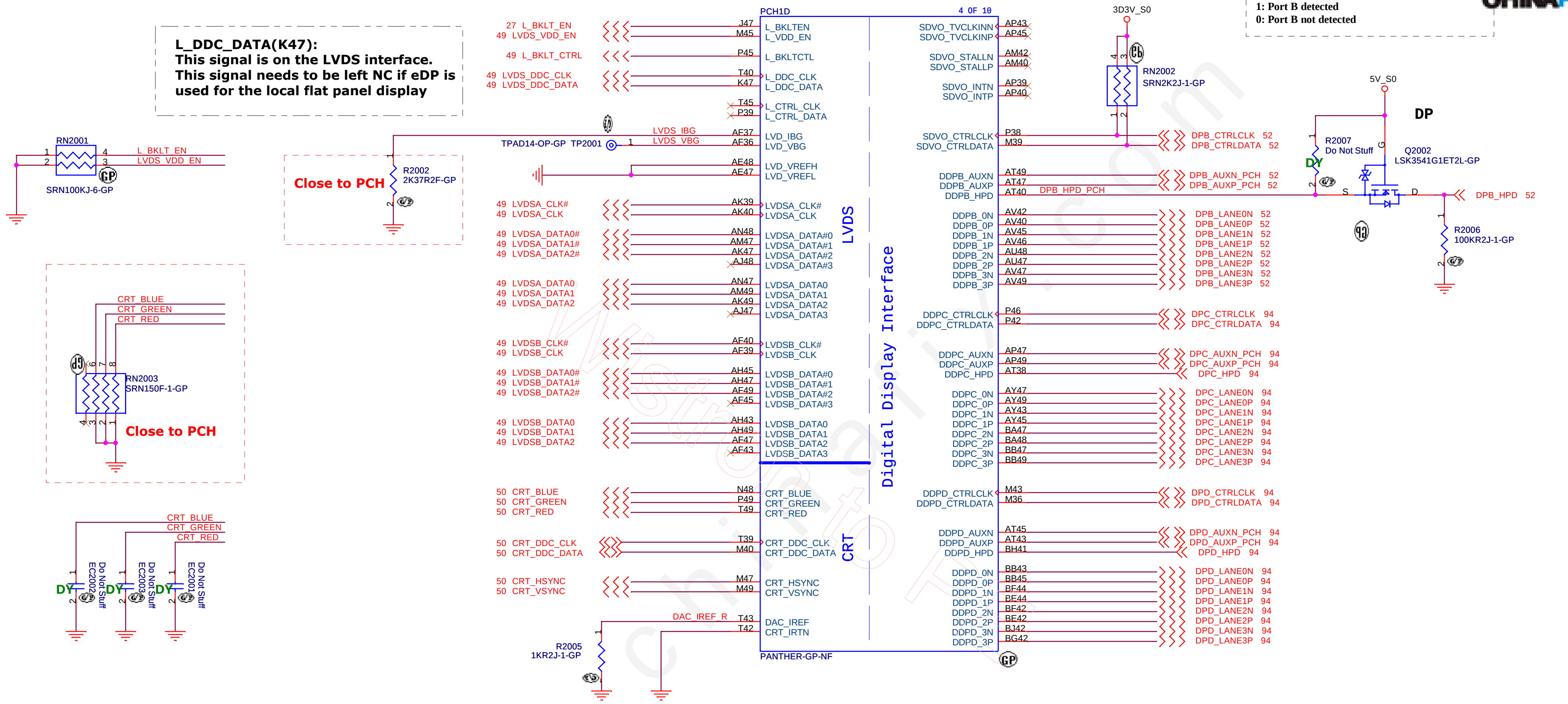
Title	PCH : DMI/FDI/PM	
Size A3	Document Number: CD1 DIS	Rev SC
Date: Tuesday, December 13, 2011	Sheet 19 of 102	

L_DDC_DATA(K47):
This signal is on the LVDS interface.
This signal needs to be left NC if eDP is used for the local flat panel display

DDI Port B Detect:(SDVO_CTRL_ DATA)
1: Port B detected
0: Port B not detected



PORT	DDI PCH Pin Names	SDVO Mapping	DisplayPort Mapping	HDMI/DVI Mapping
PORT-B	DDPB_[0]P	SDVO_RED	DDPB_[0]P	TMDSB_DATA2
	DDPB_[0]N	SDVO_RED#	DDPB_[0]N	TMDSB_DATA2#
	DDPB_[1]P	SDVO_GREEN	DDPB_[1]P	TMDSB_DATA1
	DDPB_[1]N	SDVO_GREEN#	DDPB_[1]N	TMDSB_DATA1#
	DDPB_[2]P	SDVO_BLUE	DDPB_[2]P	TMDSB_DATA0
	DDPB_[2]N	SDVO_BLUE#	DDPB_[2]N	TMDSB_DATA0#
	DDPB_[3]P	SDVO_CLK	DDPB_[3]P	TMDSB_CLK
	DDPB_[3]N	SDVO_CLK#	DDPB_[3]N	TMDSB_CLK#
	DDPB_AUXP	NA	DDPB_AUXP	NA
	DDPB_AUXN	NA	DDPB_AUXN	NA
	DDPB_HPD	NA	DDPB_HPD	HDMIB_HPD
	SDVO_CTRLCLK	SDVO_CTRLCLK	NA	HDMIB_CTRLCLK
	SDVO_CTRLDATA	SDVO_CTRLDATA	NA	HDMIB_CTRLDATA

Digital Display Ports Enable and Disable Guidelines

Port	Strap	How to Enable Port?	How to Disable Port?
LVDS	L_DDC_DATA	Pull up to 3.3 V with 2.2-kΩ ±5% resistor	No Connect
Port B	SDVO_CTRLDATA	Pull up to 3.3 V with 2.2-kΩ ±5% resistor	No Connect
Port C	DDPC_CTRLDATA	Pull up to 3.3 V with 2.2-kΩ ±5% resistor	No Connect
Port D	DDPD_CTRLDATA	Pull up to 3.3 V with 2.2-kΩ ±5% resistor	No Connect

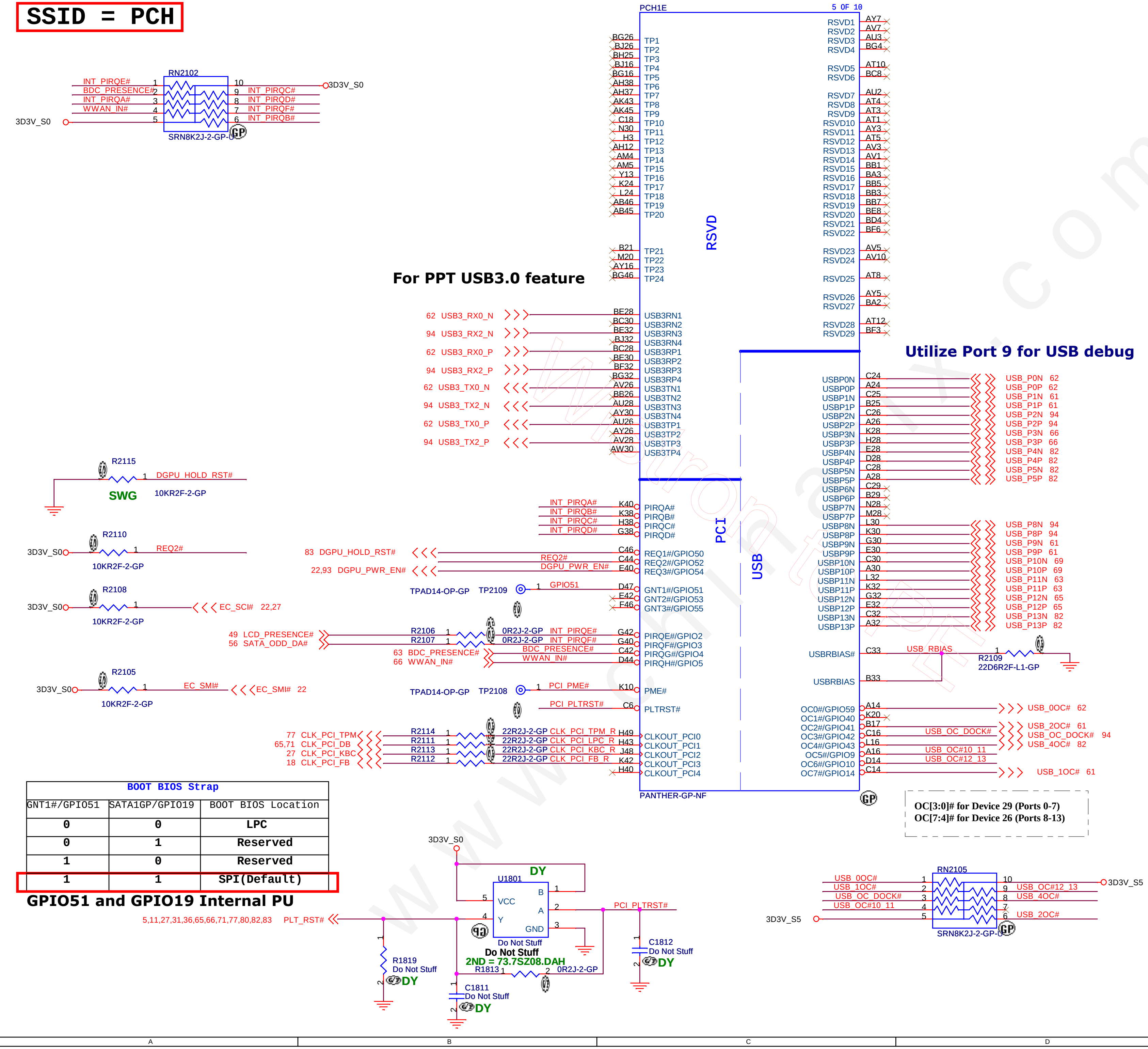
NOTE: LVDS and eDP on processor can not be enabled at the same time.

BOM1

緯創資通 **Wistron Corporation**
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih, Taipei Hsien 221, Taiwan, R.O.C.

Title		
PCH : LVDS/CRT/DDI		
Size: A3	Document Number: CD1 DIS	Rev: SC
Date: Tuesday, December 13, 2011	Sheet 20 of 102	

SSID = PCH



USB Table

Pair	Device
0	USB3.0 port 0
1	USB2.0 port 1
2	USB3.0 Docking
3	WWAN
4	USB2.0 port (AUO4)
5	New Card
6	X
7	X
8	USB2.0 Docking
9	USB2.0 port 2
10	FPR
11	BLUETOOTH
12	WLAN
13	Camera

BOOT BIOS Strap

GNT1#/GPIO51	SATA1GP/GPIO19	BOOT BIOS Location
0	0	LPC
0	1	Reserved
1	0	Reserved
1	1	SPI(Default)

GPIO51 and GPIO19 Internal PU

BOM1

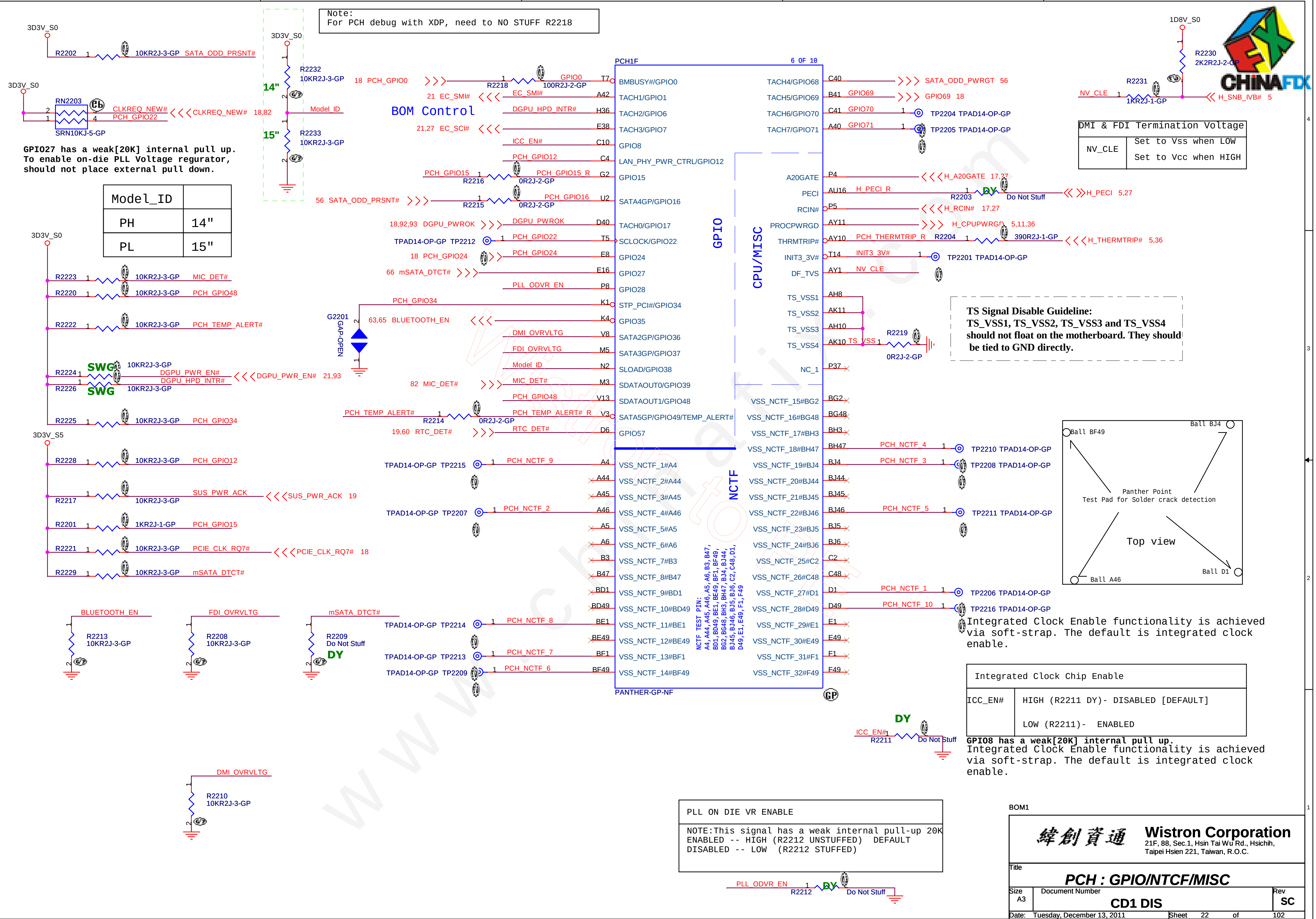
緯創資通 Wistron Corporation
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih, Taipei Hsien 221, Taiwan, R.O.C.

Title: PCH : PCI/USB/NVRAM/RSVD

Size	Document Number	Rev
A3	CD1 DIS	SC

Date: Tuesday, December 13, 2011 Sheet 21 of 102

Note:
For PCH debug with XDP, need to NO STUFF R2218



GPIO27 has a weak[20K] internal pull up.
To enable on-die PLL Voltage regurator,
should not place external pull down.

Model_ID	
PH	14"
PL	15"

DMI & FDI Termination Voltage

NV_CLE	Set to Vss when LOW Set to Vcc when HIGH

TS Signal Disable Guideline:
TS_VSS1, TS_VSS2, TS_VSS3 and TS_VSS4 should not float on the motherboard. They should be tied to GND directly.

Integrated Clock Enable functionality is achieved via soft-strap. The default is integrated clock enable.

Integrated Clock Chip Enable

ICC_EN#	HIGH (R2211 DY)- DISABLED [DEFAULT] LOW (R2211)- ENABLED

GPIO8 has a weak[20K] internal pull up.
Integrated Clock Enable functionality is achieved via soft-strap. The default is integrated clock enable.

PLL ON DIE VR ENABLE

NOTE:This signal has a weak internal pull-up 20K
ENABLED -- HIGH (R2212 UNSTUFFED) DEFAULT
DISABLED -- LOW (R2212 STUFFED)

BOM1

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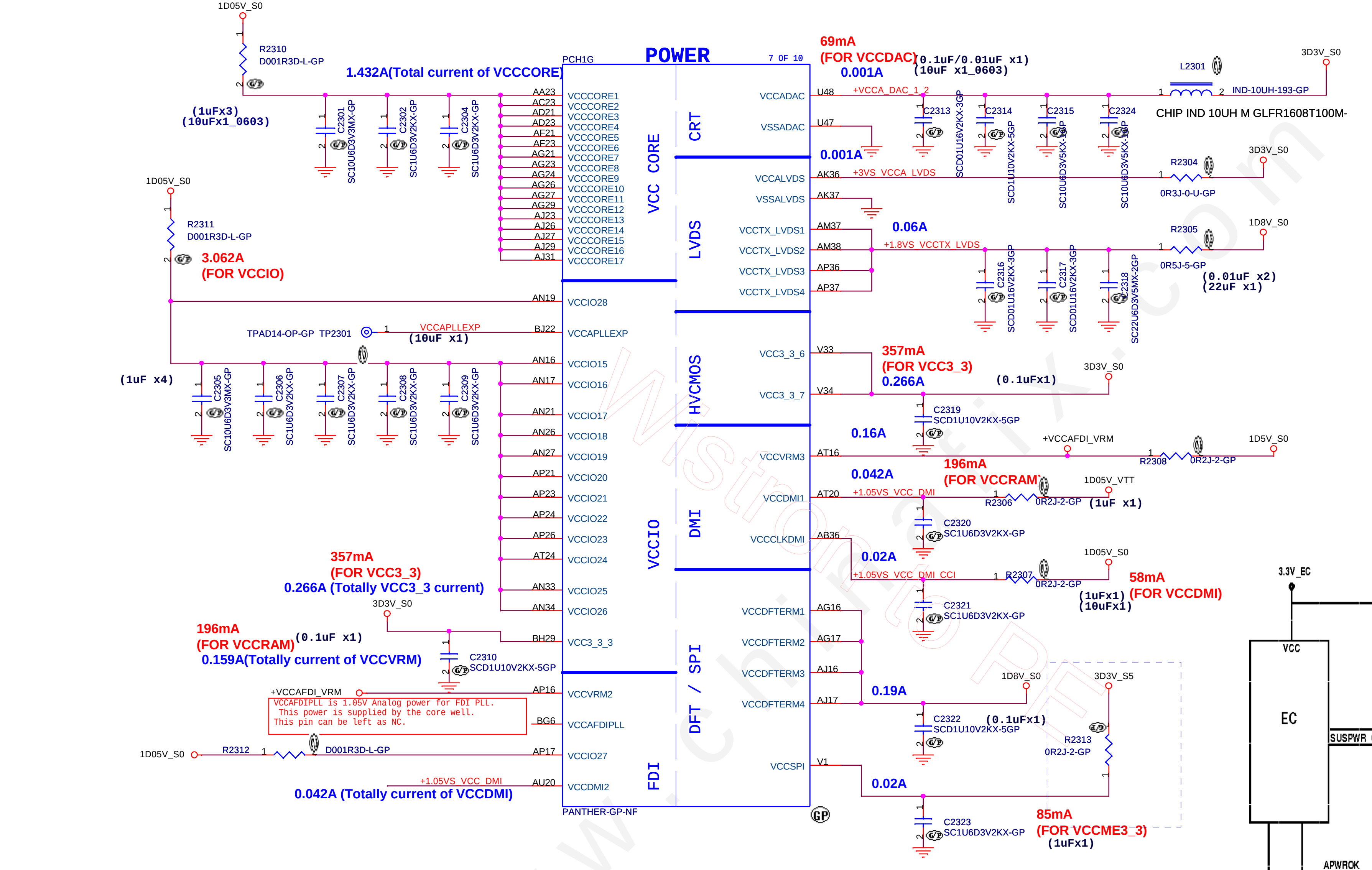
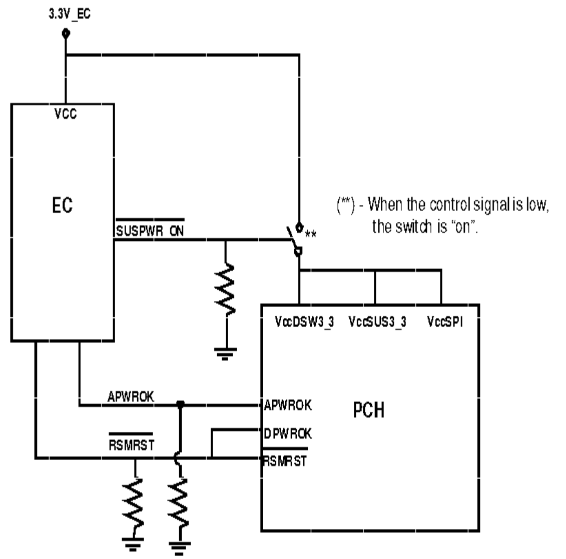
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih, Taipei Hsien 221, Taiwan, R.O.C.
Title: PCH : GPIO/NTCF/MISC
Size: A3 Document Number: CD1 DIS Rev: SC
Date: Tuesday, December 13, 2011 Sheet 22 of 102

SSID = PCH 6A



Voltage Rail	Voltage	Iccmax
V_PROC_IO	1.05	0.001
V5REF	5	0.001
V5REF_Sus	5	0.001
Vcc3_3	3.3	0.266
VccADAC3	3.3	0.001
VccADPLLA	1.05	0.08
VccADPLLB	1.05	0.08
VccCore	1.05	1.3
VccDMI	1.1	0.042
VccIO3	1.05	2.925
VccASW	1.05	1.01
VccSPI	3.3	0.02
VccDSW3_3	3.3	0.002
VccDFTERM	1.8	0.19
VccRTC	3.3	6u
VccSus3_3	3.3	0.097
VccSusHDA	3.3	0.01
VccVRM	1.5	0.16
VccClkDMI	1.05	0.02
VccSSC	1.05	0.095
VccDIFFCLKN	1.05	0.055
VccALVDS	3.3	0.001
VccTX_LVDS3	1.8	0.06

Refer to NPCE795 shared SPI flash architecture



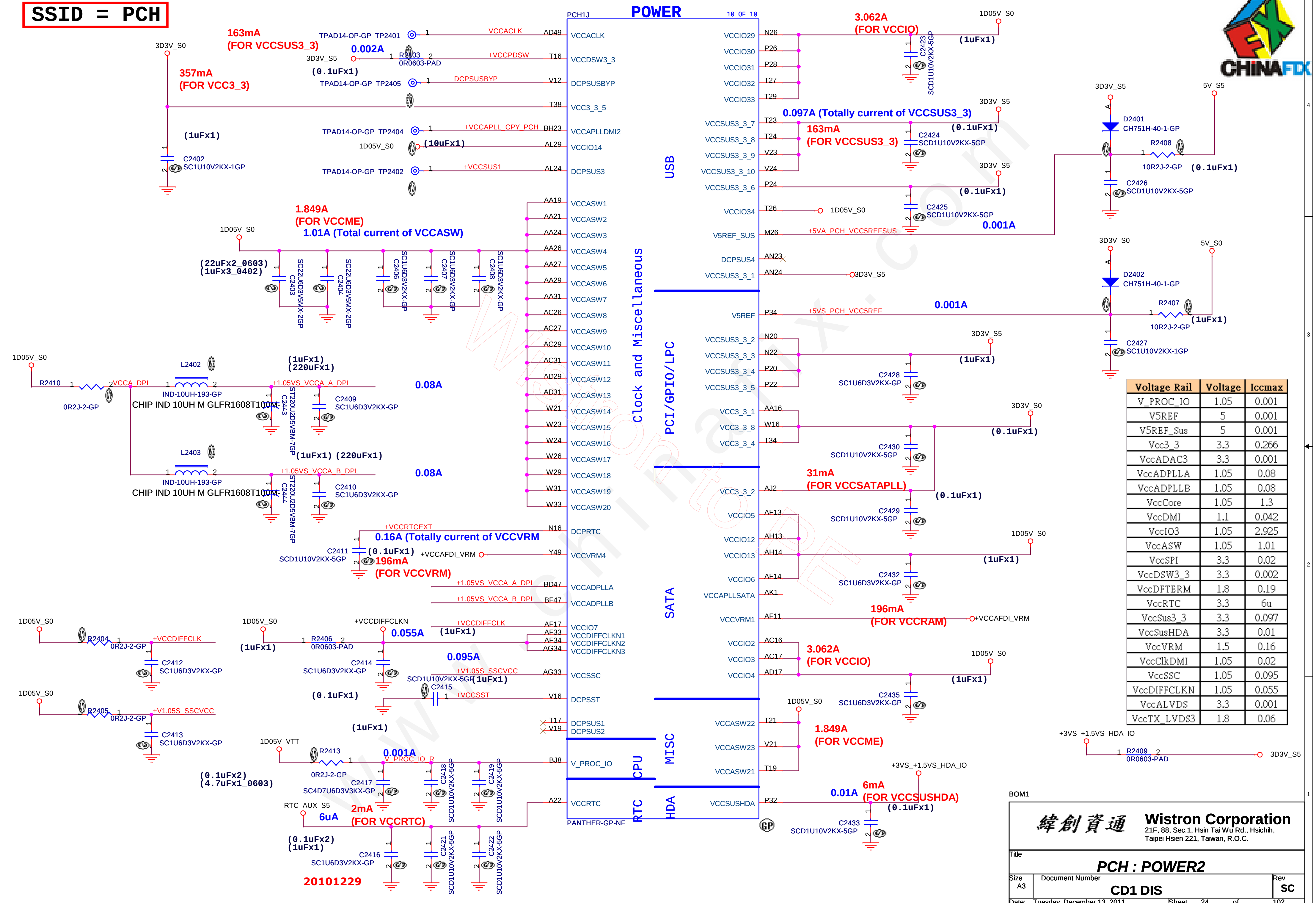
緯創資通 Wistron Corporation
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title: PCH : POWER1

Size: A3 | Document Number: CD1 DIS | Rev: SC

Date: Tuesday, December 13, 2011 | Sheet 23 of 102

SSID = PCH



Voltage Rail	Voltage	Iccmax
V_PROC_IO	1.05	0.001
V5REF	5	0.001
V5REF_Sus	5	0.001
Vcc3_3	3.3	0.266
VccADAC3	3.3	0.001
VccADPLLA	1.05	0.08
VccADPLLB	1.05	0.08
VccCore	1.05	1.3
VccDMI	1.1	0.042
VccIO3	1.05	2.925
VccASW	1.05	1.01
VccSPI	3.3	0.02
VccDSW3_3	3.3	0.002
VccDFTERM	1.8	0.19
VccRTC	3.3	6u
VccSus3_3	3.3	0.097
VccSusHDA	3.3	0.01
VccVRM	1.5	0.16
VccClkDMI	1.05	0.02
VccSSC	1.05	0.095
VccDIFFCLKN	1.05	0.055
VccALVDS	3.3	0.001
VccTX_LVDS3	1.8	0.06

BOM1

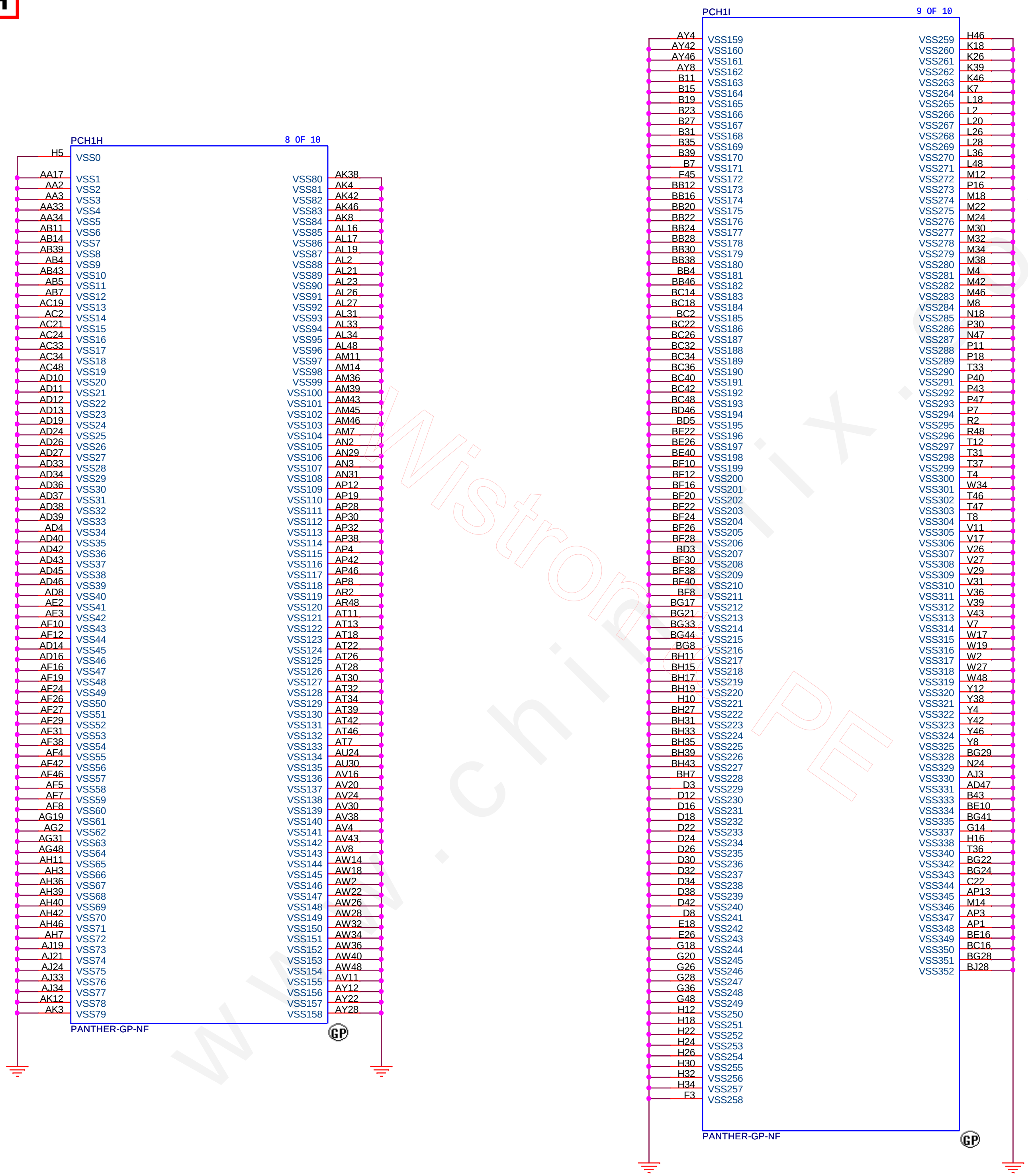
緯創資通 Wistron Corporation
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih, Taipei Hsien 221, Taiwan, R.O.C.

Title: PCH : POWER2

Size: A3 | Document Number: CD1 DIS | Rev: SC

Date: Tuesday, December 13, 2011 | Sheet 24 of 102

SSID = PCH



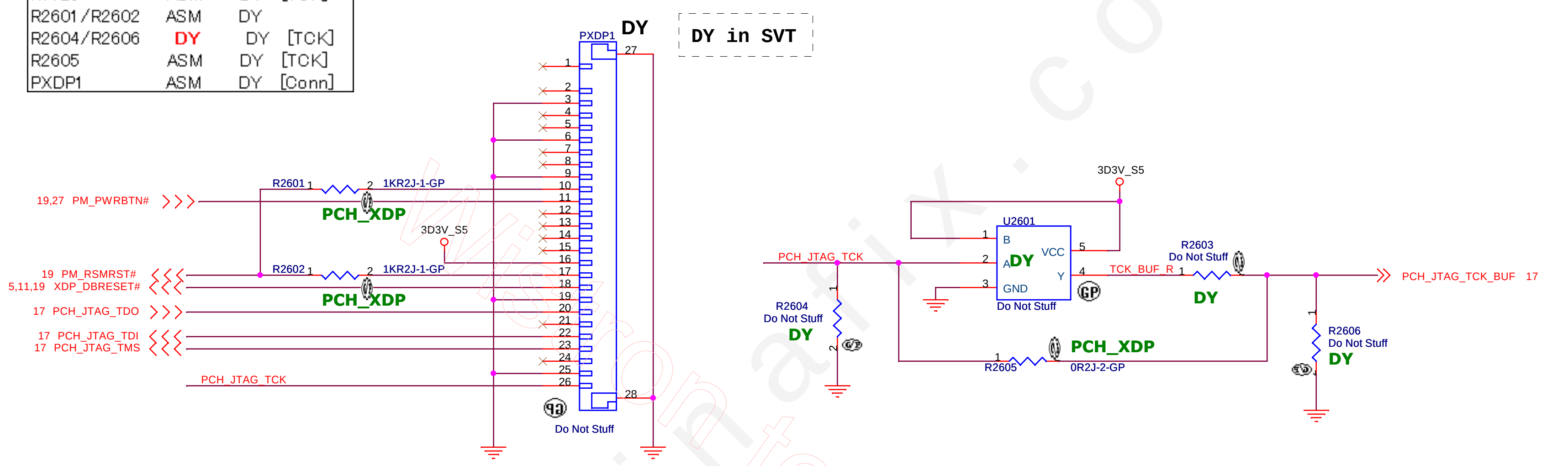
BOM1

緯創資通	**Wistron Corporation** 21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih, Taipei Hsien 221, Taiwan, R.O.C.	
Title	***PCH : VSS***	
Size A3	Document Number **CD1 DIS**	Rev **SC**
Date: Tuesday, December 13, 2011	Sheet 25 of 102	

9/2 PCH_XDP

Part	Enable	Disable	
R1726/R1730	ASM	ASM	[TMS]
R1727/R1731	ASM	ASM	[TDI]
R1728/R1732	ASM	DY	[TDO]
R1729	ASM	DY	[TCK]
R2601/R2602	ASM	DY	
R2604/R2606	DY	DY	[TCK]
R2605	ASM	DY	[TCK]
PXDP1	ASM	DY	[Conn]

PCH XDP



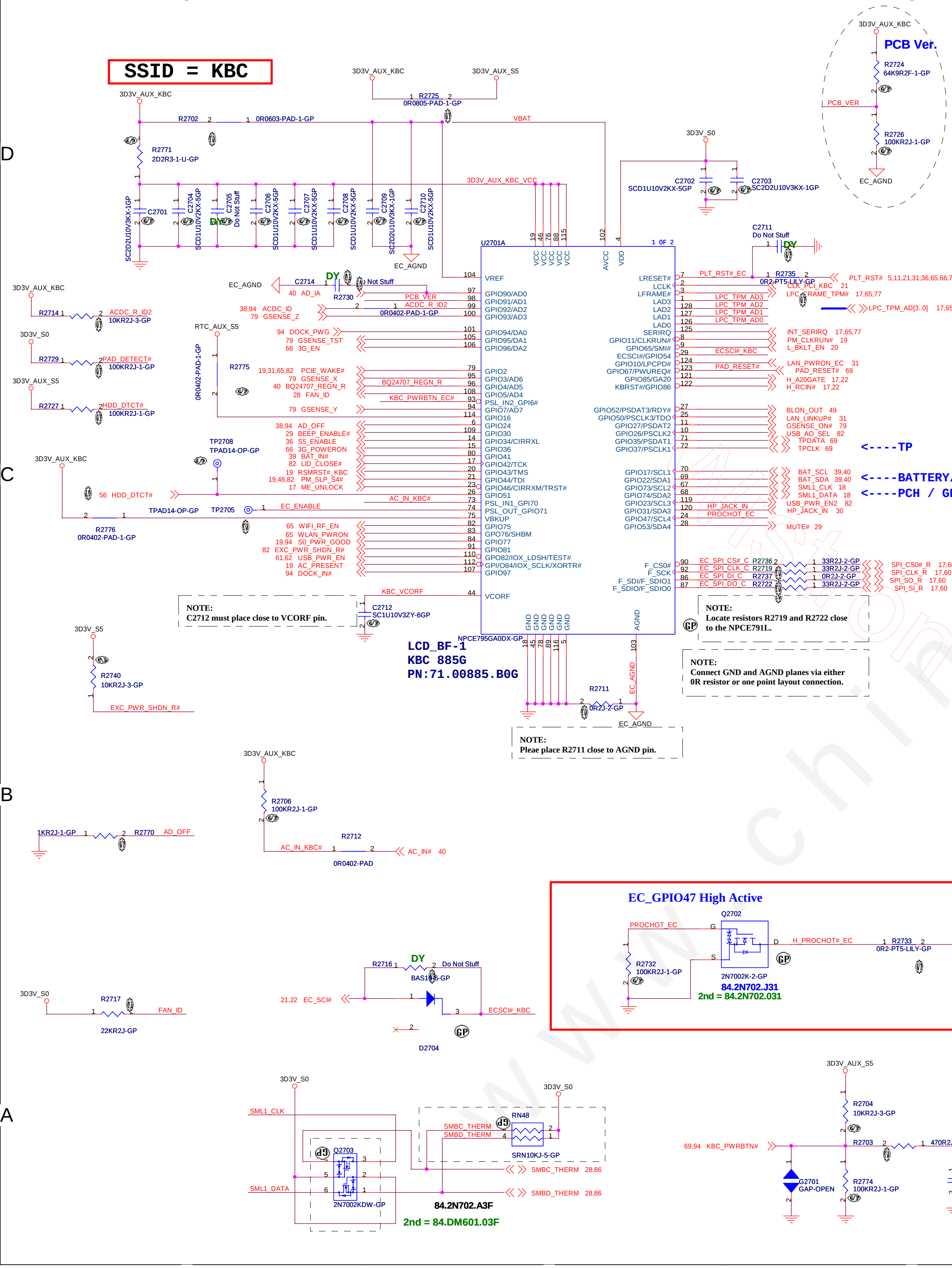
BOM1

緯創資通 Wistron Corporation
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title: PCH_XDP

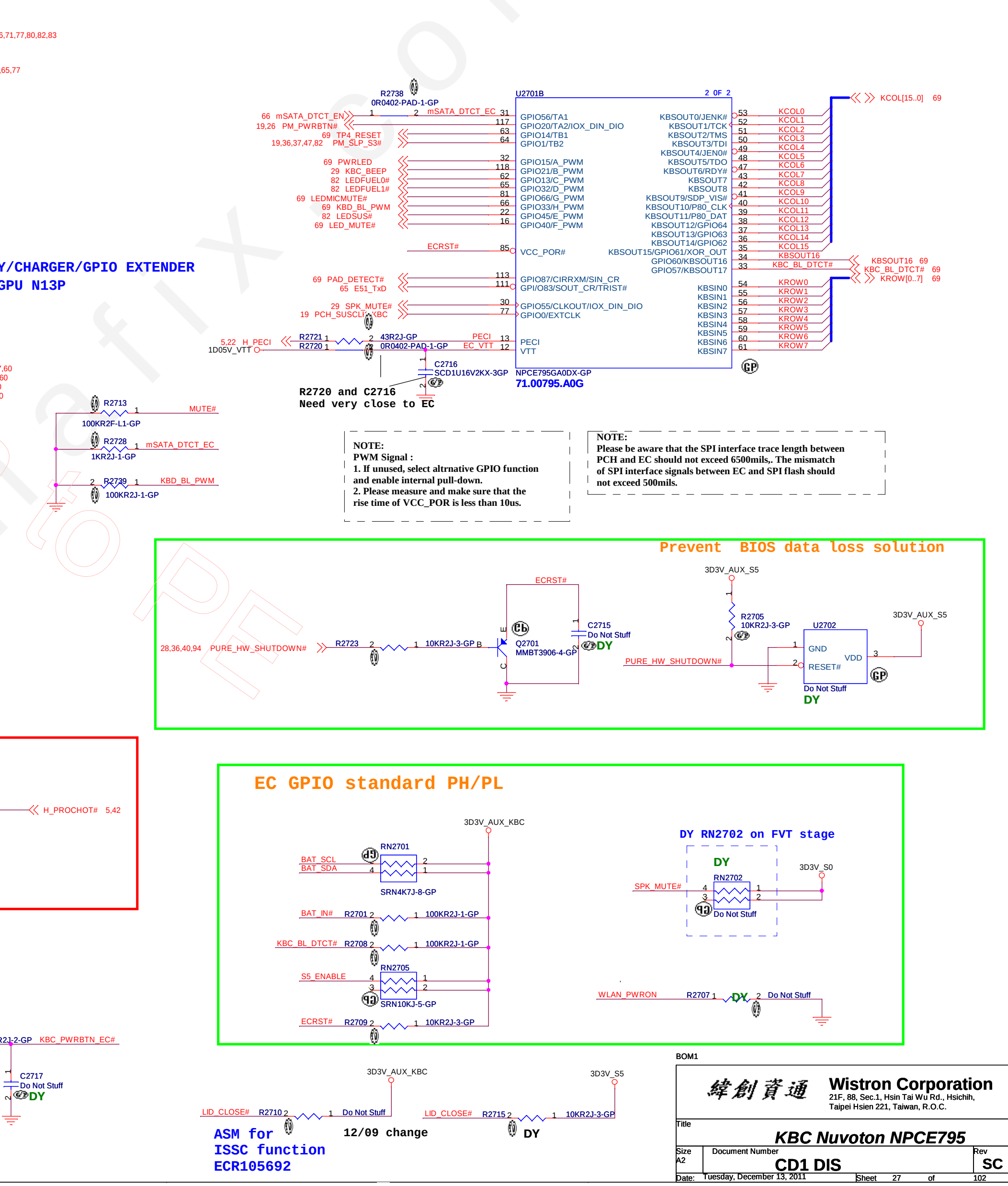
Size: A3 | Document Number: CD1 DIS | Rev: SC

Date: Tuesday, December 13, 2011



PCB Version A/D (PIN 98)		Pull-Low Resistor R2726	Pull-High Resistor (3D3V_AUX_KBC) R2724	Voltage
SA	14"	100K	10K	3.0V
	15"		20K	2.75V
SB	14"	100K	33K	2.48V
	15"		47K	2.24V
SC	14"	100K	64.9K	2.0V
	15"		76.8K	1.87V
1	14"	100K	100K	1.65V
	15"		143K	1.36V

PCB Version A/D (PIN 98)		Pull-Low Resistor	Pull-High Resistor (3D3V_AUX_KBC)	Voltage
Reserved	14"	100K	174K	1.2V
	15"		215K	1.04V
Reserved	14"			
	15"			
Reserved	14"			
	15"			
Reserved	14"			
	15"			



緯創資通 Wistron Corporation
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih, Taipei Hsien 221, Taiwan, R.O.C.

Title: KBC Nuvoton NPCE795

Size	Document Number	Rev
A2	CD1 DIS	SC

Date: Tuesday, December 13, 2011 Sheet 27 of 102

SSID = Thermal

Thermal sensor

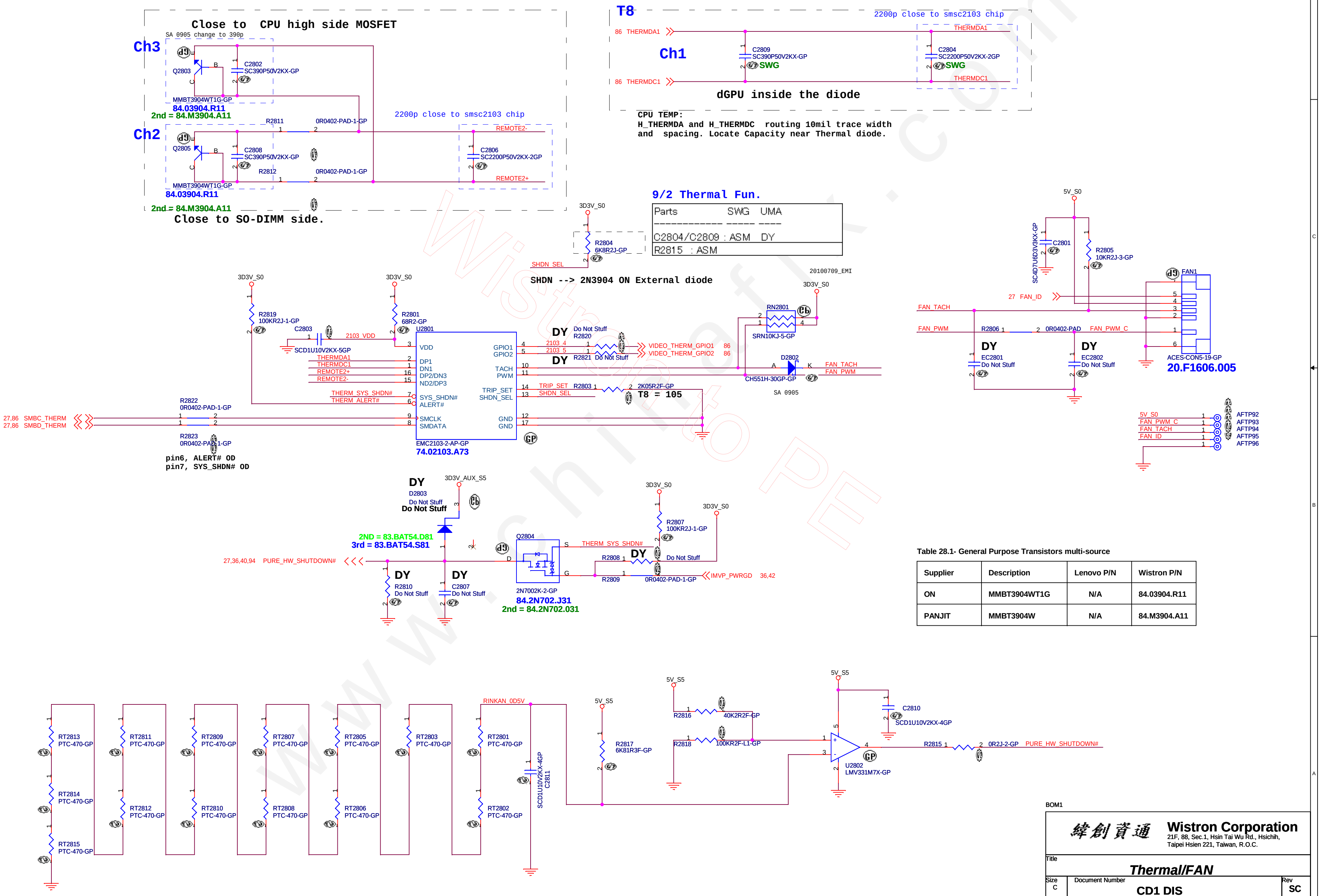


Table 28.1- General Purpose Transistors multi-source

Supplier	Description	Lenovo P/N	Wistron P/N
ON	MMBT3904WT1G	N/A	84.03904.R11
PANJIT	MMBT3904W	N/A	84.M3904.A11

BOM1

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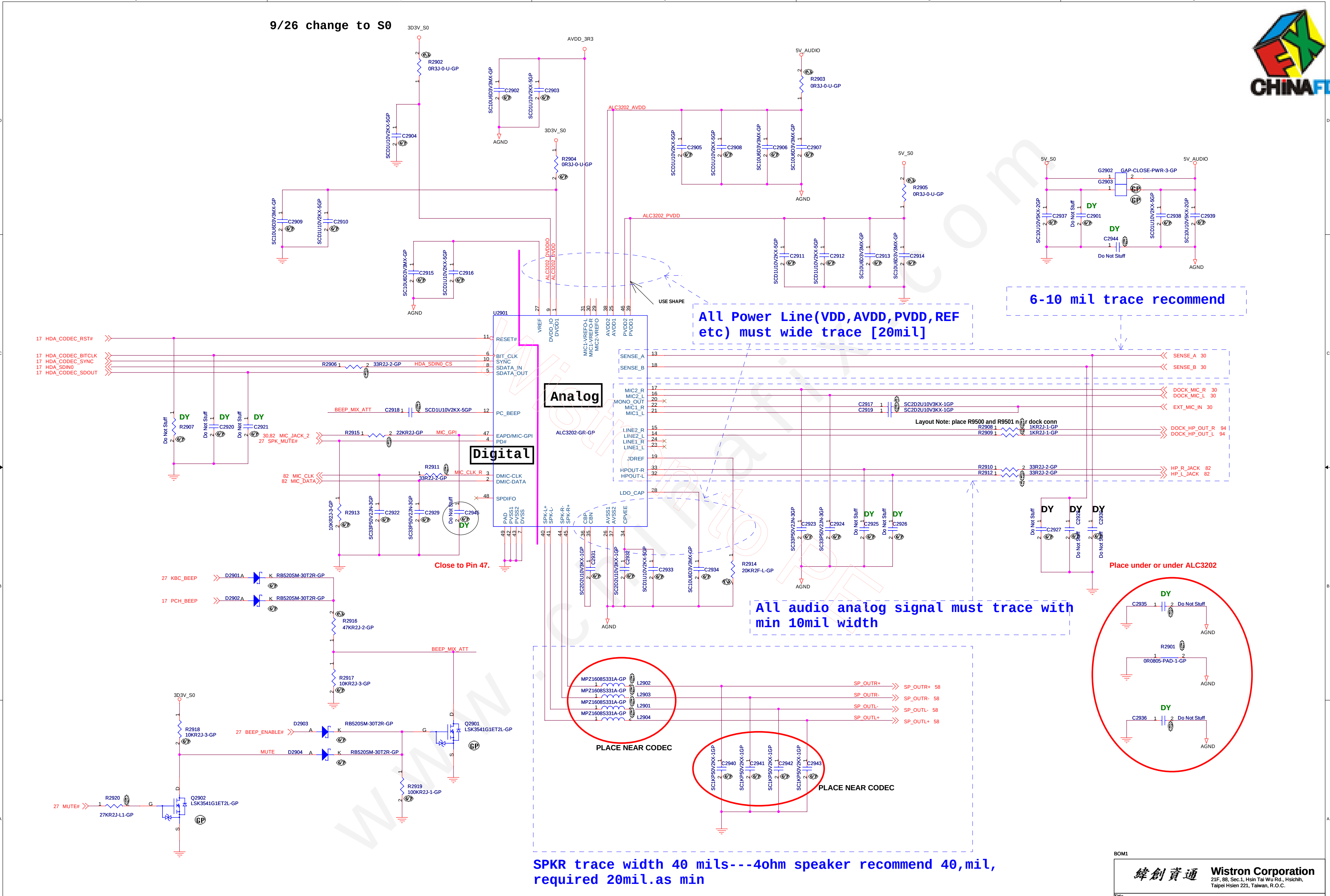
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title: ***Thermal/FAN***

Size: C | Document Number: **CD1 DIS** | Rev: **SC**

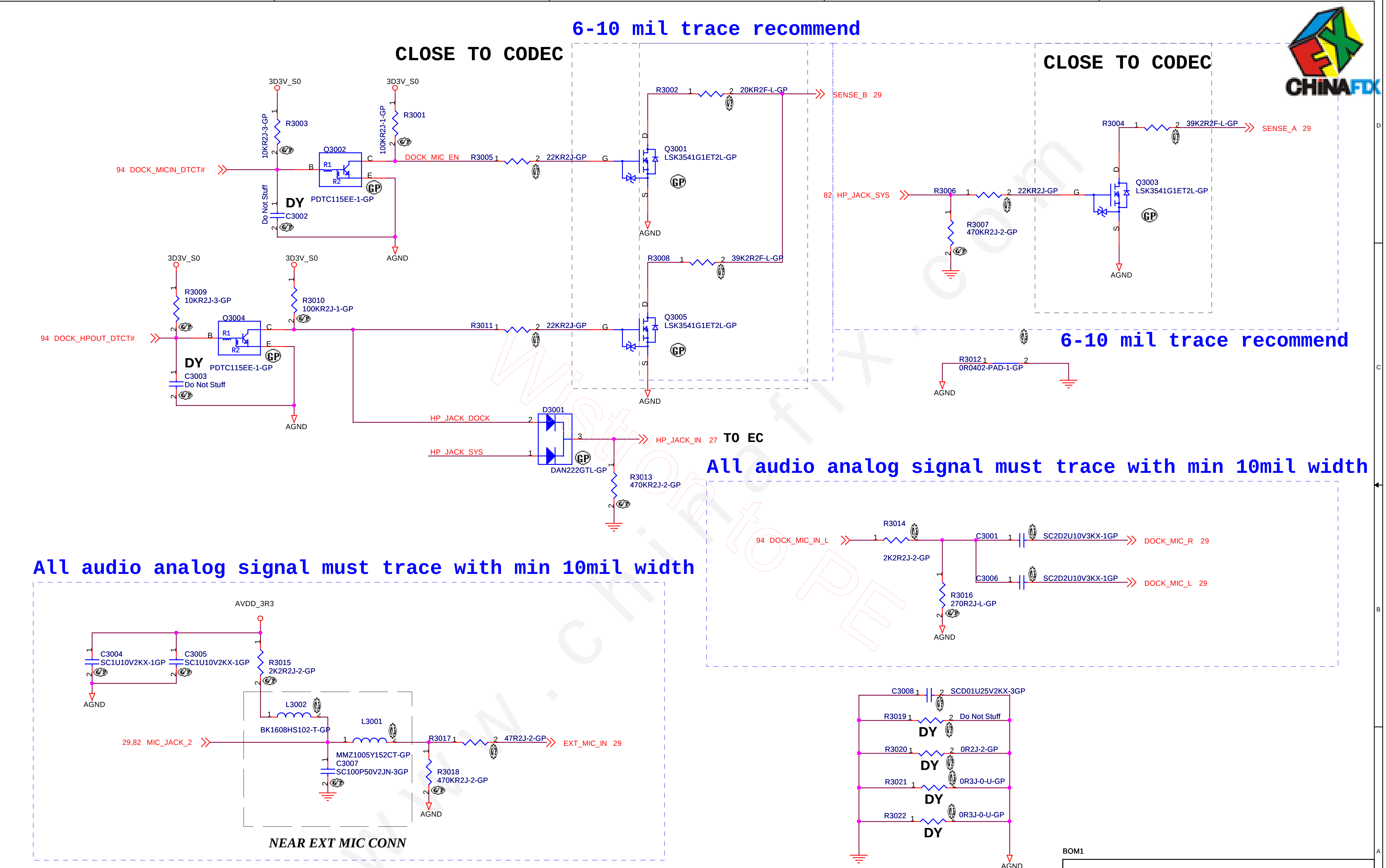
Date: Tuesday, December 13, 2011 | Sheet 28 of 102



BOM1

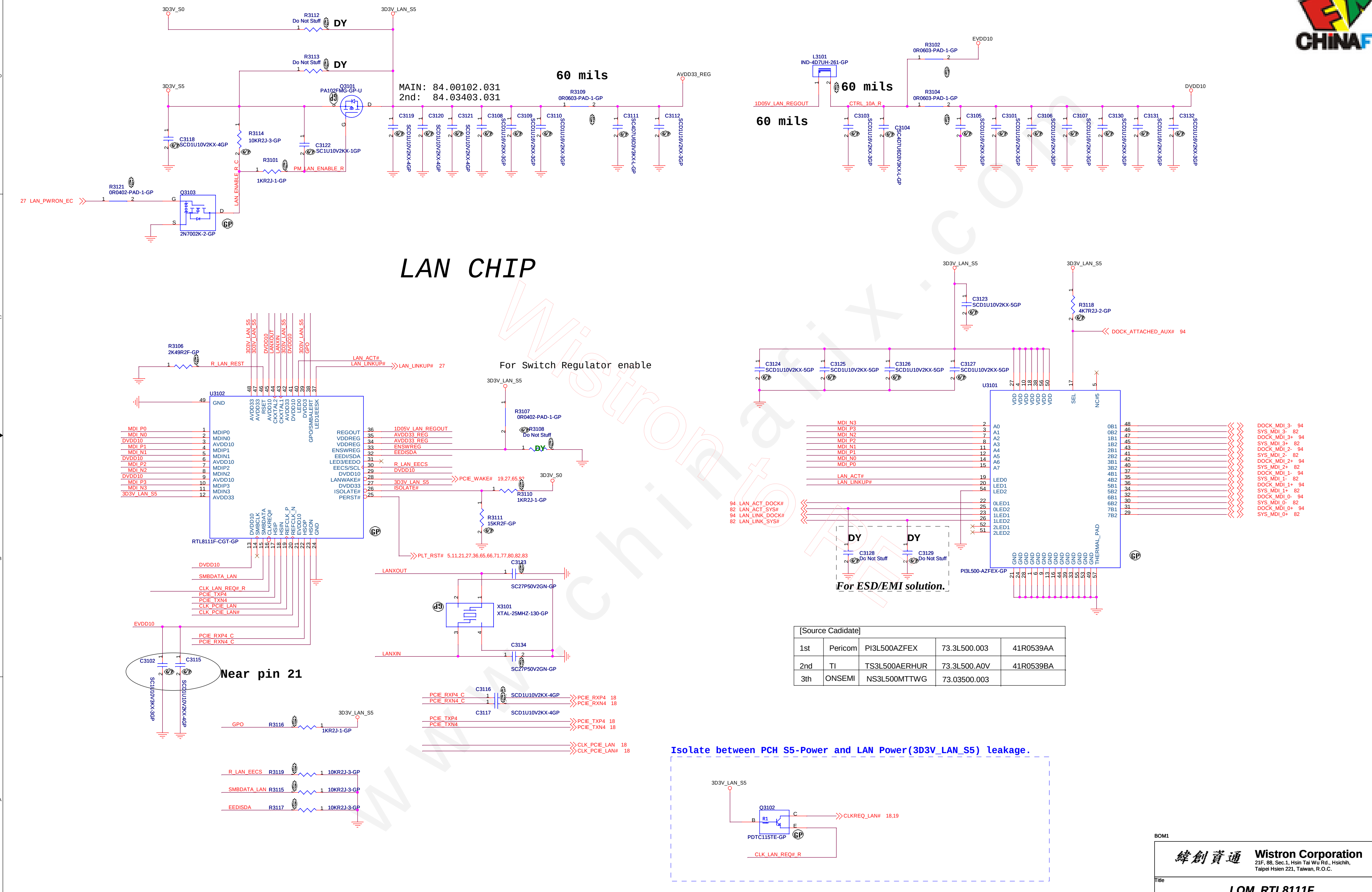
Wistron Corporation
緯創資通
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih, Taipei Hsien 221, Taiwan, R.O.C.

Title	Audio Codec ALC3202	
Size A2	Document Number CD1 DIS	Rev SC
Date: Tuesday, December 13, 2011	Sheet 29 of 102	



緯創資通 **Wistron Corporation**
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih, Taipei Hsien 221, Taiwan, R.O.C.

Title	Speaker Conn	
Size: A3	Document Number: CD1 DIS	Rev: SC
Date: Tuesday, December 13, 2011	Sheet 30 of 102	



[Source Cadidate]				
1st	Pericom	PI3L500AZFEX	73.3L500.003	41R0539AA
2nd	TI	TS3L500AERHUR	73.3L500.A0V	41R0539BA
3th	ONSEMI	NS3L500MTTWG	73.03500.003	

BOM1

緯創資通 **Wistron Corporation**

21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih, Taipei Hsien 221, Taiwan, R.O.C.

Title		
LOM_RTL8111F		
Size: A2	Document Number: CD1 DIS	Rev: SC
Date: Tuesday, December 13, 2011	Sheet 31 of 102	

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D C B A

CHINAFIX

(Blanking)

BOM1

緯創資通	Wistron Corporation 21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih, Taipei Hsien 221, Taiwan, R.O.C.	
Title	Reserved	
Size A3	Document Number CD1 DIS	Rev SC
Date: Tuesday, December 13, 2011	Sheet 33 of 102	

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BOM1

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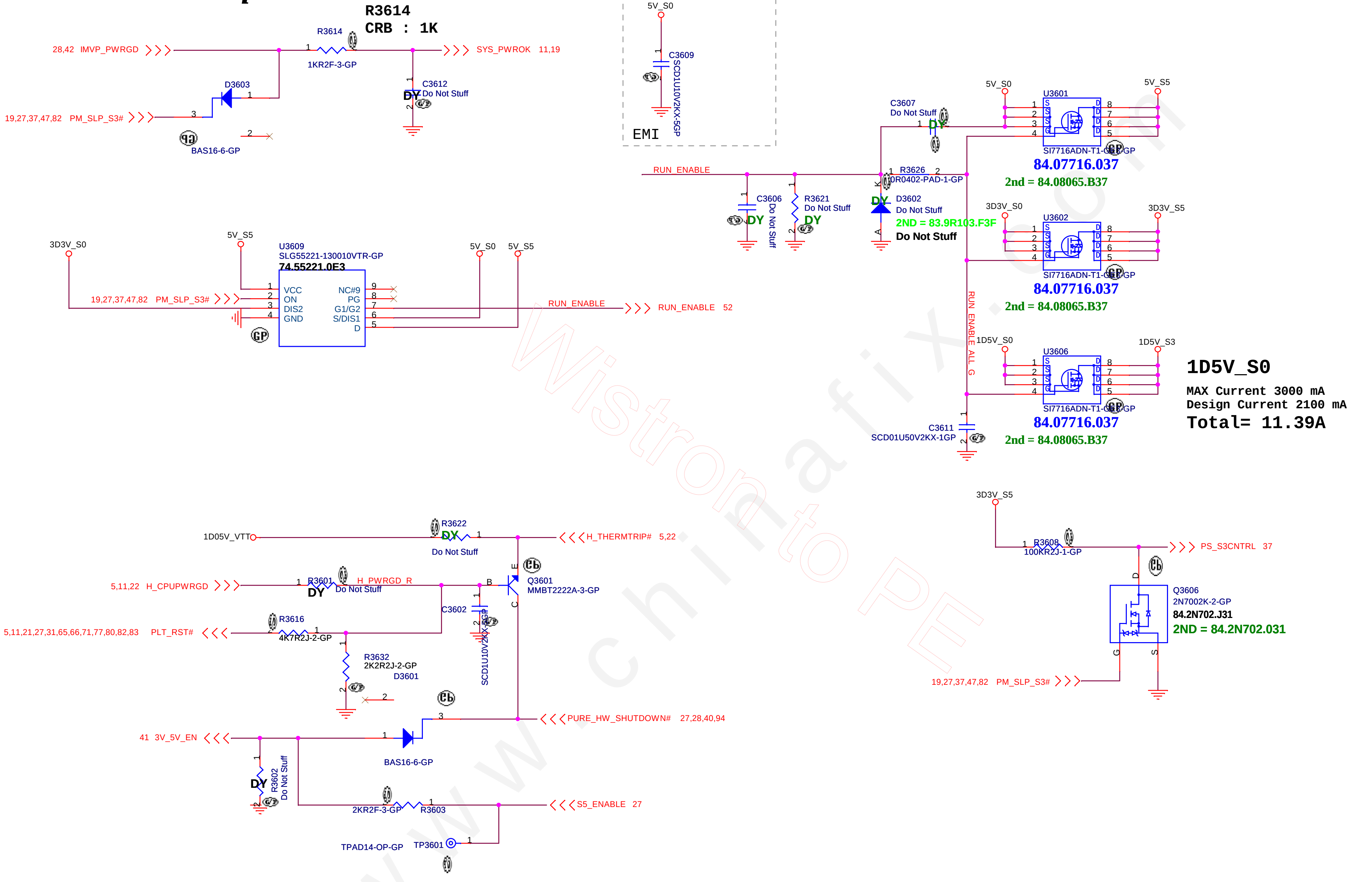
BOM1

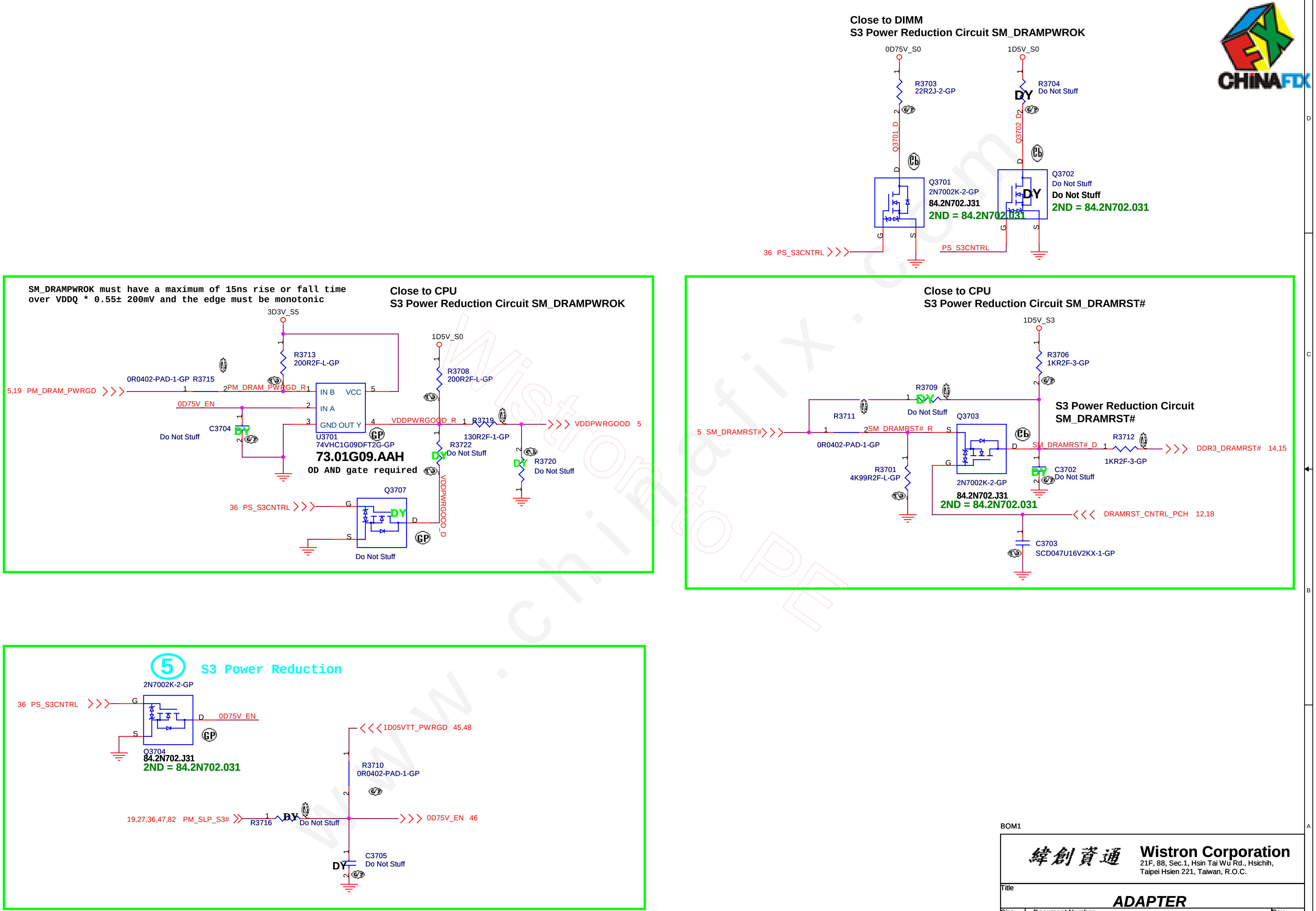
Power Sequence



BOM1

緯創資通 **Wistron Corporation**
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih, Taipei Hsien 221, Taiwan, R.O.C.

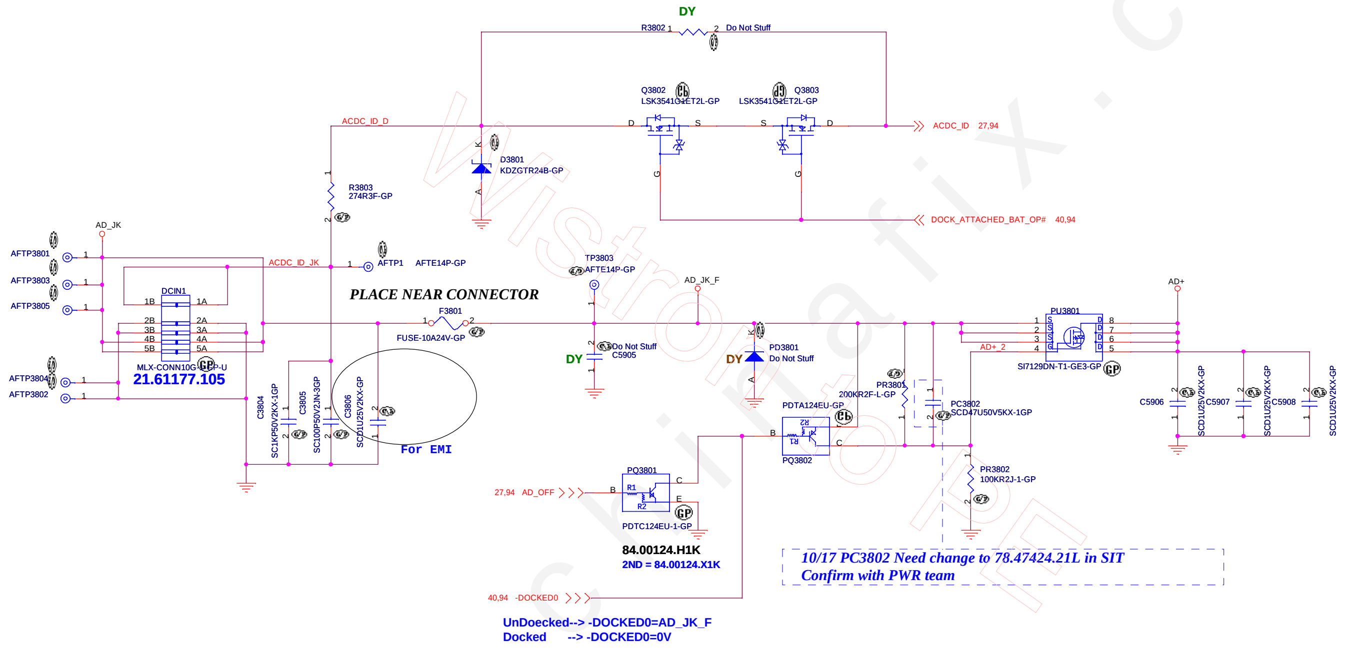
Title	*Power Plane Enable*	
Size A3	Document Number CD1 DIS	Rev SC
Date: Tuesday, December 13, 2011	Sheet 36 of 102	



BOM1

緯創資通 **Wistron Corporation**
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

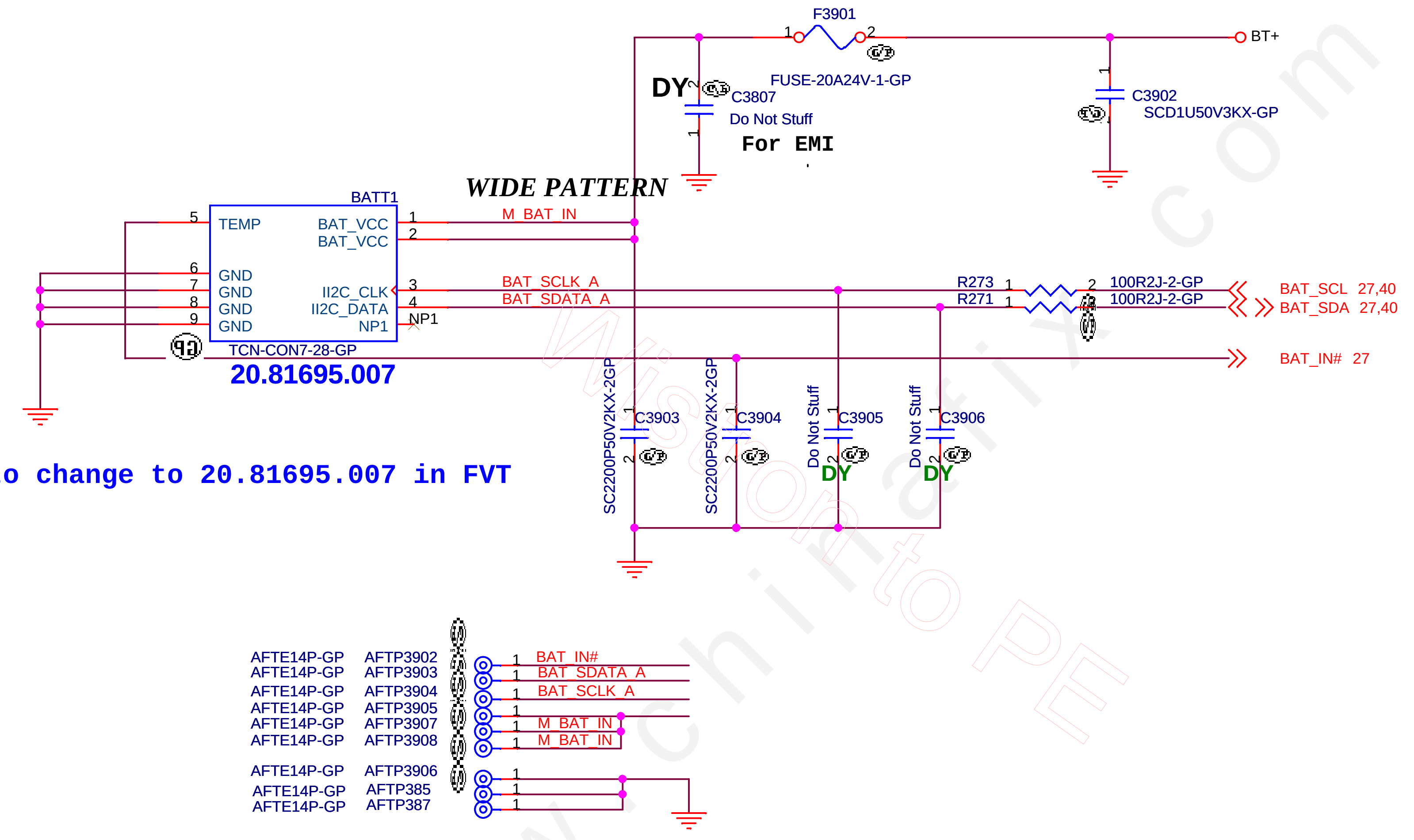
Title			
ADAPTER			
Size: A3	Document Number: CD1 DIS		Rev: SC
Date: Tuesday, December 13, 2011		Sheet 37 of 102	



BOM1

緯創資通	**Wistron Corporation** 21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih, Taipei Hsien 221, Taiwan, R.O.C.
Title	**DCIN JACK**
Size A2	Document Number **CD1 DIS** — Rev **SC**
Date: Tuesday, December 13, 2011	Sheet 38 of 102

BATT Connecter



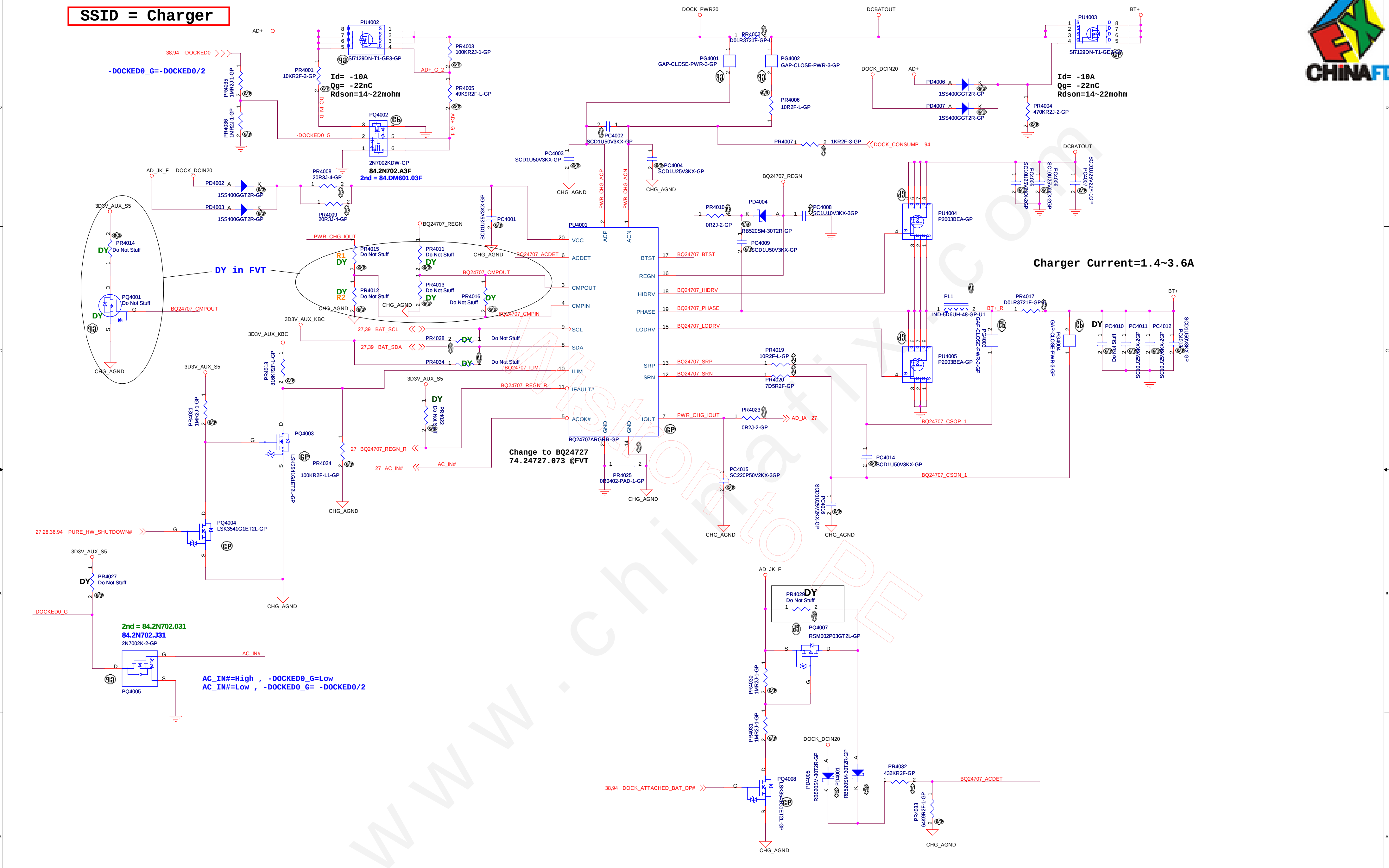
Need to change to 20.81695.007 in FVT

BOM1

緯創資通 Wistron Corporation
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title: BATT CONN		
Size: A4	Document Number: CD1 DIS	Rev: SC
Date: Tuesday, December 13, 2011	Sheet 39 of 102	

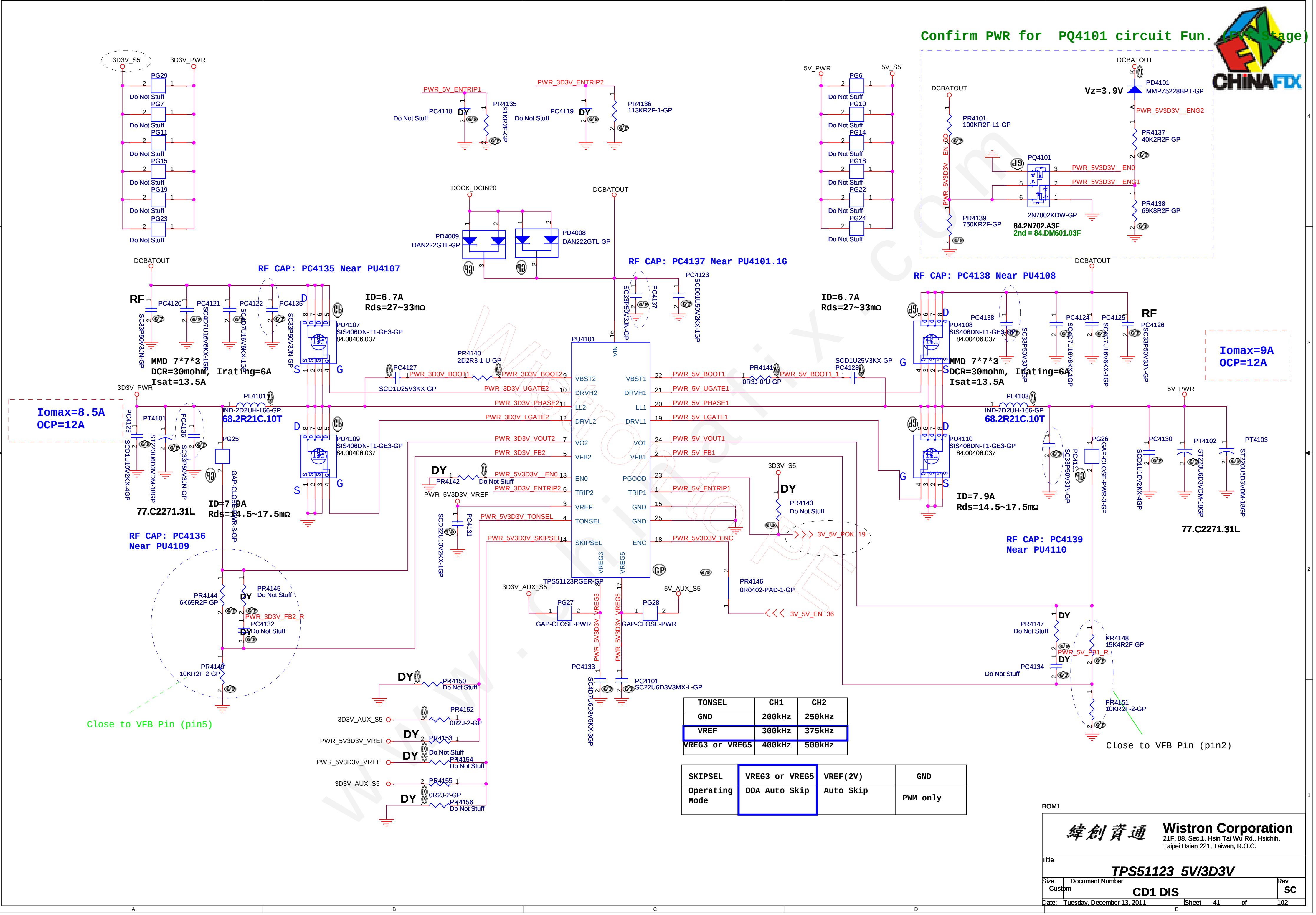
SSID = Charger



BOM1

緯創資通 Wistron Corporation
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title	CHARGER	
Size SC	Document Number CD1 DIS	Rev SC
Date: Tuesday, December 13, 2011	Sheet 40 of 102	



TONSEL	CH1	CH2
GND	200kHz	250kHz
VREF	300kHz	375kHz
VREG3 or VREG5	400kHz	500kHz

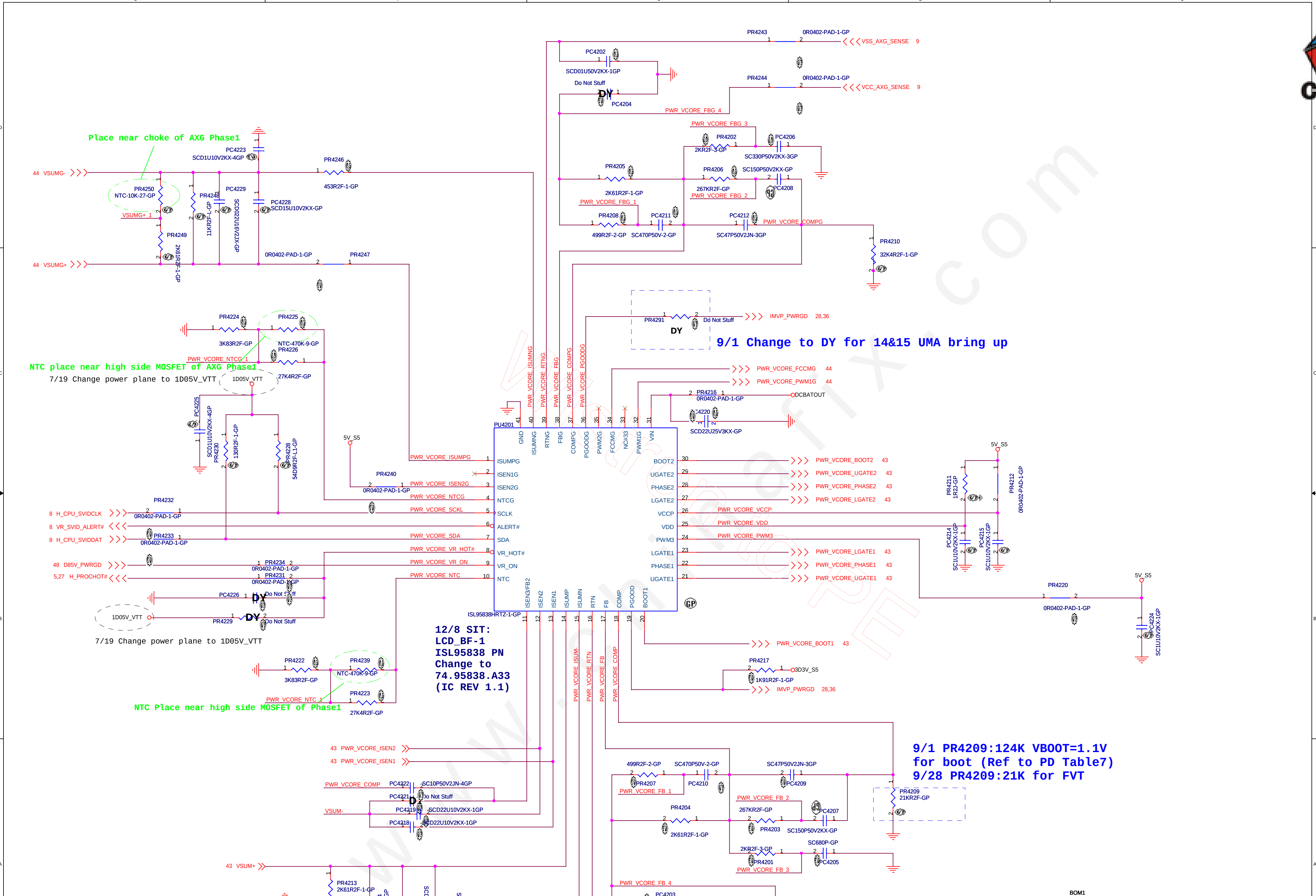
SKIPSEL	VREG3 or VREG5	VREF(2V)	GND
Operating Mode	OOA Auto Skip	Auto Skip	PWM only

緯創資通 Wistron Corporation
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih, Taipei Hsien 221, Taiwan, R.O.C.

Title: TPS51123 5V/3D3V

Size: Custom | Document Number: CD1 DIS | Rev: SC

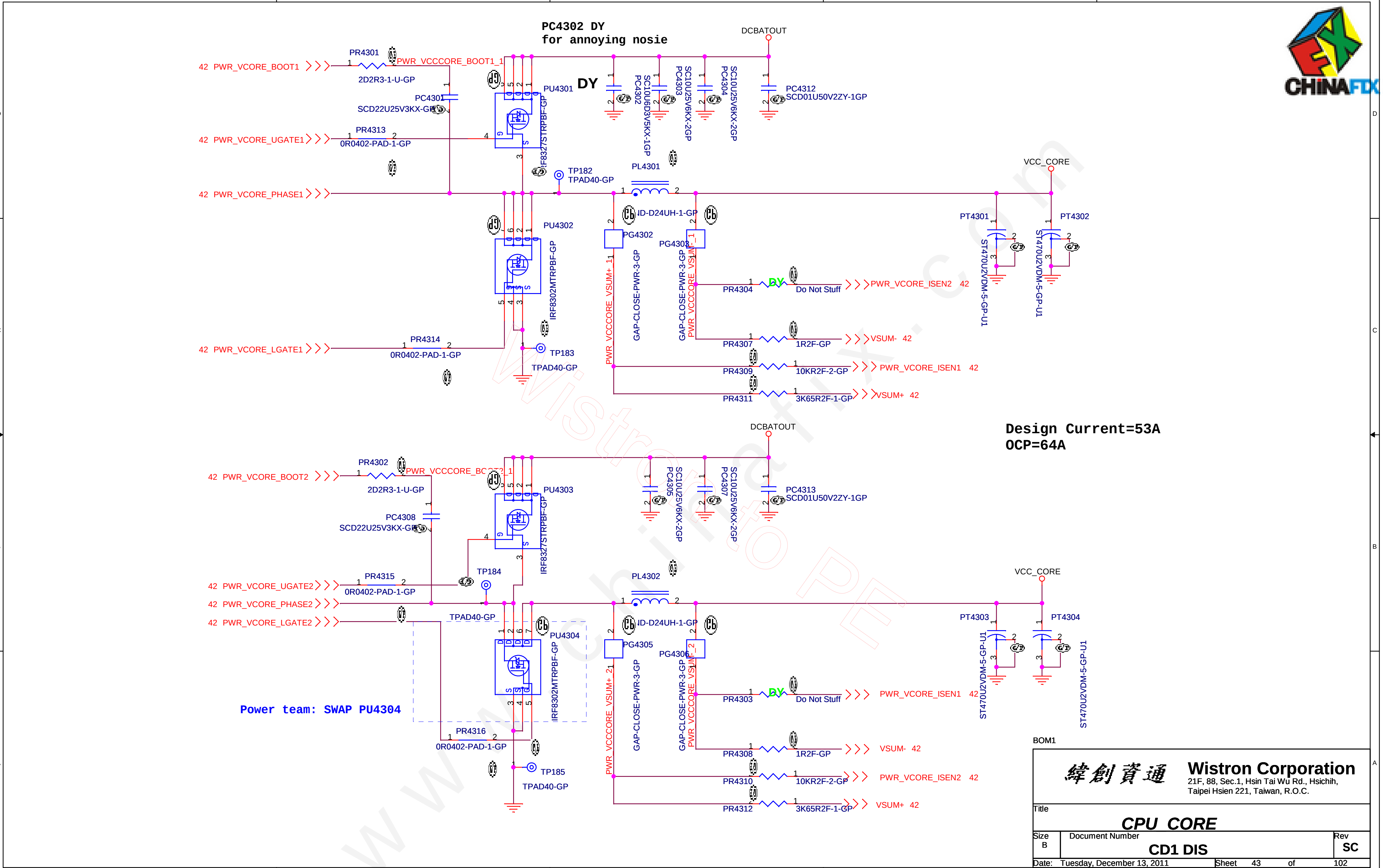
Date: Tuesday, December 13, 2011 | Sheet 41 of 102



BOM1

緯創資通 **Wistron Corporation**
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih, Taipei Hsien 221, Taiwan, R.O.C.

Title	CPU_CORE		
Size C	Document Number: CD1 DIS		Rev SC
Date: Tuesday, December 13, 2011	Sheet 42 of 102		



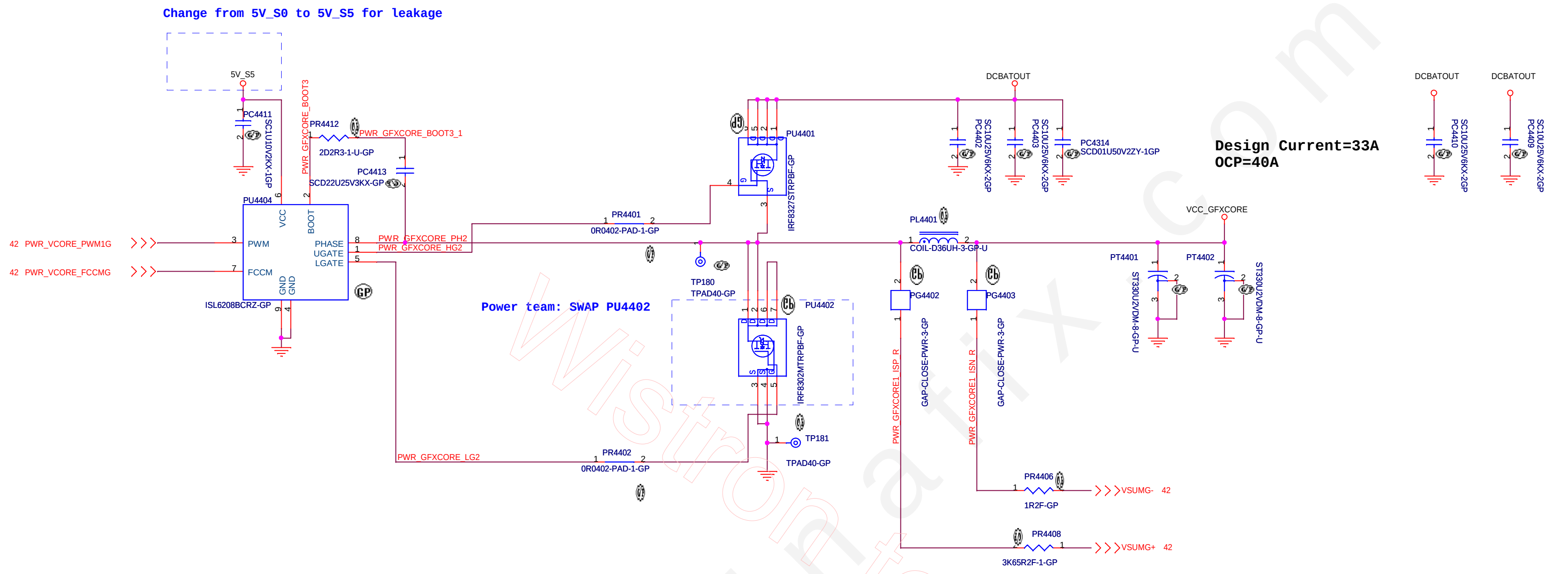
緯創資通 **Wistron Corporation**
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title: **CPU_CORE**

Size	Document Number	Rev
B	CD1 DIS	SC

Date: Tuesday, December 13, 2011 Sheet 43 of 102

SSID = AXG.Regulator



BOM1

緯創資通	Wistron Corporation 21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih, Taipei Hsien 221, Taiwan, R.O.C.	
Title	CPU_CORE	
Size C	Document Number CD1 DIS	Rev SC
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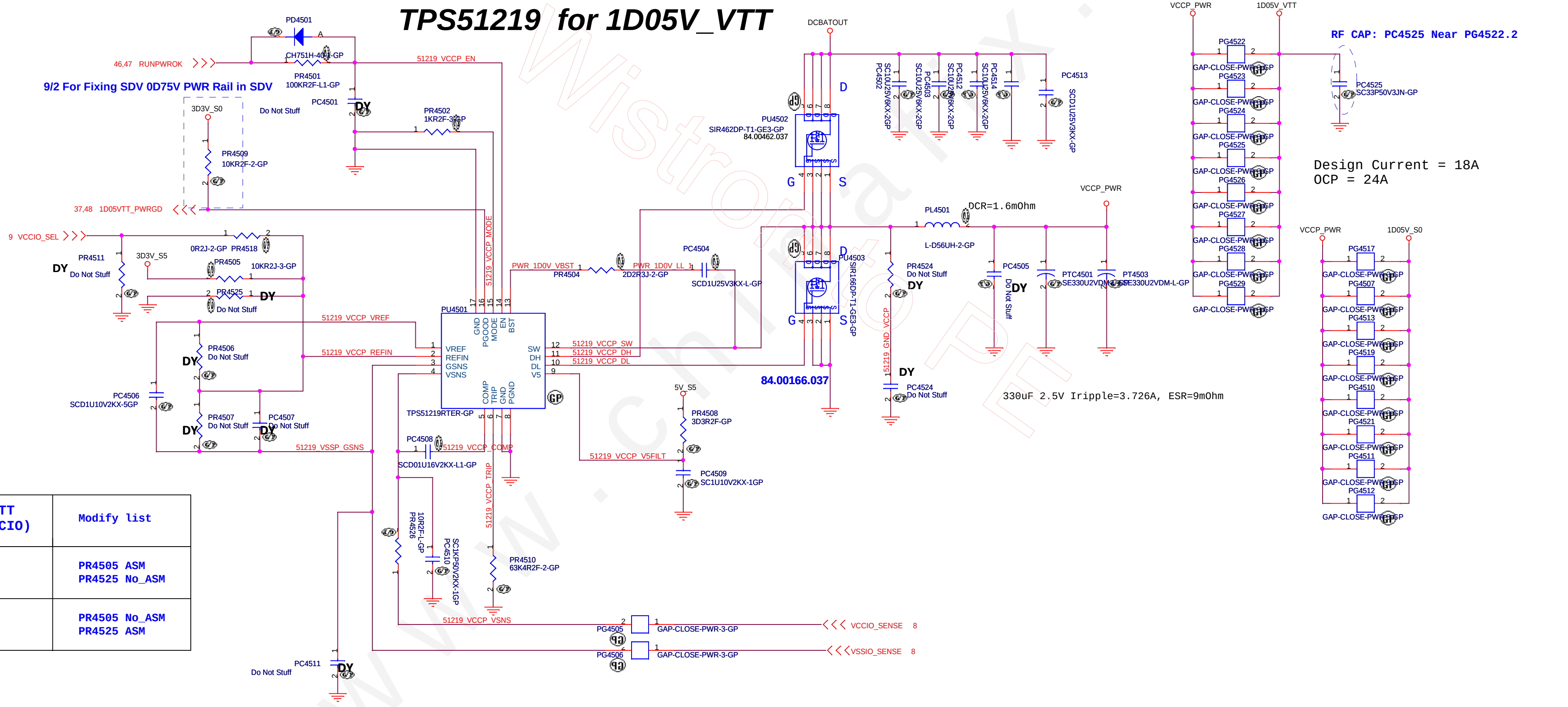
TPS51219 for 1D05V_VTT



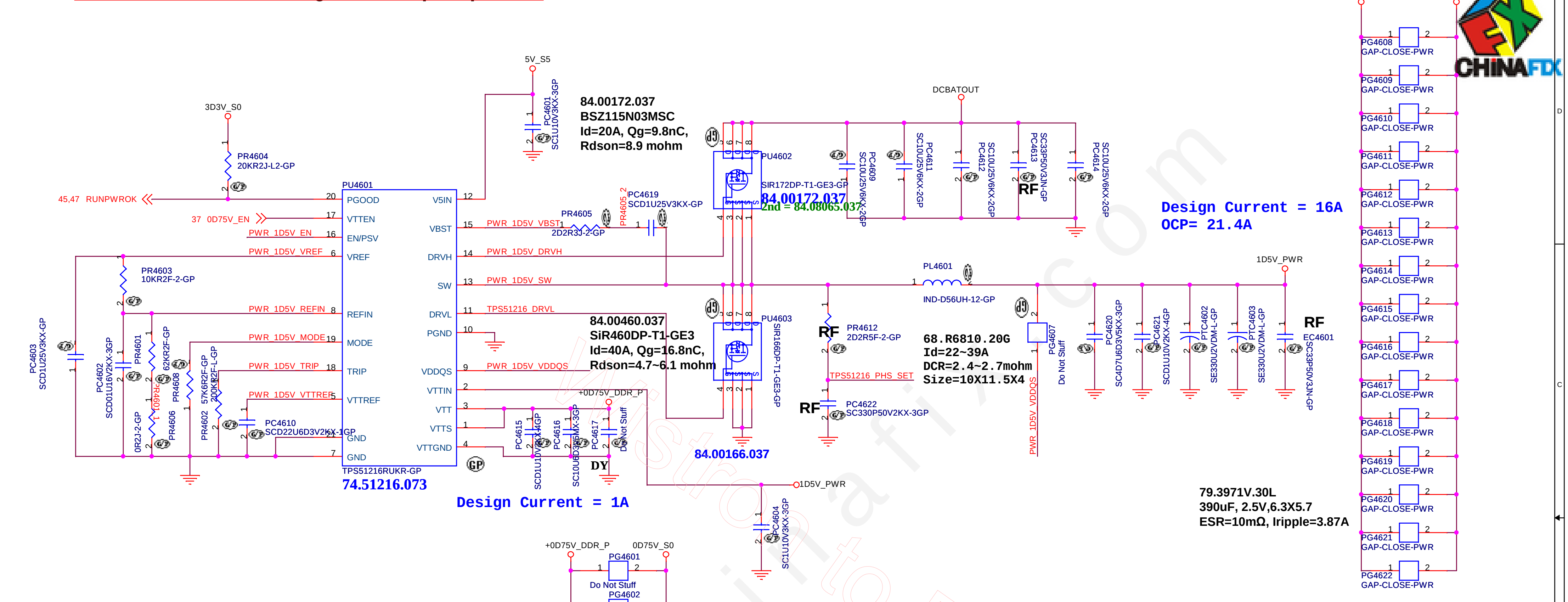
1D05V_VTT (CPU VCCIO)	Modify list
1.05V	PR4505 ASM PR4525 No_ASM
1.0V	PR4505 No_ASM PR4525 ASM

BOM1

緯創資通 **Wistron Corporation**
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title	TPS51219_1D05V	
Size	Document Number	Rev SC
Date: Tuesday, December 13, 2011	Sheet 45 of 102	

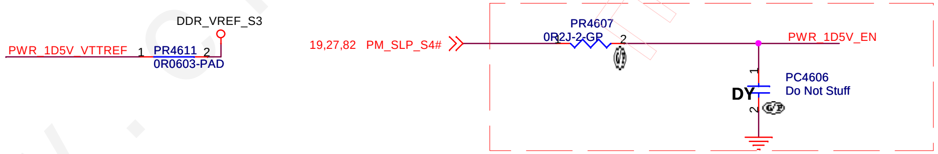
SSID = PWR.Plane.Regulator 1p5v0p75v



State	S3	S5	VDDR	VTTREF	VTT
S0	Hi	Hi	On	On	On
S3	Lo	Hi	On	On	Off(Hi-Z)
S4/S5	Lo	Lo	Off	Off	Off

MODE

PR5003	Frequency	Discharge Mode
200k ohm	400kHz	Tracking Discharge
100k ohm	300kHz	
68k ohm	300kHz	Non-tracking Discharge
47k ohm	400kHz	



BOM1

緯創資通 Wistron Corporation
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title	TPS51216_1D5V&0D75V	
Size A3	Document Number CD1 DIS	Rev SC
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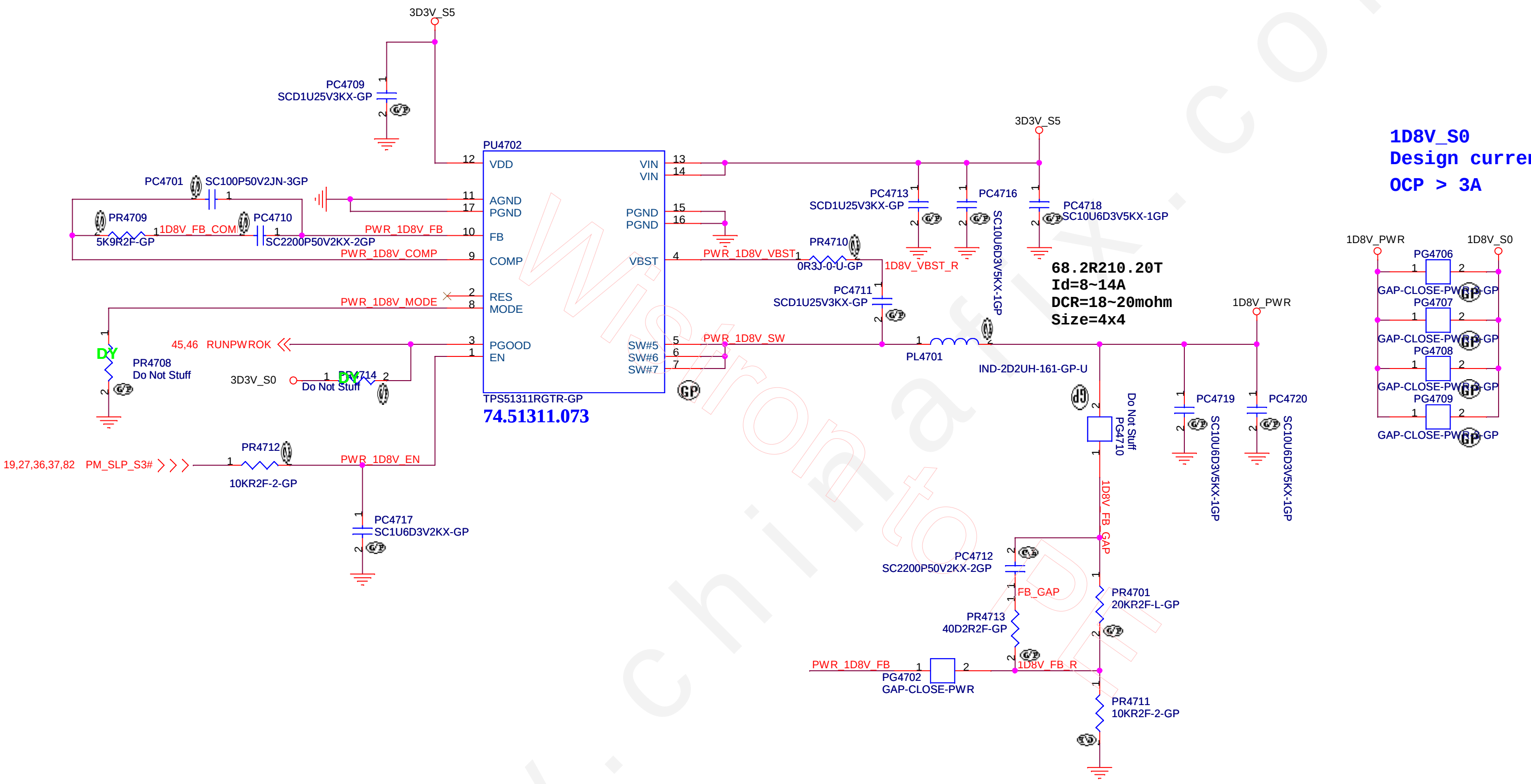
TPS51311 for 1D8V_S0

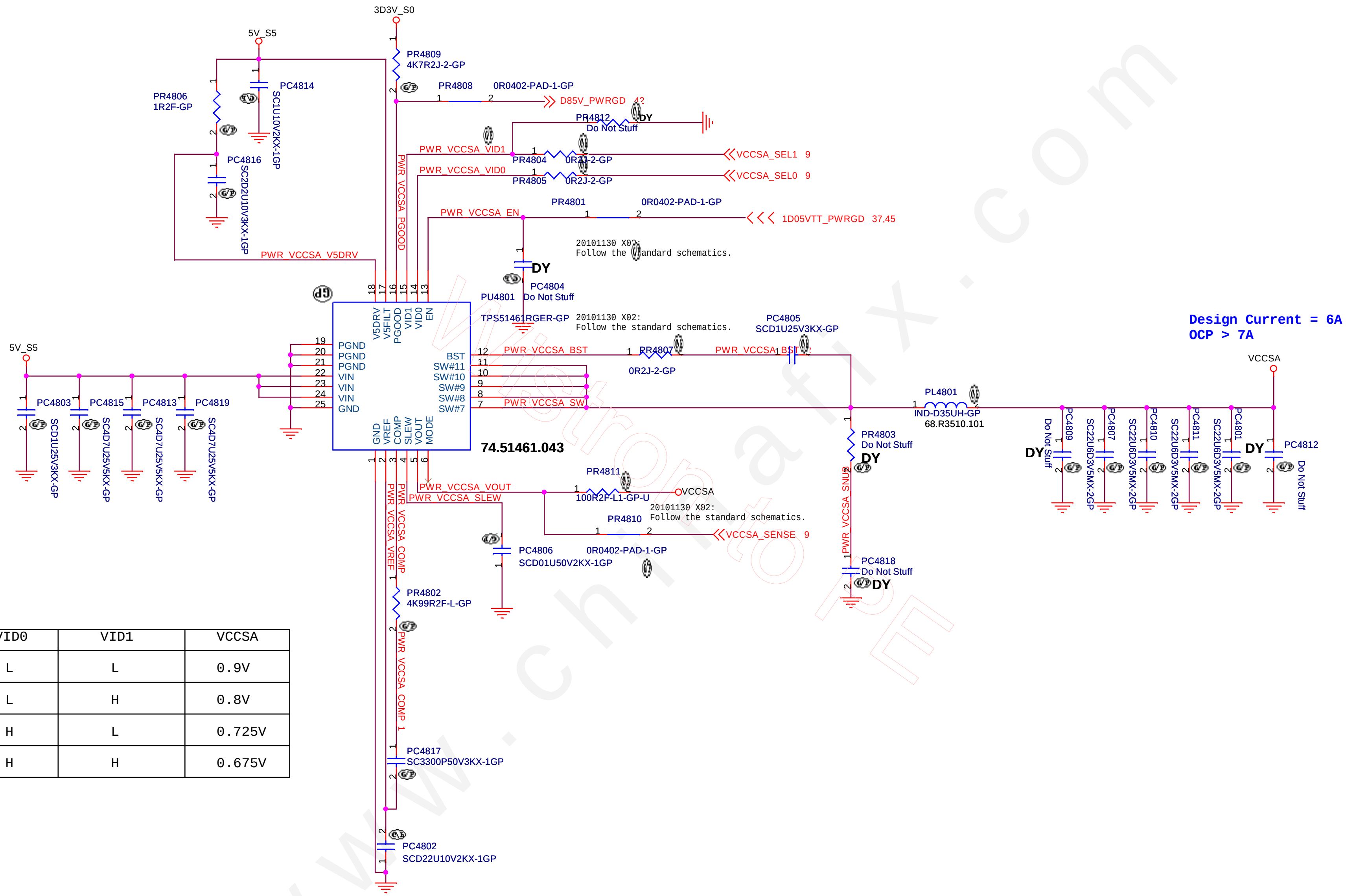


1D8V_S0
Design current = 1.5A
OCP > 3A

BOM1

緯創資通	**Wistron Corporation** 21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih, Taipei Hsien 221, Taiwan, R.O.C.	
Title	**PWM_1D8V_RT8015B**	
Size	Document Number **CD1 DIS**	Rev **SC**
Date: Tuesday, December 13, 2011	Sheet 47 of 102	

TPS51461 for VCCSA



VID0	VID1	VCCSA
L	L	0.9V
L	H	0.8V
H	L	0.725V
H	H	0.675V

BOM1

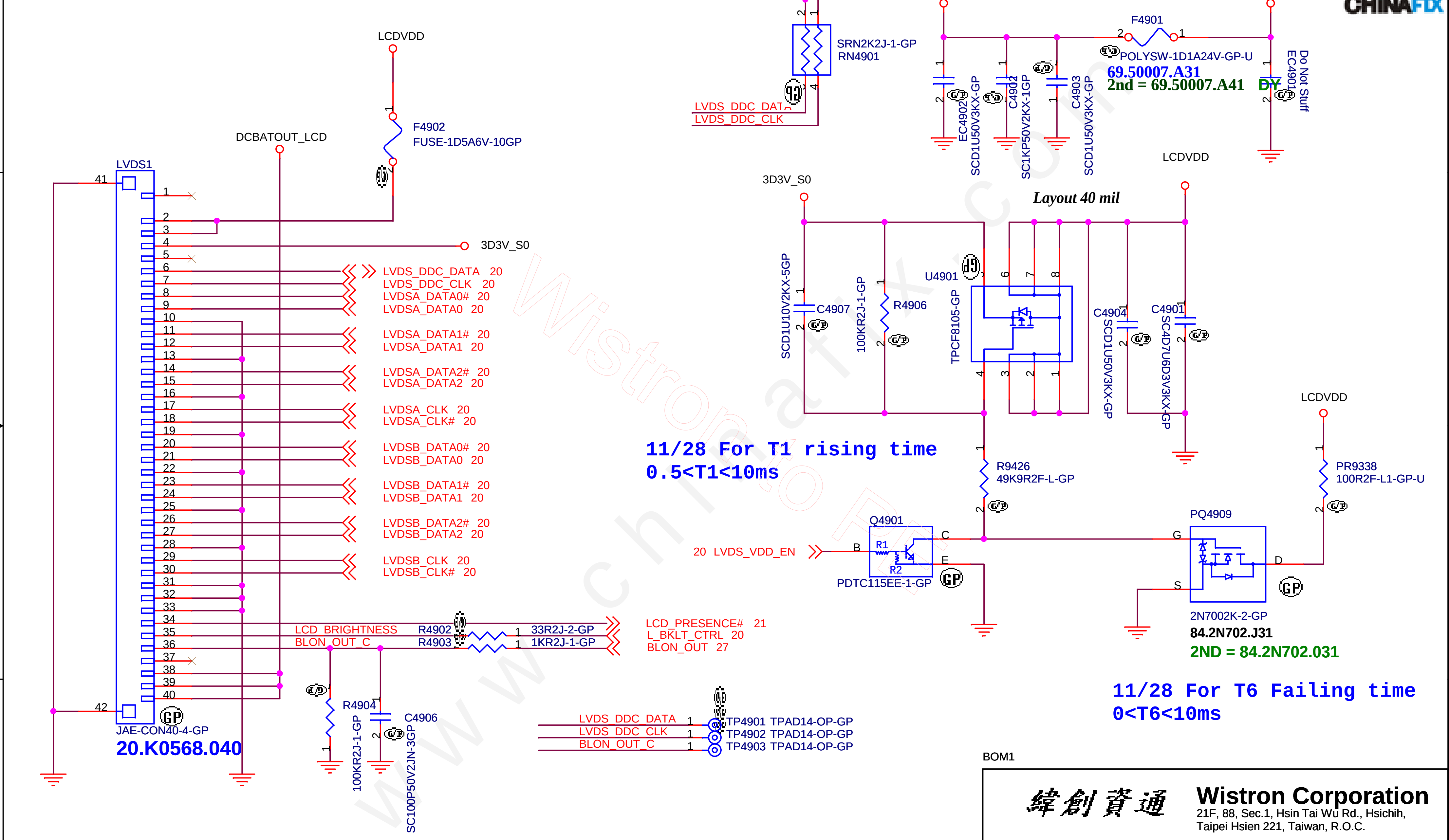
緯創資通 **Wistron Corporation**
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih, Taipei Hsien 221, Taiwan, R.O.C.

Title: **VCCSA_TPS51461**

Size | Document Number | Rev: SC

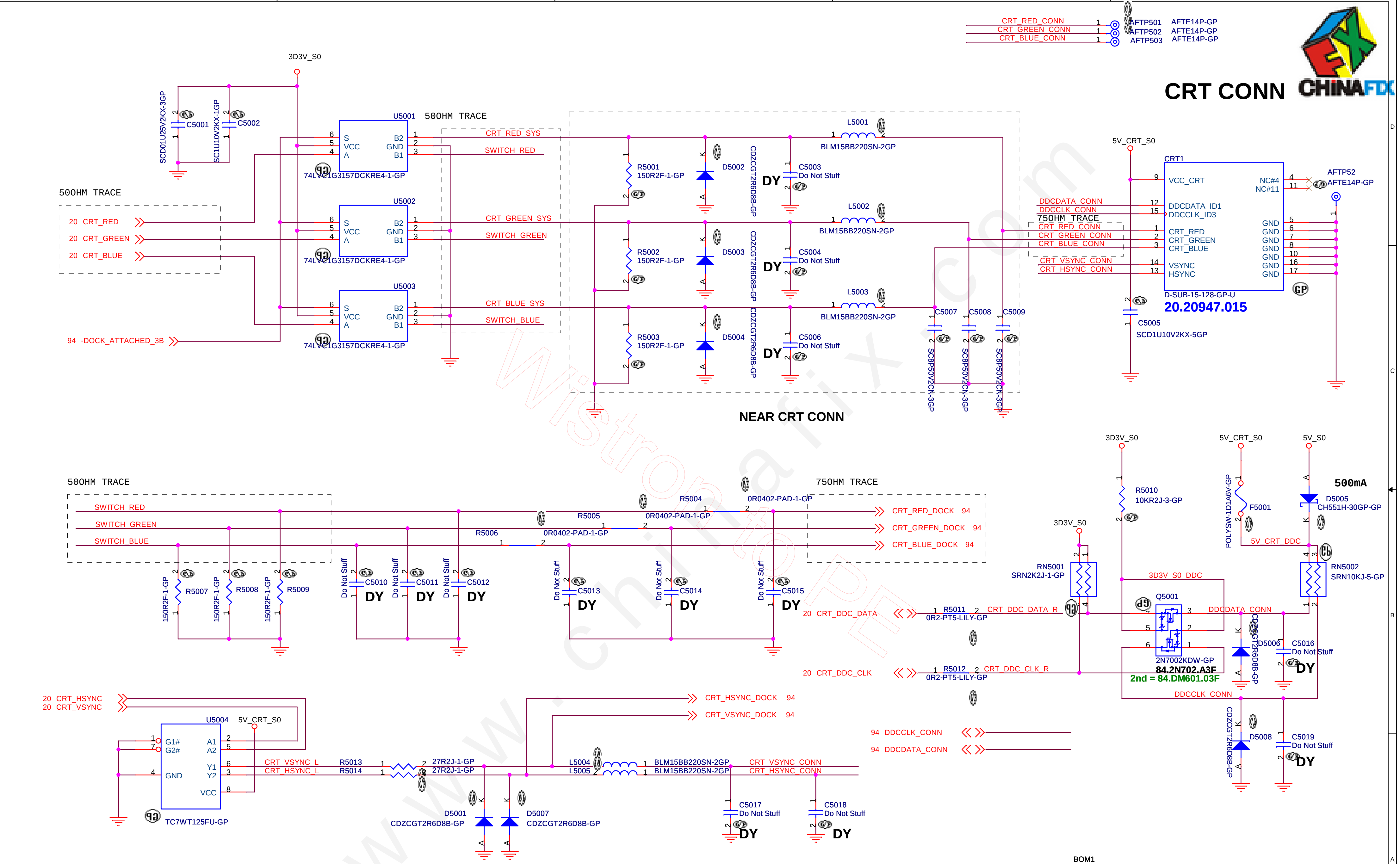
Date: Tuesday, December 13, 2011 | Sheet 48 of 102

LVDS CONNECTOR



緯創資通 **Wistron Corporation**
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title: LCD Connector		
Size: A4	Document Number: CD1 DIS	Rev: SC
Date: Tuesday, December 13, 2011	Sheet 49 of 102	



BOM1

緯創資通 **Wistron Corporation**
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title	CRT	
Size A3	Document Number CD1 DIS	Rev SC
Date: Tuesday, December 13, 2011	Sheet 50 of 102	

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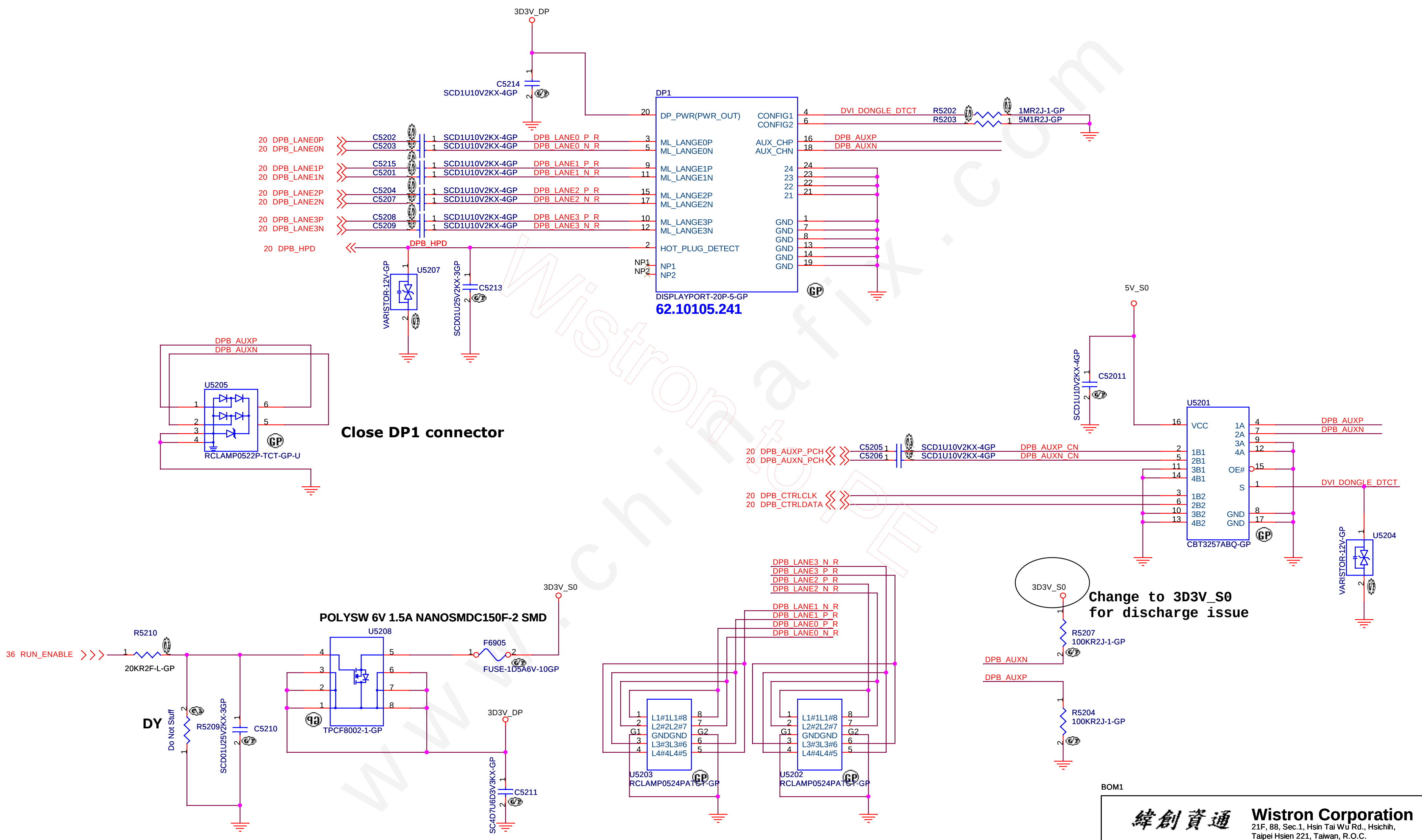
BOM1

Mini Display Port Connector



BOM1

緯創資通	**Wistron Corporation** 21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih, Taipei Hsien 221, Taiwan, R.O.C.
Title	***Display Port***
Size A3	Document Number **CD1 DIS** — Rev **SC**
Date: Tuesday, December 13, 2011	Sheet 52 of 102

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BOM1

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BOM1

緯創資通 Wistron Corporation
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title: ***Reserved***		
Size: A3	Document Number: **CD1 DIS**	Rev: **SC**
Date: Tuesday, December 13, 2011	Sheet 54 of 102	

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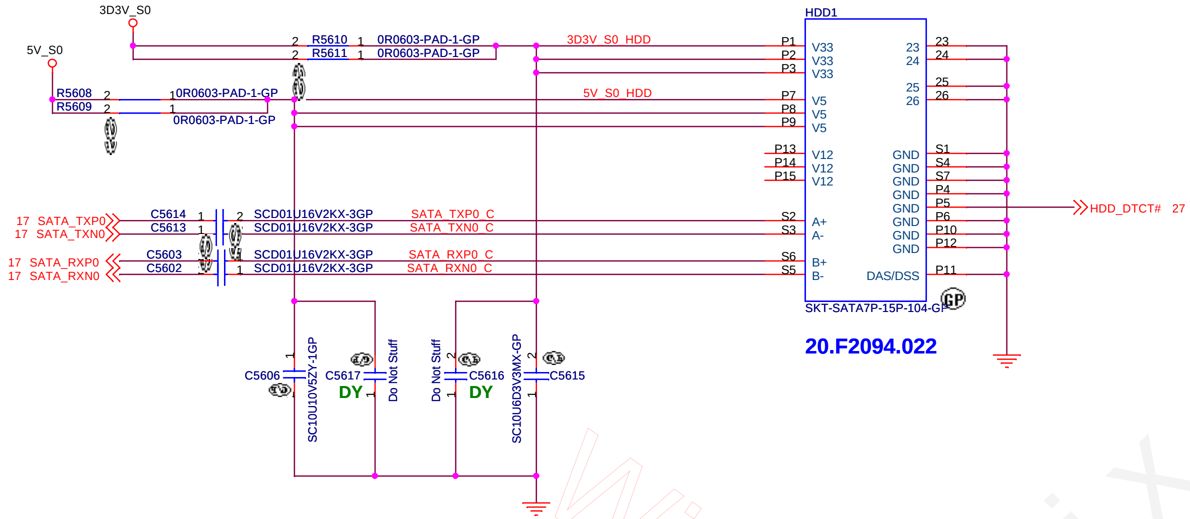
BOM1

緯創資通 **Wistron Corporation**
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

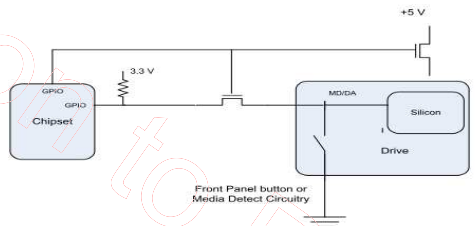
Title	***Reserved***	
Size A3	Document Number **CD1 DIS**	Rev **SC**
Date: Tuesday, December 13, 2011	Sheet 55 of 102	

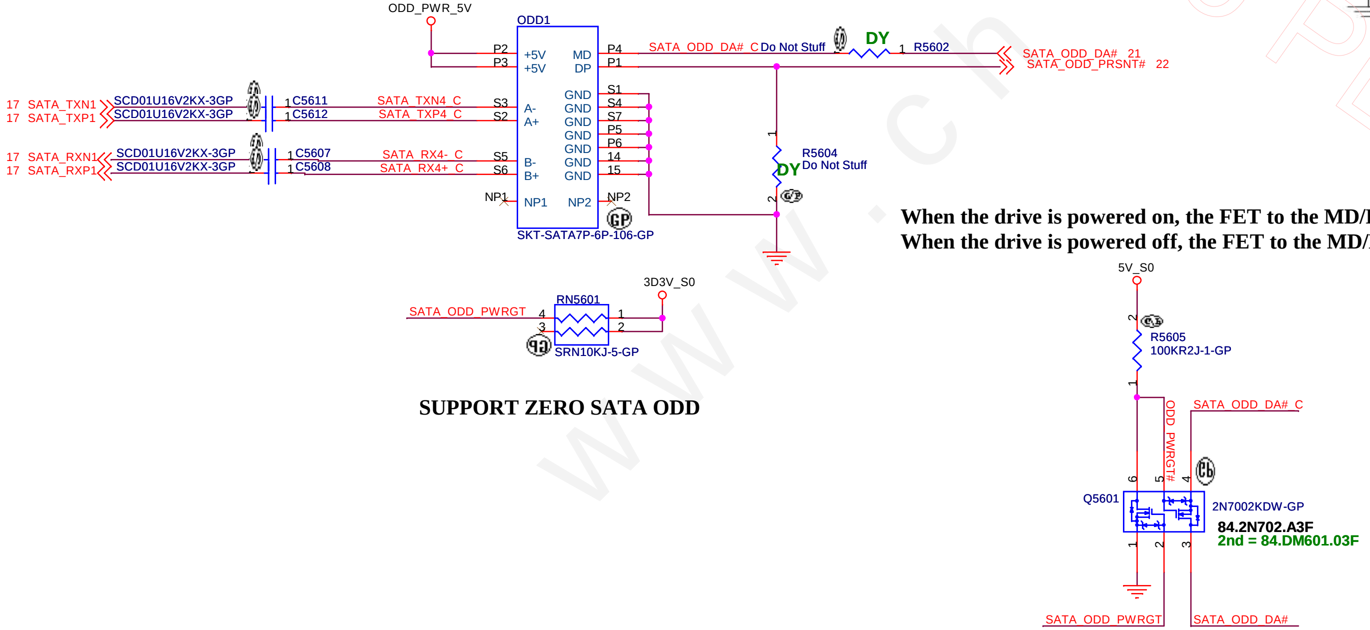
SATA HDD Connector



ODD Connector

SATA_RX- and SATA_RX+ Trace
Length match within 20 mil

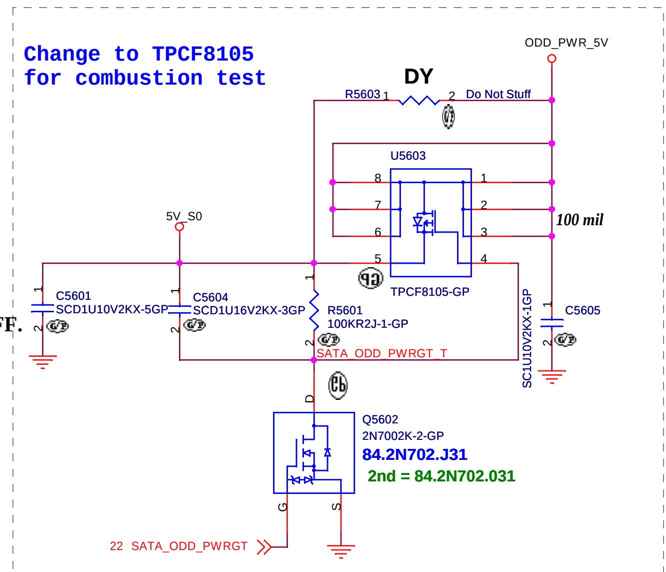




When the drive is powered on, the FET to the MD/DA pin drive is OFF.
When the drive is powered off, the FET to the MD/DA pin is ON

SUPPORT ZERO SATA ODD

SATA Zero Power ODD



BOM1

緯創資通	Wistron Corporation 21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih, Taipei Hsien 221, Taiwan, R.O.C.	
Title	HDD/ODD	
Size A3	Document Number CD1 DIS	Rev SC
Date: Tuesday, December 13, 2011	Sheet 56 of 102	

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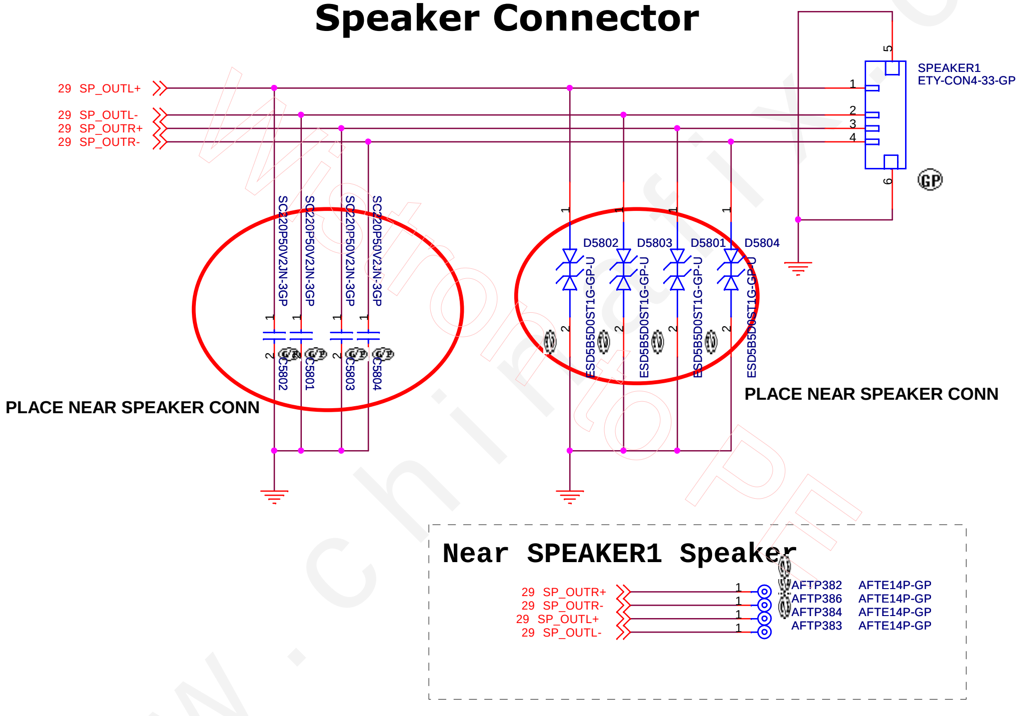
BOM1

緯創資通 Wistron Corporation
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title		
RESERVED		
Size: A3	Document Number: CD1 DIS	Rev: SC
Date: Tuesday, December 13, 2011	Sheet 57 of 102	

Speaker Connector



BOM1

緯創資通 Wistron Corporation 21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih, Taipei Hsien 221, Taiwan, R.O.C.		
Title: SPEAKER CONN		
Size: A3	Document Number: CD1 DIS	Rev: SC
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BOM1

緯創資通 **Wistron Corporation**
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title	Reserved	
Size: A3	Document Number: CD1 DIS	Rev: SC
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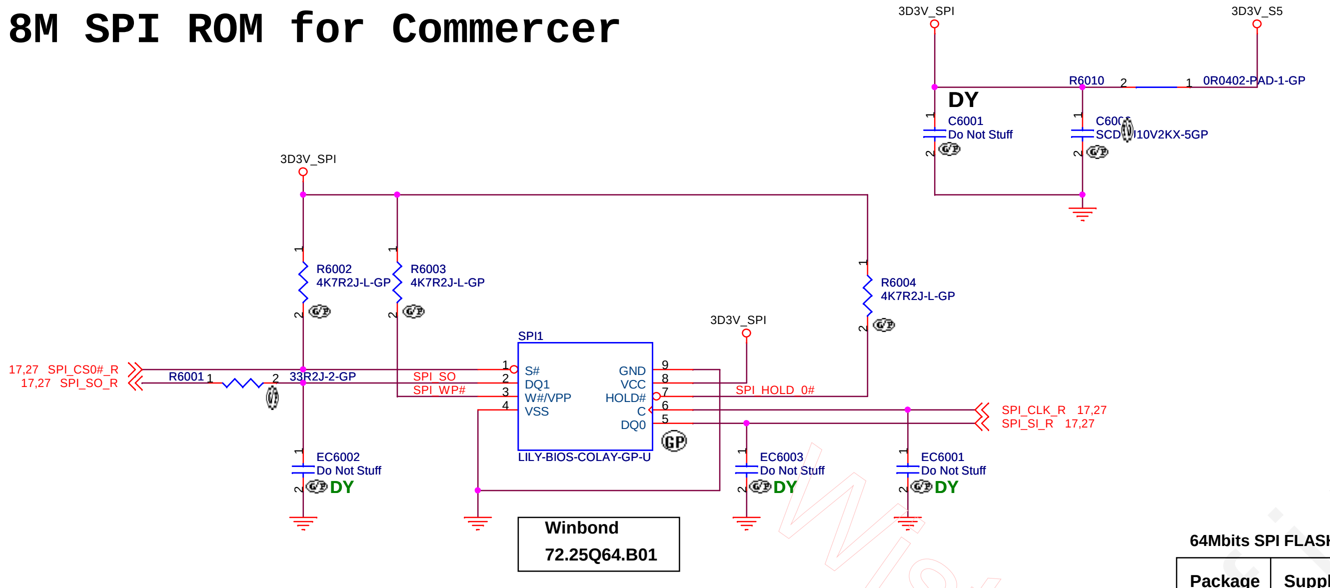
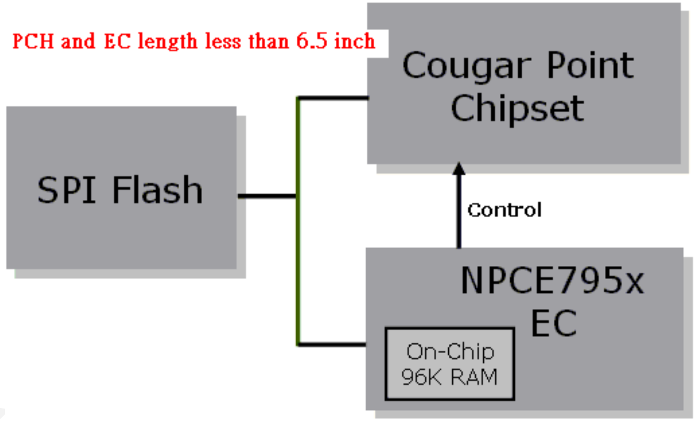
SSID = Flash.ROM

8M SPI ROM for Commercer



Notes:
The total SPI interface signal between EC and PCH can't not exceed 6500mil. The mismatch between SPI signal must be within 500mil

PCH and EC length less than 6.5 inch



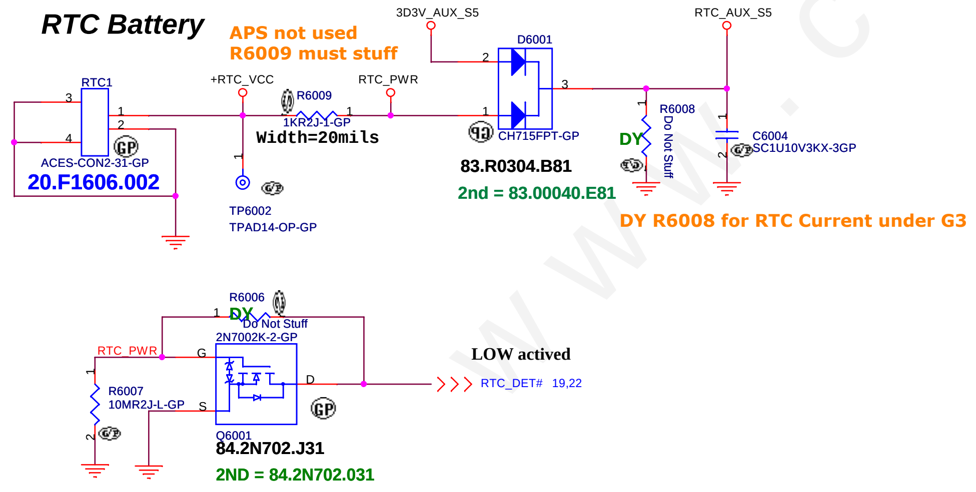
64Mbits SPI FLASH (SPI1):

Package	Supplier	Vendor P/N	Wistron P/N
SO8	Macronix	MX25L6406EM2I-12G	72.25640.D01
	Winbond	W25Q64CVSSIG	72.25Q64.B01
	Numonyx	N25Q064A13ESE40F	72.25Q64.D01

SSID = RBATT

RTC Battery

APS not used
R6009 must stuff



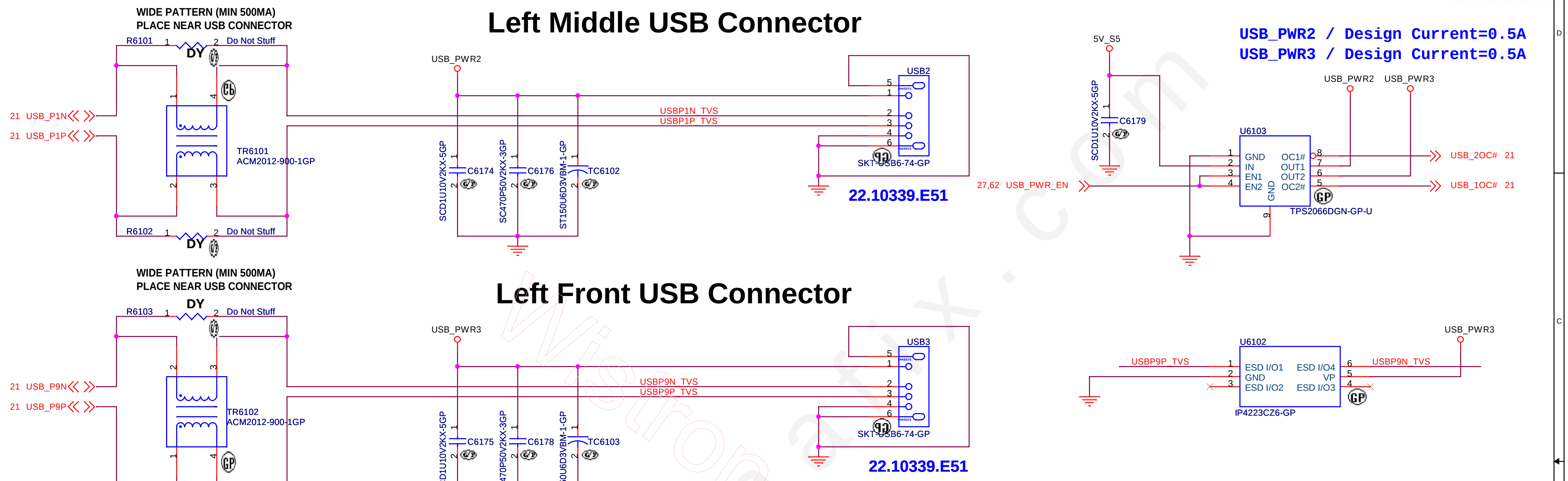
DY R6008 for RTC Current under G3

BOM1

緯創資通 **Wistron Corporation**
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih, Taipei Hsien 221, Taiwan, R.O.C.

Title		
Flash/RTC		
Size: A3	Document Number: CD1 DIS	Rev: SC
Date: Tuesday, December 13, 2011	Sheet 60 of 102	

SSID = USB



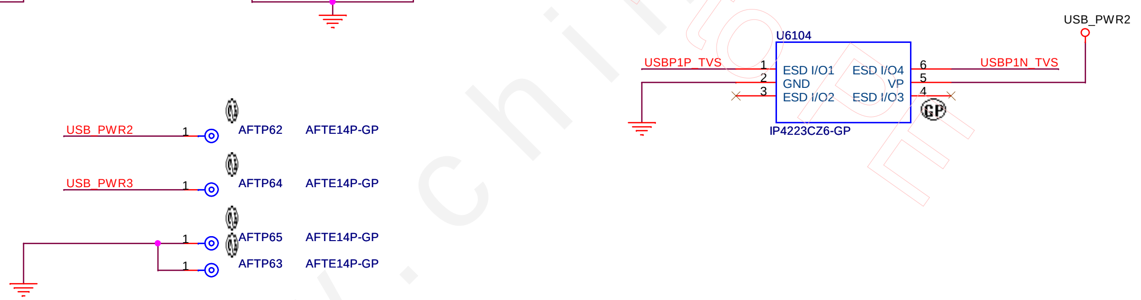


Table 61.1- USB2.0 PWR SW multi-source

Supplier	Description	Lenovo P/N	Wistron P/N
TI	TPS2066DGN	41R0511AA	74.02066.A71
TI	TPS2066DGN-1	N/A	74.02066.B71

Table 61.2- 150U 6.3V POSCAP multi-source

Supplier	Description	Lenovo P/N	Wistron P/N
NEC-TOKIN	TEPSLB20J157M	N/A	77.C1571.09L
SANYO	6TPE150MAZB	N/A	77.21571.111
HPC	TNCB0J157MTRZTF	N/A	80.15715.12L

Table 61.3- ESD protection

Supplier	Description	Lenovo P/N	Wistron P/N
NXP	IP4223CZ6	N/A	83.42236.0AE
AOS	AOZ8904CIL	N/A	83.08904.0AE
AMC	AZC099-04S	N/A	83.09904.AAE

BOM1

緯創資通 Wistron Corporation
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title: USB Power SW

Size: A3 | Document Number: CD1 DIS | Rev: SC

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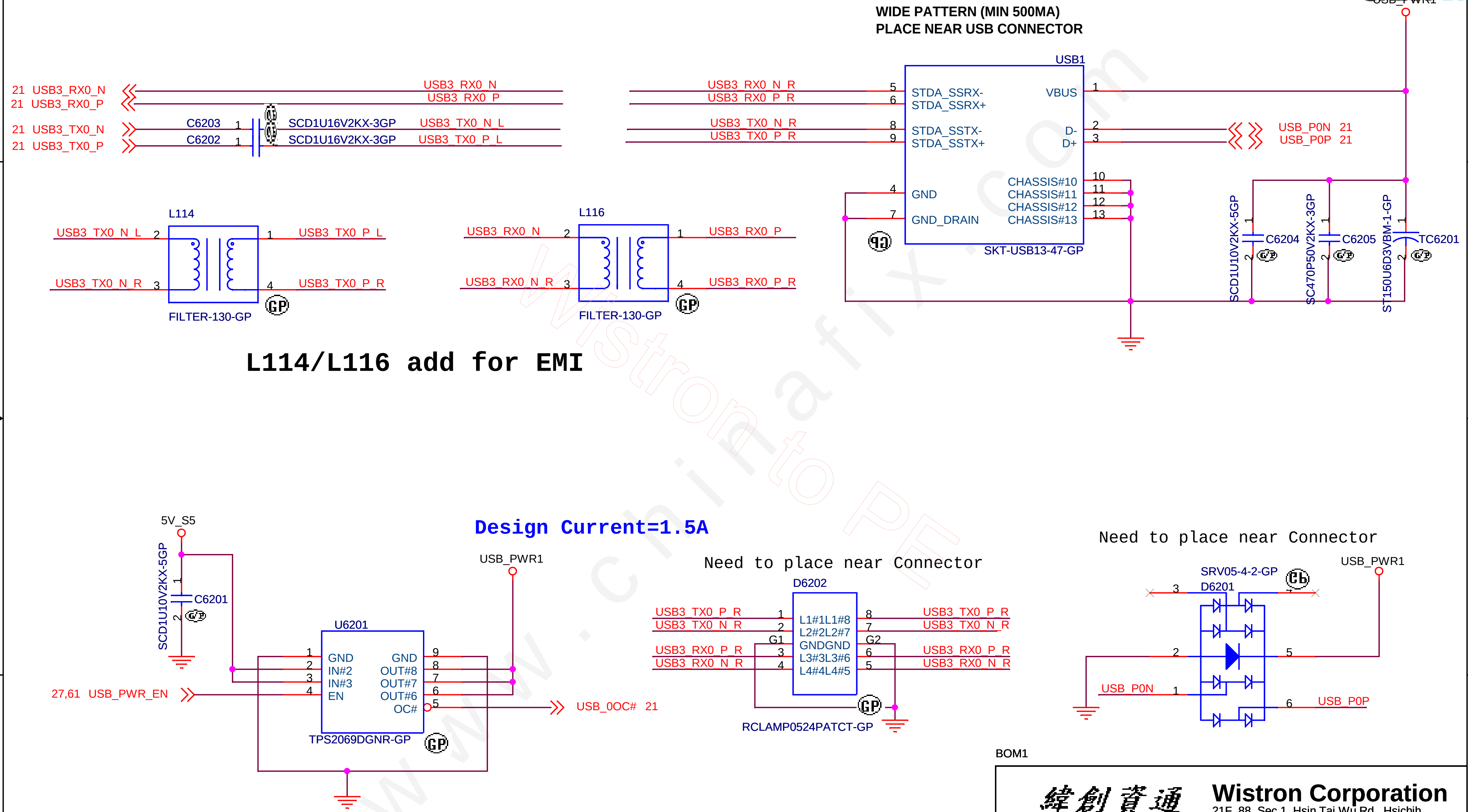
Left Rear USB Connector



BOM1

緯創資通 **Wistron Corporation**

21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih, Taipei Hsien 221, Taiwan, R.O.C.

Title	USB3.0	
Size A4	Document Number CD1 DIS	Rev SC
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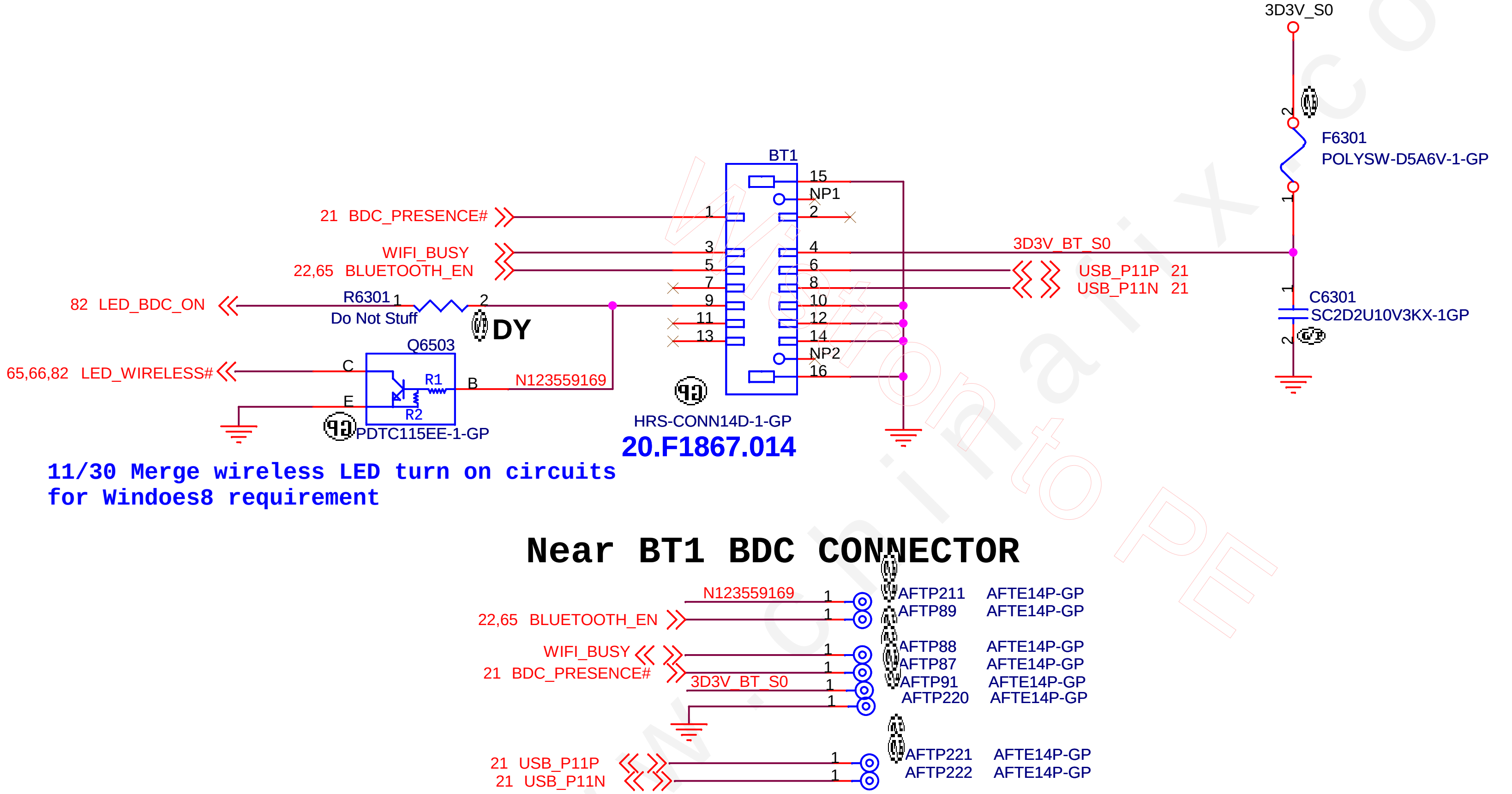
Bluetooth Module conn.



BOM1

緯創資通	**Wistron Corporation** 21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih, Taipei Hsien 221, Taiwan, R.O.C.
Title	***Bluetooth***
Size: A4	Document Number: **CD1 DIS** — Rev: **SC**
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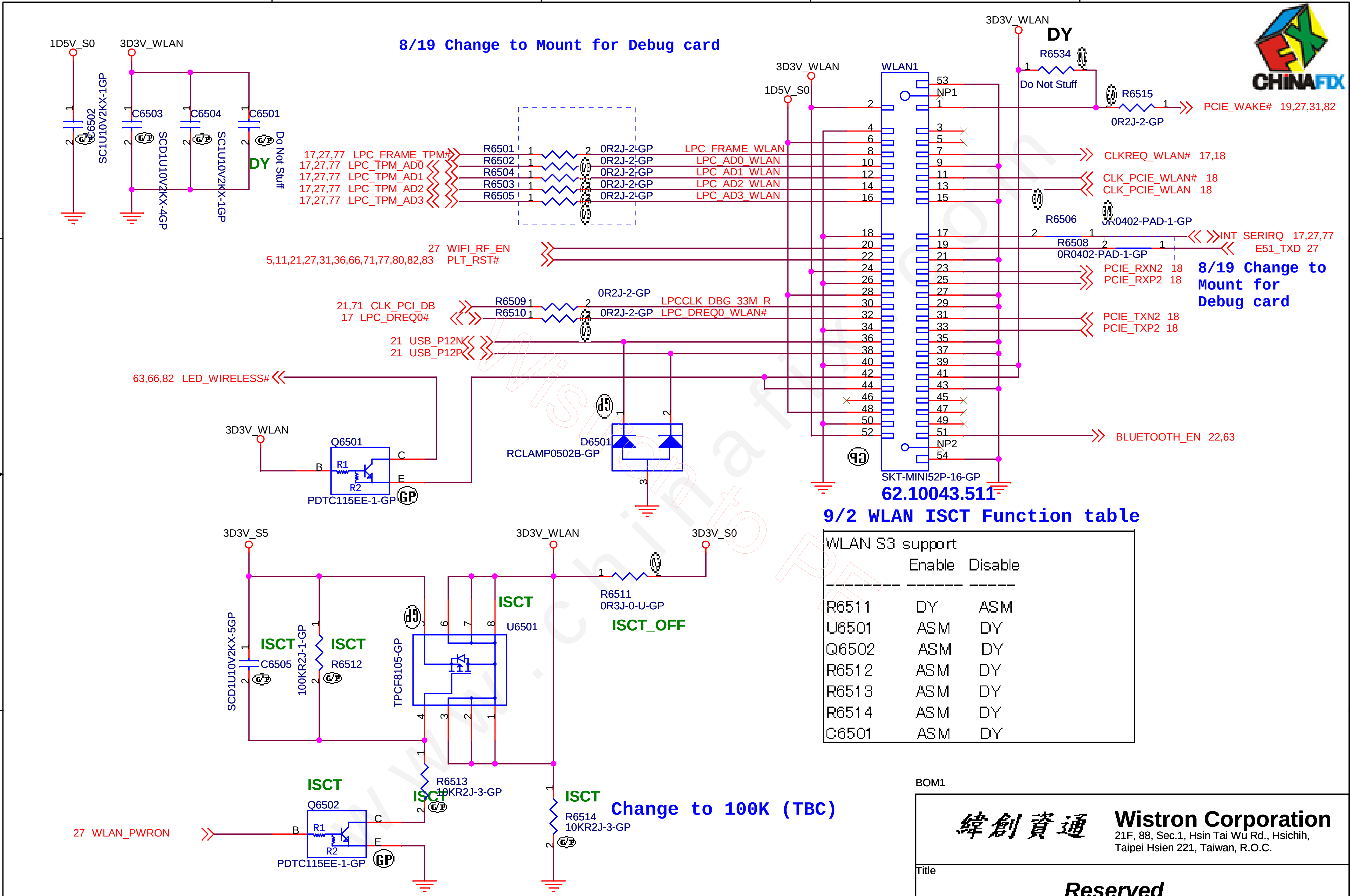
BOM1

緯創資通 Wistron Corporation
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title: *Finger Print Conn*		
Size: A3	Document Number: CD1 DIS	Rev: SC
Date: Tuesday, December 13, 2011	Sheet 64 of 102	



9/2 WLAN ISCT Function table

WLAN S3 support

	Enable	Disable
R6511	DY	ASM
U6501	ASM	DY
Q6502	ASM	DY
R6512	ASM	DY
R6513	ASM	DY
R6514	ASM	DY
C6501	ASM	DY

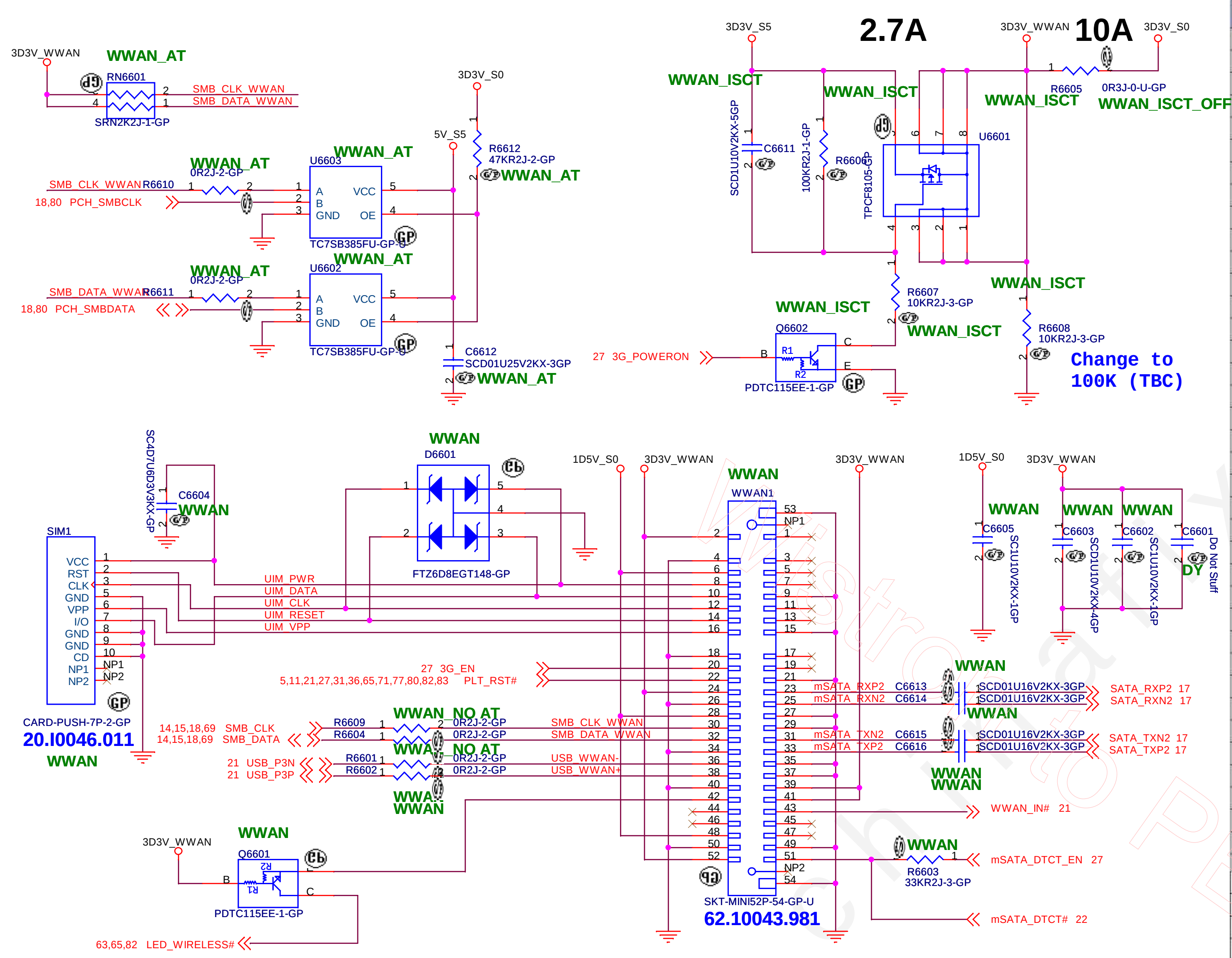
BOM1

緯創資通 Wistron Corporation
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title: **Reserved**

Size	Document Number	Rev
A4	CD1 DIS	SC

Date: Tuesday, December 13, 2011 Sheet 65 of 102



Pin	Function	Definition
P1	Reserved	No Connect
P2	+3.3 V	3.3 V Source
P3	Reserved	No Connect
P4	GND	Return Current Path
P5	Reserved	No Connect
P6	+1.5 V	1.5 V Source
P7	Reserved	No Connect
P8	Reserved	No Connect
P9	GND	Return Current Path
P10	Reserved	No Connect
P11	Reserved	No Connect
P12	Reserved	No Connect
P13	Reserved	No Connect
P14	Reserved	No Connect
P15	GND	Return Current Path
P16	Reserved	No Connect
P17	Reserved	No Connect
P18	GND	Return Current Path
P19	Reserved	No Connect
P20	Reserved	No Connect
P21	GND	Return Current Path
P22	Reserved	No Connect
P23	+B	Host Receiver Differential Signal Pair
P24	+3.3 V	3.3 V Source
P25	-B	Host Receiver Differential Signal Pair
P26	GND	Return Current Path
P27	GND	Return Current Path
P28	+1.5 V	1.5 V Source
P29	GND	Return Current Path
P30	Two Wire Interface	Two Wire Interface Clock[2]
P31	-A	Host Transmitter Differential Signal Pair
P32	Two Wire Interface	Two Wire Interface Data[2]
P33	+A	Host Transmitter Differential Signal Pair
P34	GND	Return Current Path
P35	GND	Return Current Path
P36	Reserved	No Connect
P37	GND	Return Current Path
P38	Reserved	No Connect
P39	+3.3 V	3.3 V Source
P40	GND	Return Current Path
P41	+3.3 V	3.3 V Source
P42	Reserved	No Connect
P43	Reserved	No Connect
P44	Reserved	No Connect
P45	Vendor	Vendor Specific / Manufacturing Pin[3]
P46	Reserved	No Connect
P47	Vendor	Vendor Specific / Manufacturing Pin[3]
P48	+1.5 V	1.5 V Source
P49	DA/DSS	Device Activity Signal / Disable Staggered Spin-up
P50	GND	Return Current Path
P51	Presence Detection	Shall be pulled to GND by device[1]
P52	+3.3 V	3.3 V Source

WWAN ASM/NON ASM

WWAN	YES	NO
WWAN1	ASM	No_ASM
SIM1	ASM	No_ASM
C6604	ASM	No_ASM
C6602	ASM	No_ASM
C6603	ASM	No_ASM

LOGIC

WWAN ISCT Funtion

Location	Disable 3G_PWRON	Enable 3G_PWRON
R6605	ASM	No_ASM
R6606	No_ASM	ASM
R6607	No_ASM	ASM
R6608	No_ASM	ASM
C6611	No_ASM	ASM
U6601	No_ASM	ASM
Q6602	No_ASM	ASM

LOGIC

Anti Theft Function

Location	AT No suport	AT suport
R6604	ASM	No_ASM
R6609	ASM	ASM
R6610	No_ASM	ASM
R6611	No_ASM	ASM
U6602	No_ASM	ASM
U6603	No_ASM	ASM
RN6601	No_ASM	ASM
R6612	No_ASM	ASM

LOGIC

BOM1

緯創資通 Wistron Corporation
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title: **WWAN/mSATA**

Size: A3 | Document Number: CD1 DIS | Rev: SC

Date: Tuesday, December 13, 2011 |

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BOM1

緯創資通	**Wistron Corporation** 21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih, Taipei Hsien 221, Taiwan, R.O.C.	
Title: **Reserved**		
Size: A3	Document Number: **CD1 DIS**	Rev: **SC**
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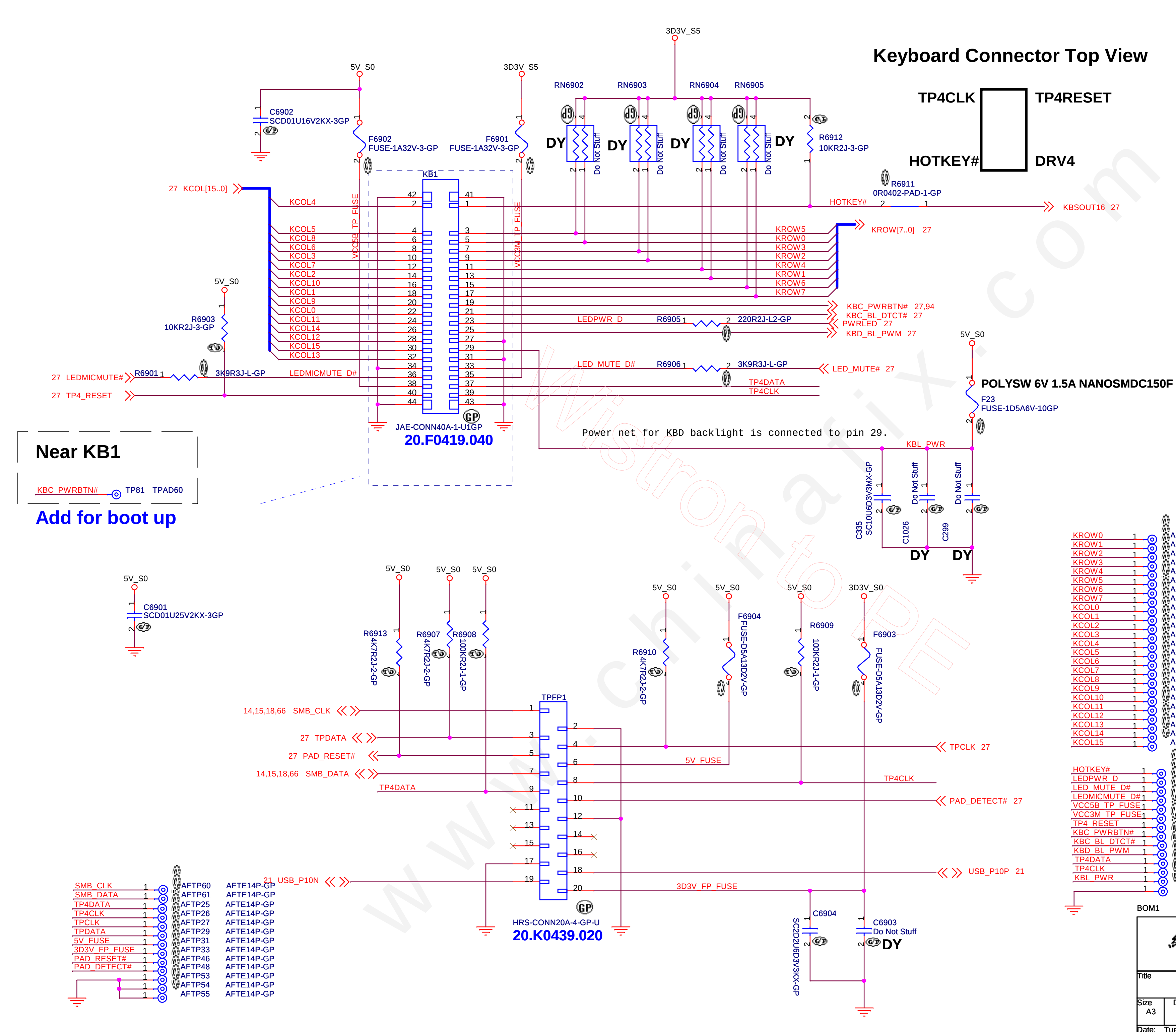
BOM1



MB Pin	Description
1	Diag_Loop3 = GPIO_1 (TPC)
2	KSI[7] = KBD S8
3	KSI[6] = KBD S7
4	KSI[4] = KBD S5
5	KSI[2] = KBD S3
6	KSI[5] = KBD S6
7	KSI[1] = KBD S2
8	KSI[3] = KBD S4
9	KSI[0] = KBD S1
10	KSO[5] = KBD D6
11	KSO[4] = KBD D5
12	KSO[7] = KBD D8
13	KSO[6] = KBD D7
14	KSO[8] = KBD D9
15	KSO[3] = KBD D4
16	KSO[1] = KBD D2
17	KSO[2] = KBD D3
18	KSO[0] = KBD D1
19	KSO[12] = KBD D13
20	KSO[16] = KBD D17
21	KSO[15] = KBD D16
22	KSO[13] = KBD D14
23	KSO[14] = KBD D15
24	KSO[9] = KBD D10
25	KSO[11] = KBD D12
26	KSO[10] = KBD D11
27	NC (reserved for Caps LK LED)
28	NC (reserved for Num LK LED)
29	NC (reserved for Scroll LK LED)
30	GND

緯創資通 **Wistron Corporation**
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title: **Key Board/Touch Pad**

Size: A3 | Document Number: CD1 DIS | Rev: SC

Date: Tuesday, December 13, 2011 | Sheet 69 of 102

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BOM1

緯創資通 Wistron Corporation
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title	Hall Sensor	
Size A4	Document Number CD1 DIS	Rev SC
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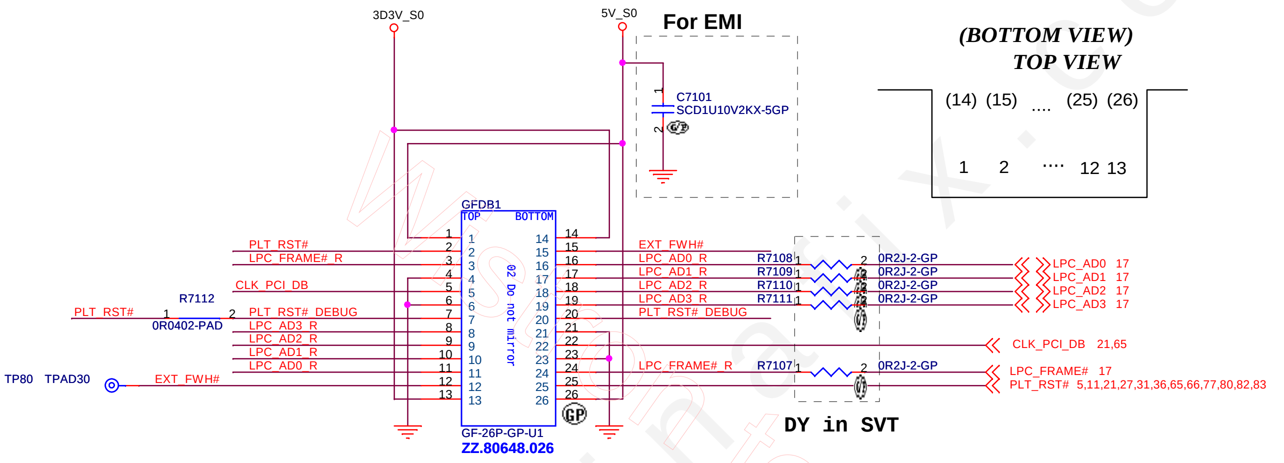
Golden Finger for Debug Board



BOM1

緯創資通 **Wistron Corporation**
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title	Dubug connector		
Size: A3	Document Number: CD1 DIS		Rev: SC
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BOM1

緯創資通	Wistron Corporation 21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih, Taipei Hsien 221, Taiwan, R.O.C.		
Title	Reserved		
Size A3	Document Number CD1 DIS		Rev SC
Date: Tuesday, December 13, 2011	Sheet 72 of 102		

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BOM1

緯創資通 Wistron Corporation
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title	Reserved	
Size A3	Document Number CD1 DIS	Rev SC
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20.I0129.001

PIN	TYPE	FUNCTION	RTS5138 NET
P1	SD	SD-CD	SP6
P2	SD	SD-WP	SP1
P3	SD	SD-DAT1	SP3
P4	SD	SD-DAT0	SP4
P5	MMC_PLUS	MMC-DATA7	SP5
P6	MemoryStick	MS-GND	GND
P7	SD	SD-GND	GND
P8	MMC_PLUS	MMC-DATA6	SP7
P9	MemoryStick	MS-BS	SP14
P10	SD	SD-CLK	SP8
P11	MemoryStick	MS-DATA1	SP12
P12	MemoryStick	MS-DATA0	SP9
P13	SD	SD-VCC	3D3V_CARD_S0
P14	MemoryStick	MS-DATA2	SP8
P15	SD	SD-GND	GND
P16	MemoryStick	MS-INS	SP2
P17	MMC_PLUS	MMC-DATA5	SP9
P18	MemoryStick	MS-DATA3	SP5
P19	SD	SD-CMD	SP10
P20	MemoryStick	MS-SCLK	SP1
P21	MMC_PLUS	MMC-DATA4	SP11
P22	MemoryStick	MS-VCC	3D3V_CARD_S0
P23	SD	SD-DATA3	SP12
P24	MemoryStick	MS-GND	GND
P25	SD	SD-DAT2	SP13
P26	SD	SD-WP COM /SDIO GND	GND
P27	SD	SD-CD COM /SDIO GND	GND
#1	XD	XD-CD	XD_CD#
#2	XD	XD-R/B	SP1
#3	XD	XD-RE	SP2
#4	XD	XD-CE	SP3
#5	XD	XD-CLE	SP4
#6	XD	XD-ALE	SP5
#7	XD	XD-WE	SP6
#8	XD	XD-WP-IN	SP7
#9	XD	XD-GND	GND
#10	XD	XD-D0	SP8
#11	XD	XD-D1	SP9
#12	XD	XD-D2	SP10
#13	XD	XD-D3	SP11
#14	XD	XD-D4	SP12
#15	XD	XD-D5	SP13
#16	XD	XD-D6	SP14
#17	XD	XD-D7	XD-D7
#18	XD	XD-VCC	3D3V_CARD_S0
#19	XD	XD-GND	GND

BOM1

緯創資通 Wistron Corporation
21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title: SD/XD/MS/MMC Card CONN

Size: A3 | Document Number: CD1 DIS | Rev: SC

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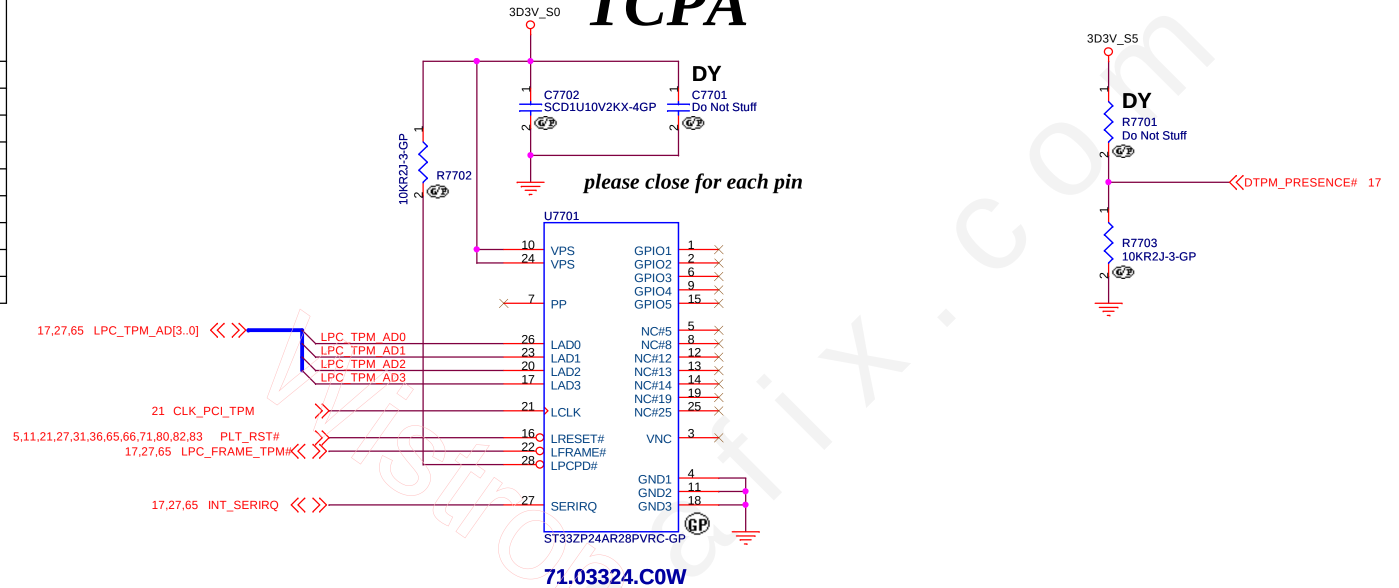
	NO TPM	ST Micro ST19NP18ER28PVMK
U39	NO_ASM	ASM
C379	NO_ASM	ASM
C9203	NO_ASM	NO_ASM
R149	NO_ASM	ASM
R212	ASM	NO_ASM
R270	NO_ASM	ASM

LOGIC

TCPA



BOM1

緯創資通 **Wistron Corporation**

21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih, Taipei Hsien 221, Taiwan, R.O.C.

Title	TPM	
Size A3	Document Number CD1 DIS	Rev SC
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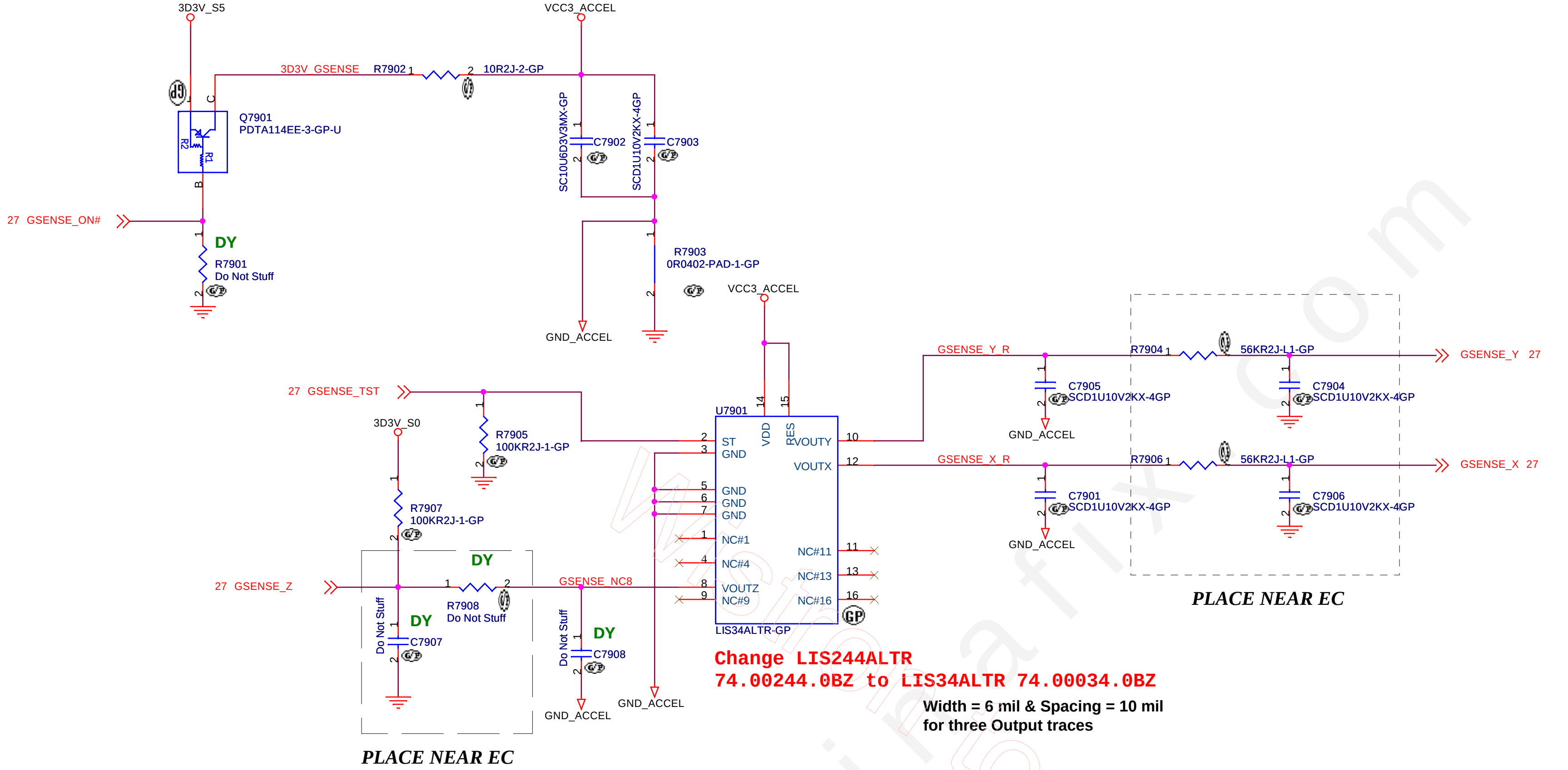
BOM1



	LIS244AL LIS34AL	NO ACC.
R401	NO-ASM	ASM
R957	ASM	ASM
U65	ASM	NO-ASM
Q105	ASM	NO-ASM
R885	10-OHM	NO-ASM
C829	ASM	NO-ASM
C969	ASM	NO-ASM
C830	ASM	NO-ASM
C847	ASM	NO-ASM
R970	56K	NO-ASM
C956	ASM	NO-ASM
R969	56K	NO-ASM
C938	ASM	NO-ASM
C704	NO-ASM	NO-ASM
R344	NO-ASM	NO-ASM
C703	NO-ASM	NO-ASM
R125	ASM	ASM

Table

	Supplier	Vendo P/N	WISTRON P/N
1	ST	LIS34ALTR	74.00034.0BZ 41R0828AA
2	Kionix	KXTC8-2850	74.KXTC8.0BZ

Layout Comment :

(1) Place C586, C588, Q17, R415, R417, C584, C585, R420 close to U34.

(2) Avoid routing under DCDC switching area.

BOM1

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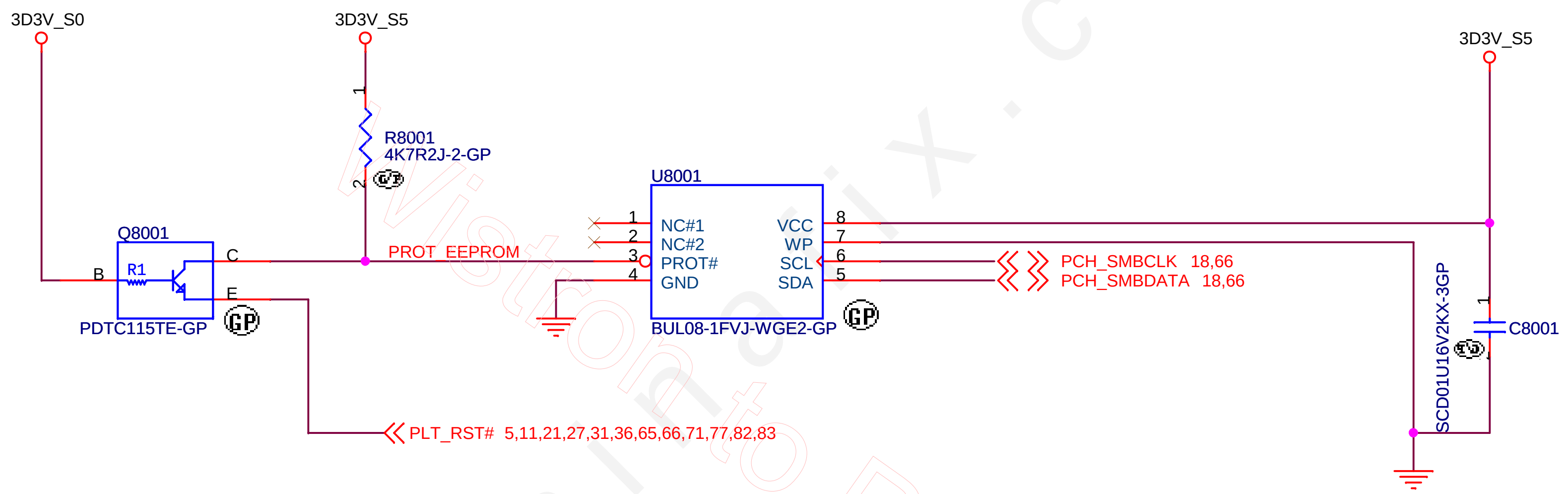
Title	Reserved		
Size: A3	Document Number: CD1 DIS		Rev: SC
Date: Tuesday, December 13, 2011	Sheet 79 of 102		



Supplier	Description	Wistron P/N
ROHM	BUL08-1FVJ-WGE2	72.BUL08.A0Q
NXP	PCA24S08ADP	72.24S08.A0Q
SANYO	LE26CAP08TT-TLM-H	72.26C08.00R

BOM1

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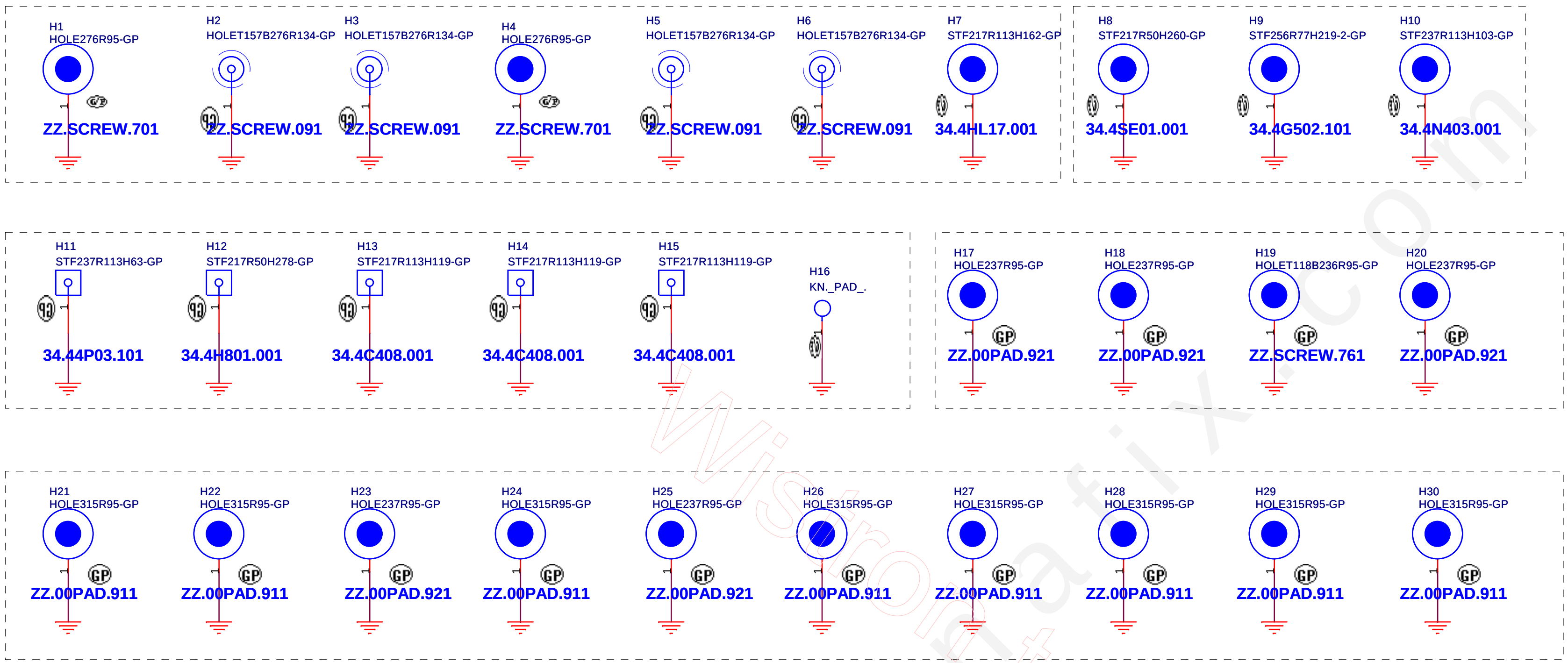
Title
RFID

Size A4 | Document Number **CD1 DIS** | Rev **SC**



BOM1

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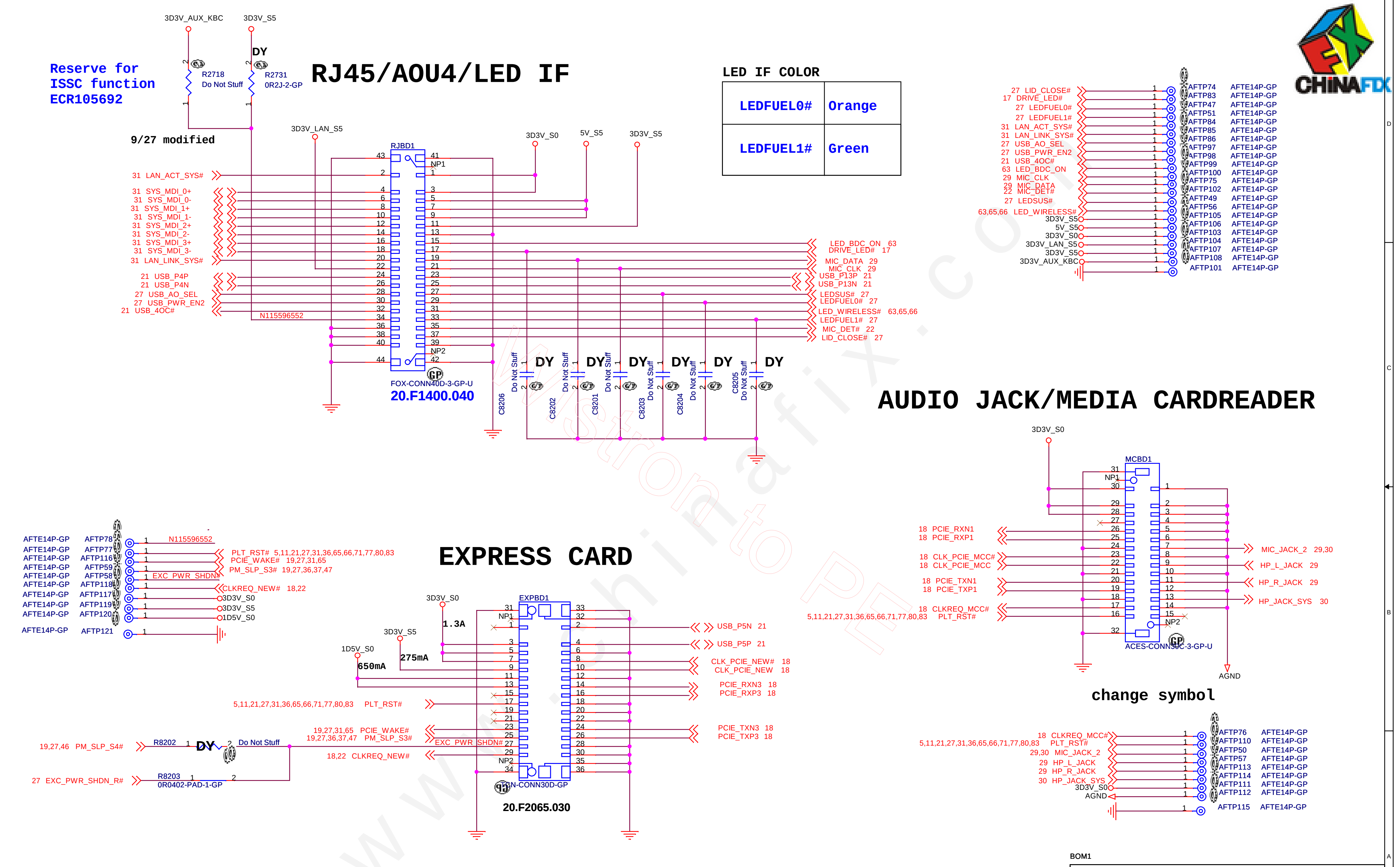
Taipei Hsien 221, Taiwan, R.O.C.

Title			
Reserved			
Size A3	Document Number CD1 DIS		Rev SC
Date: Tuesday, December 13, 2011		Sheet 81 of 102	



LED IF COLOR

LEDFUEL0#	Orange
LEDFUEL1#	Green

BOM1

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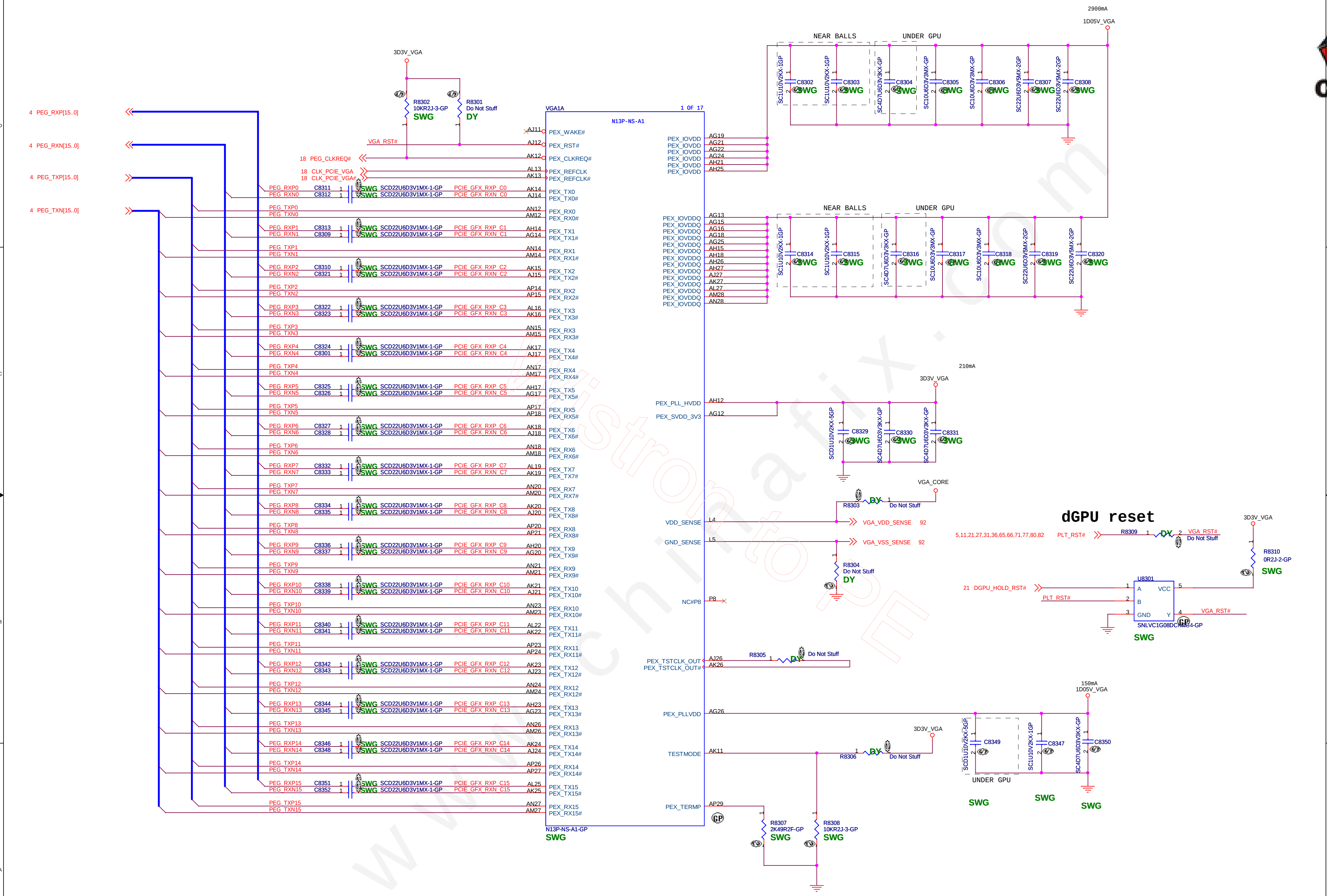
Title		
IO Board Connector		
Size A3	Document Number CD1 DIS	Rev SC
Date: Tuesday, December 13, 2011	Sheet 82 of 102	



BOM1

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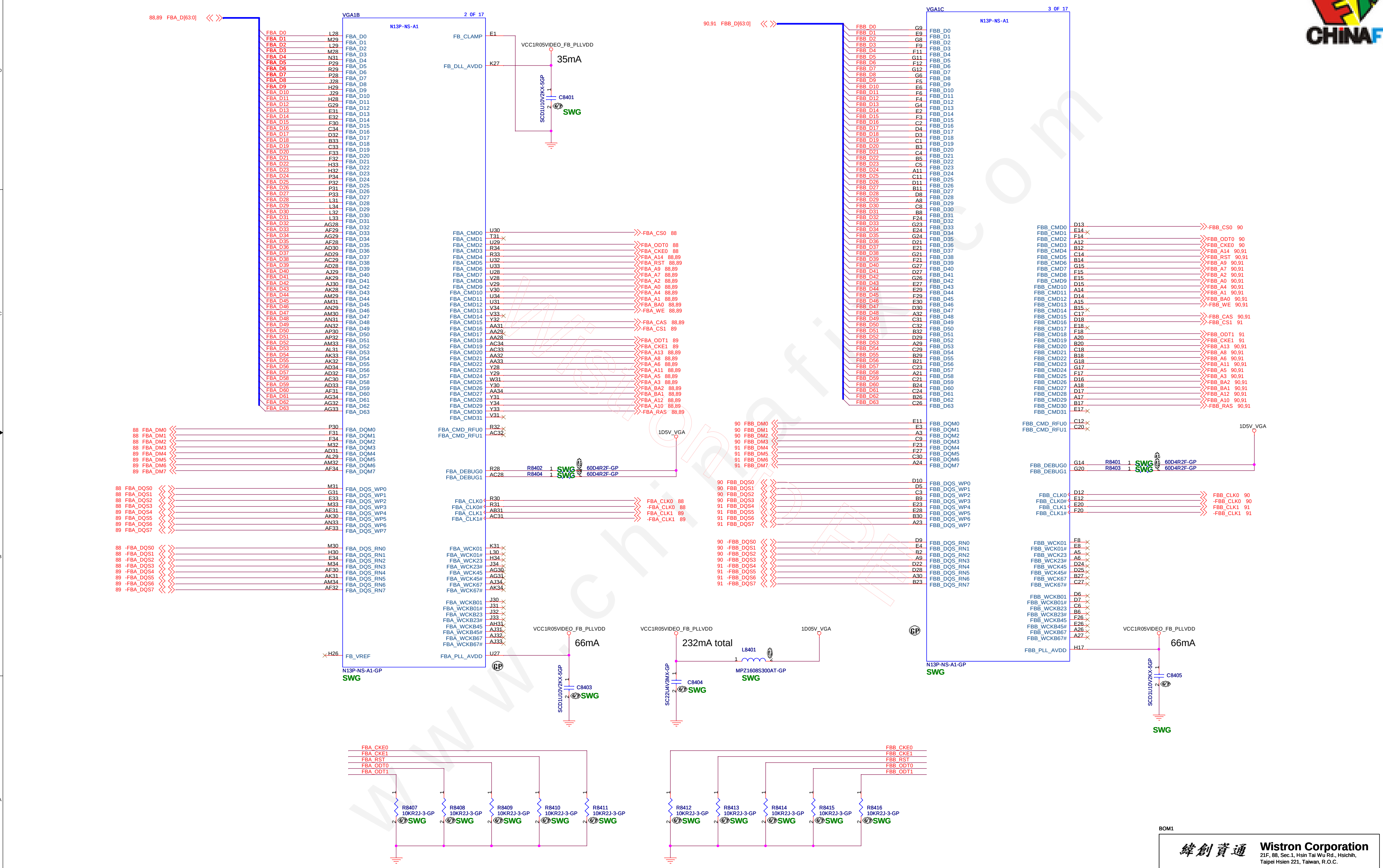
Taipei Hsien 221, Taiwan, R.O.C.

Title
GPU (1/5): PCIE/STRAPPING

Size	Document Number	Rev
C	CD1 DIS	SC

Date: Tuesday, December 13, 2011 Sheet 83 of 102



緯創資通 Wistron Corporation
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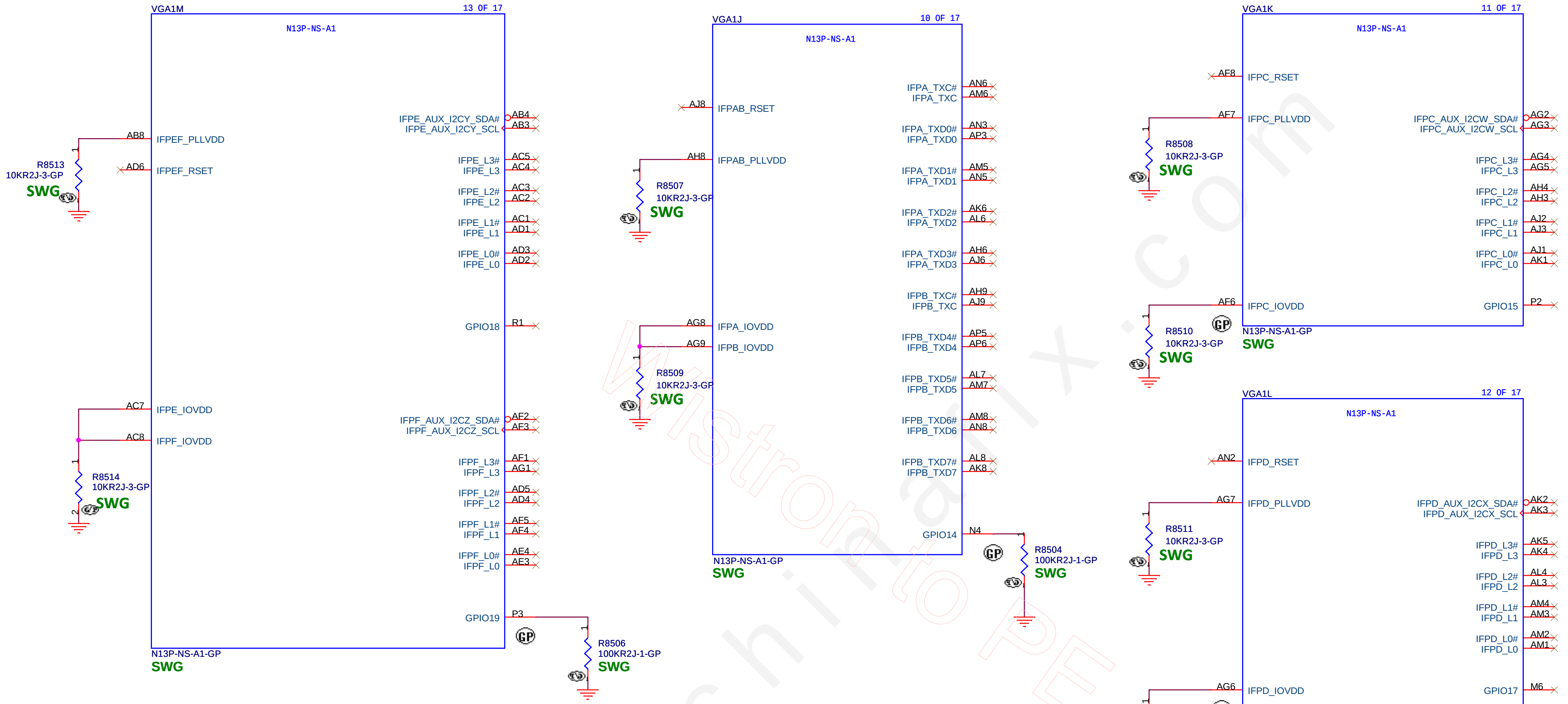
Title		
GPU (2/5): Memory		
Size A2	Document Number CD1 DIS	Rev SC
Date: Tuesday, December 13, 2011	Sheet 84 of 102	



GPIO14&19 will be NC on SIV satge.
DG-05587-001_v03_secured.pdf"
page180 12.2 says "Unused GPIOs may
be left as NC or floating

BOM1

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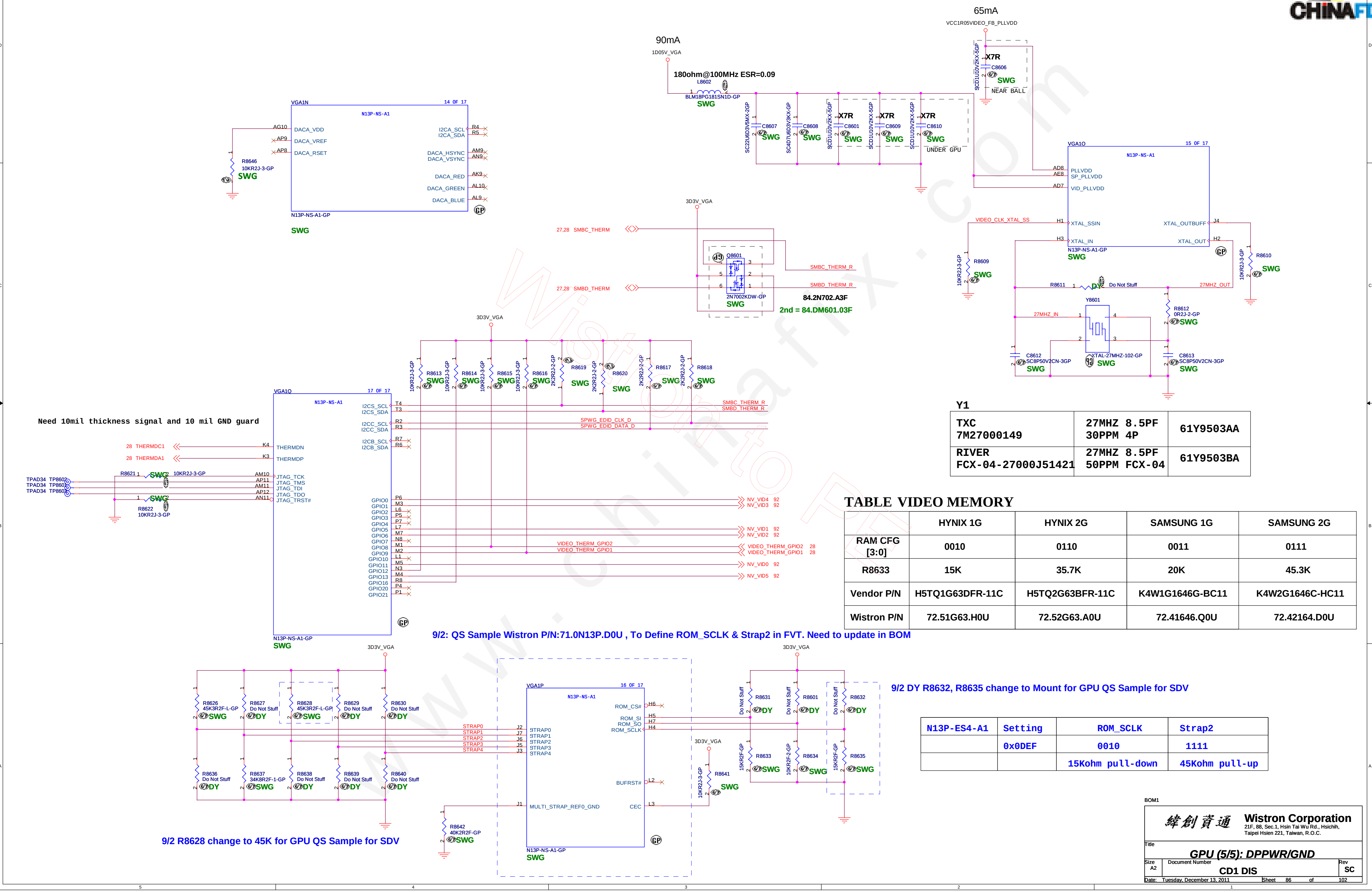
Title		
GPU (3/5): DP/LVDS/CRT/GPIO		
Size A3	Document Number CD1 DIS	Rev SC
Date: Tuesday, December 13, 2011	Sheet 85 of 102	



Y1

TXC 7M27000149	27MHZ 8.5PF 30PPM 4P	61Y9503AA
RIVER FCX-04-27000J51421	27MHZ 8.5PF 50PPM FCX-04	61Y9503BA

TABLE VIDEO MEMORY

	HYNIX 1G	HYNIX 2G	SAMSUNG 1G	SAMSUNG 2G
RAM CFG [3:0]	0010	0110	0011	0111
R8633	15K	35.7K	20K	45.3K
Vendor P/N	H5TQ1G63DFR-11C	H5TQ2G63BFR-11C	K4W1G1646G-BC11	K4W2G1646C-HC11
Wistron P/N	72.51G63.H0U	72.52G63.A0U	72.41646.Q0U	72.42164.D0U

N13P-ES4-A1	Setting	ROM_SCLK	Strap2
	0x0DEF	0010	1111
		15Kohm pull-down	45Kohm pull-up

BOM1

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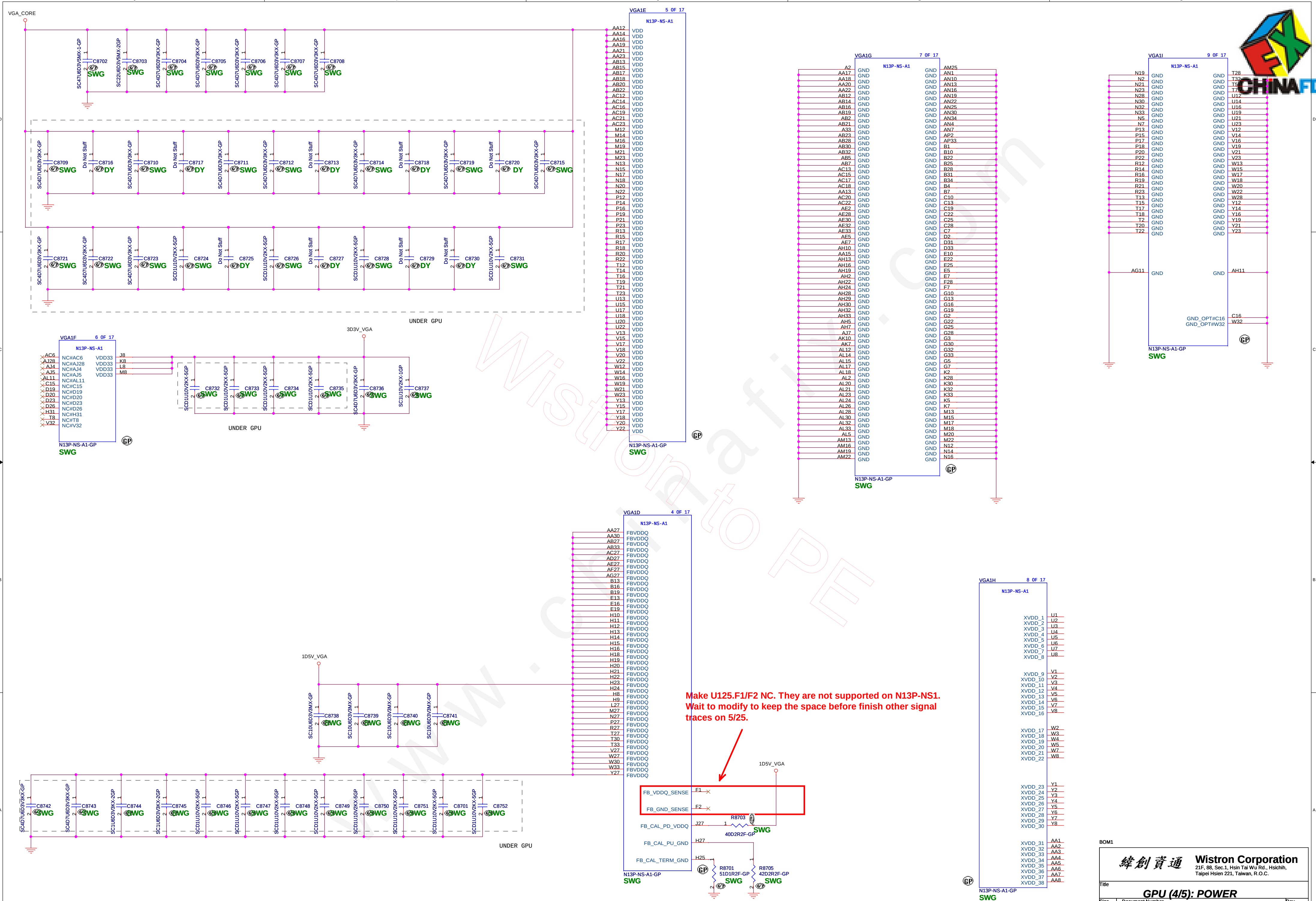
Title: GPU (5/5): DPPWR/GND

Size: A2 | Document Number: CD1 DIS | Rev: SC

Date: Tuesday, December 13, 2011 | Sheet 86 of 102



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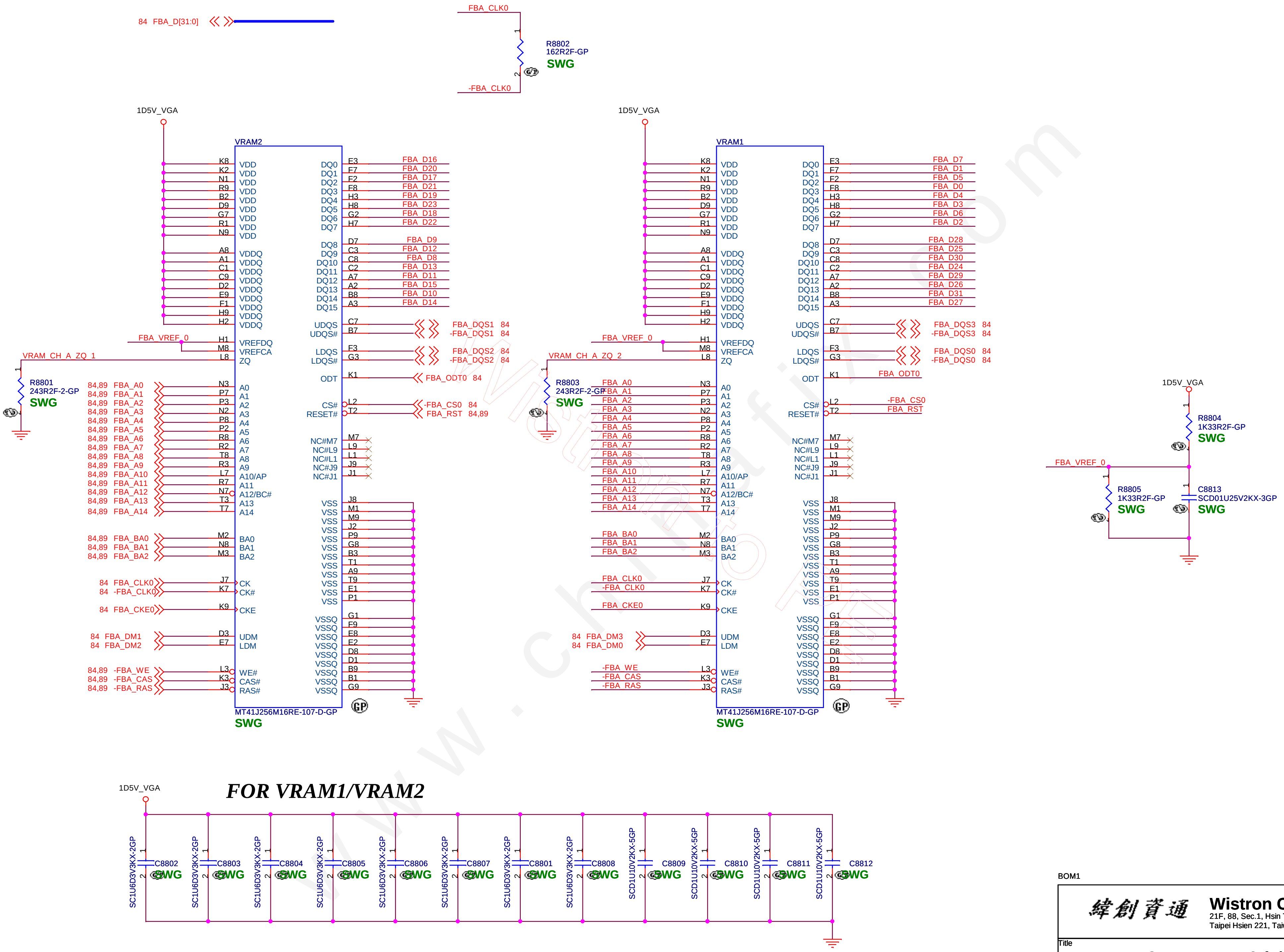
Title: GPU (4/5): POWER

Size	Document Number	Rev
A2	CD1 DIS	SC

Date: Tuesday, December 13, 2011 — Sheet 87 of 102



BOM1

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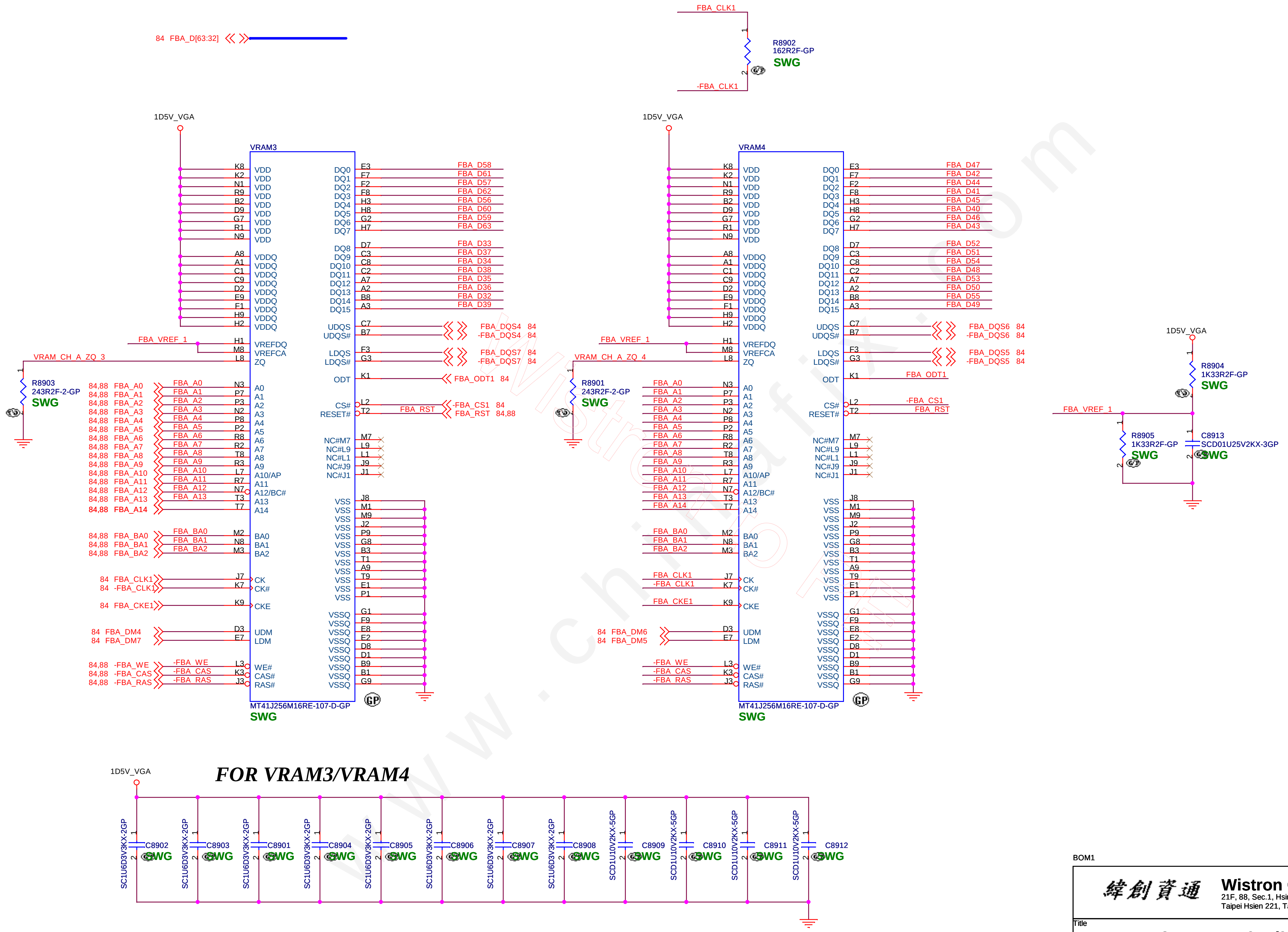
Taipei Hsien 221, Taiwan, R.O.C.

Title			
	GPU-VRAM1,2 (1/4)		
Size A3	Document Number CD1 DIS		Rev SC
Date: Tuesday, December 13, 2011		Sheet 88 of 102	



BOM1

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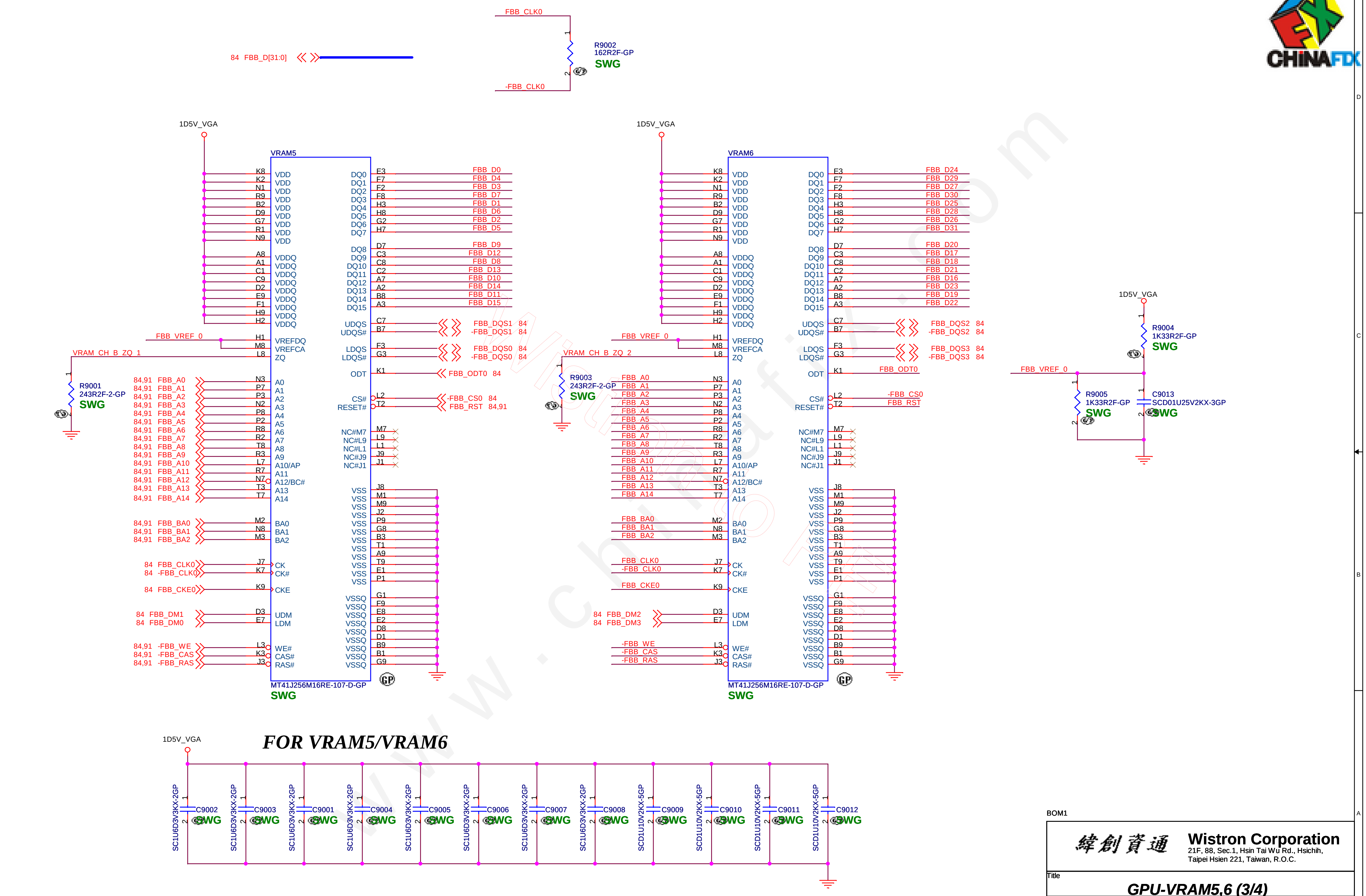
Title		
GPU-VRAM3,4 (2/4)		
Size: A3	Document Number: CD1 DIS	Rev: SC
Date: Tuesday, December 13, 2011	Sheet 89 of 102	



BOM1

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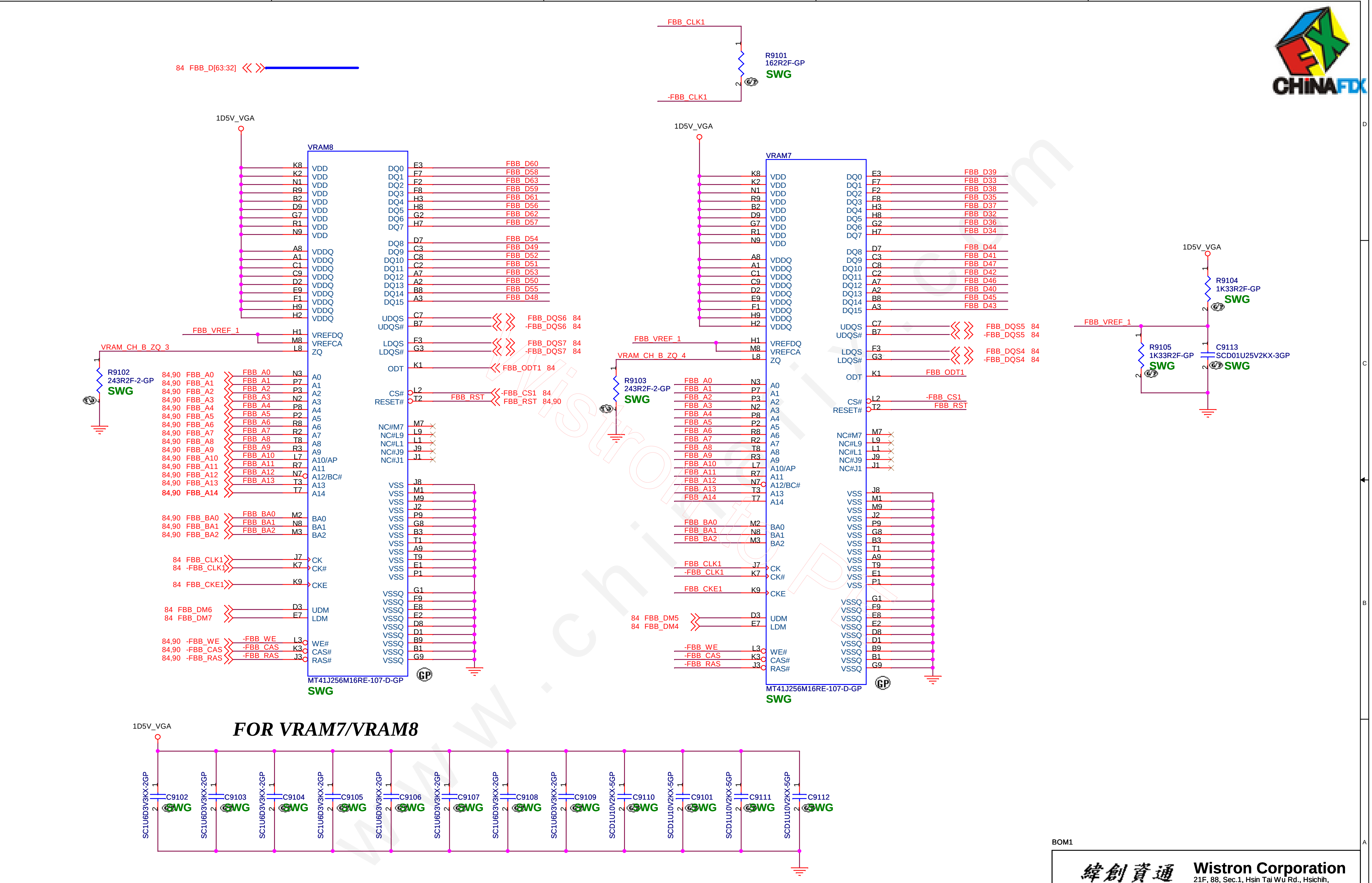
Taipei Hsien 221, Taiwan, R.O.C.

Title		
GPU-VRAM5,6 (3/4)		
Size: A3	Document Number: CD1 DIS	Rev: SC
Date: Tuesday, December 13, 2011	Sheet 90 of 102	



BOM1

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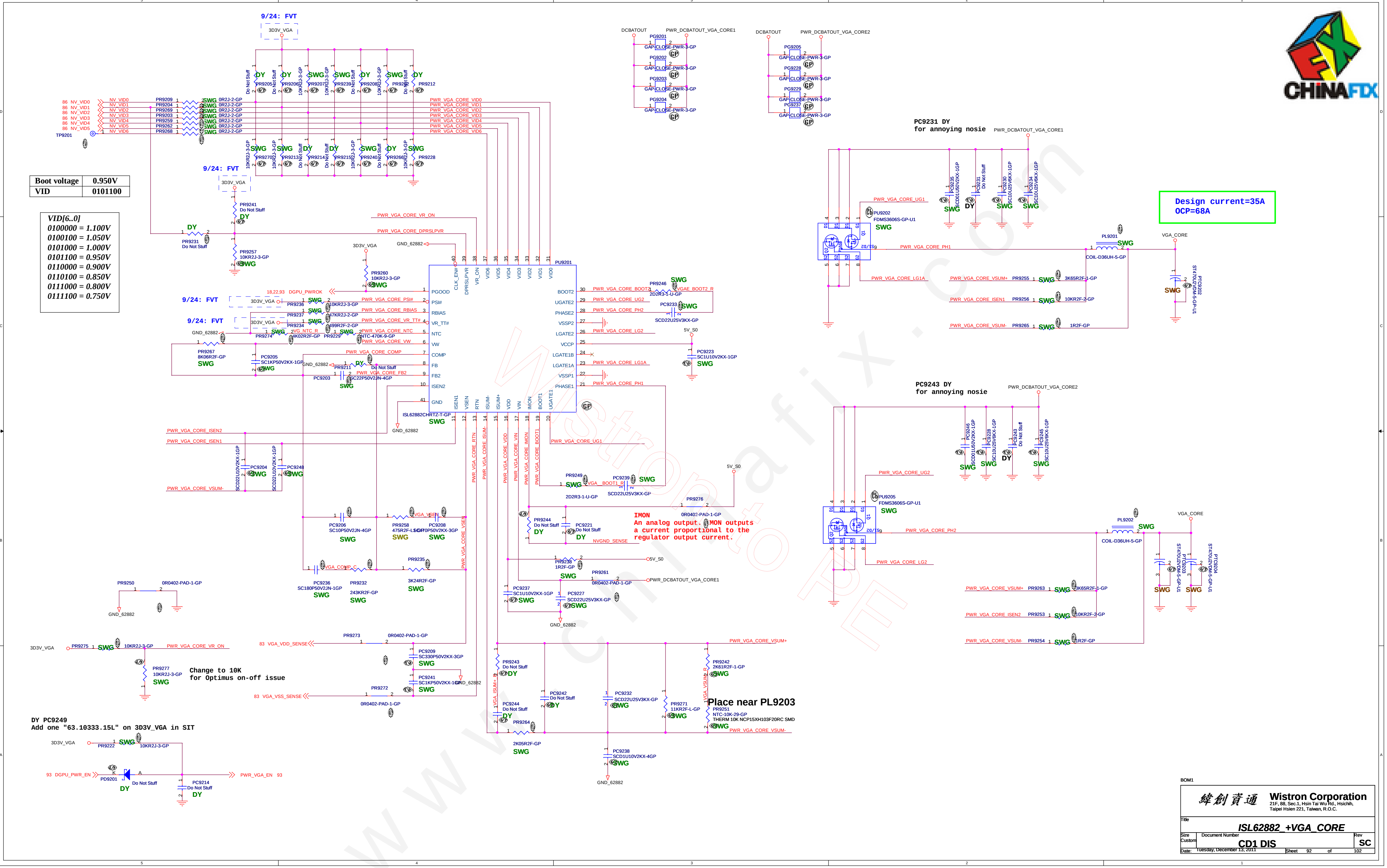
Title		
GPU-VRAM7,8 (4/4)		
Size: A3	Document Number: CD1 DIS	Rev: SC
Date: Tuesday, December 13, 2011	Sheet 91 of 102	



9/24: FVT
3D3V_VGA
Boot voltage 0.950V
VID 0101100
VID[6..0]
0100000 = 1.100V
0100100 = 1.050V
0101000 = 1.000V
0101100 = 0.950V
0110000 = 0.900V
0110100 = 0.850V
0111000 = 0.800V
0111100 = 0.750V
PU9201
ISL62882CHRTZ-T-GP
PC9231 DY
for annoying nosie
PC9243 DY
for annoying nosie
Design current=35A
OCP=68A
PWR_DCBATOUT_VGA_CORE1
PWR_DCBATOUT_VGA_CORE2
VGA_CORE
PU9202
FDMS3606S-GP-U1
PU9205
FDMS3606S-GP-U1
PL9201
COIL-D36UH-5-GP
PL9202
COIL-D36UH-5-GP
IMON
An analog output. IMON outputs
a current proportional to the
regulator output current.
Place near PL9203
PR9251
NTC-10K-29-GP
THERM 10K NCP15XH103F20RC SMD
Change to 10K
for Optimus on-off issue
DY PC9249
Add one "63.10333.15L" on 3D3V_VGA in SIT
93 DGPU_PWR_EN
PWR_VGA_EN 93
83 VGA_VDD_SENSE
83 VGA_VSS_SENSE
緯創資通 Wistron Corporation
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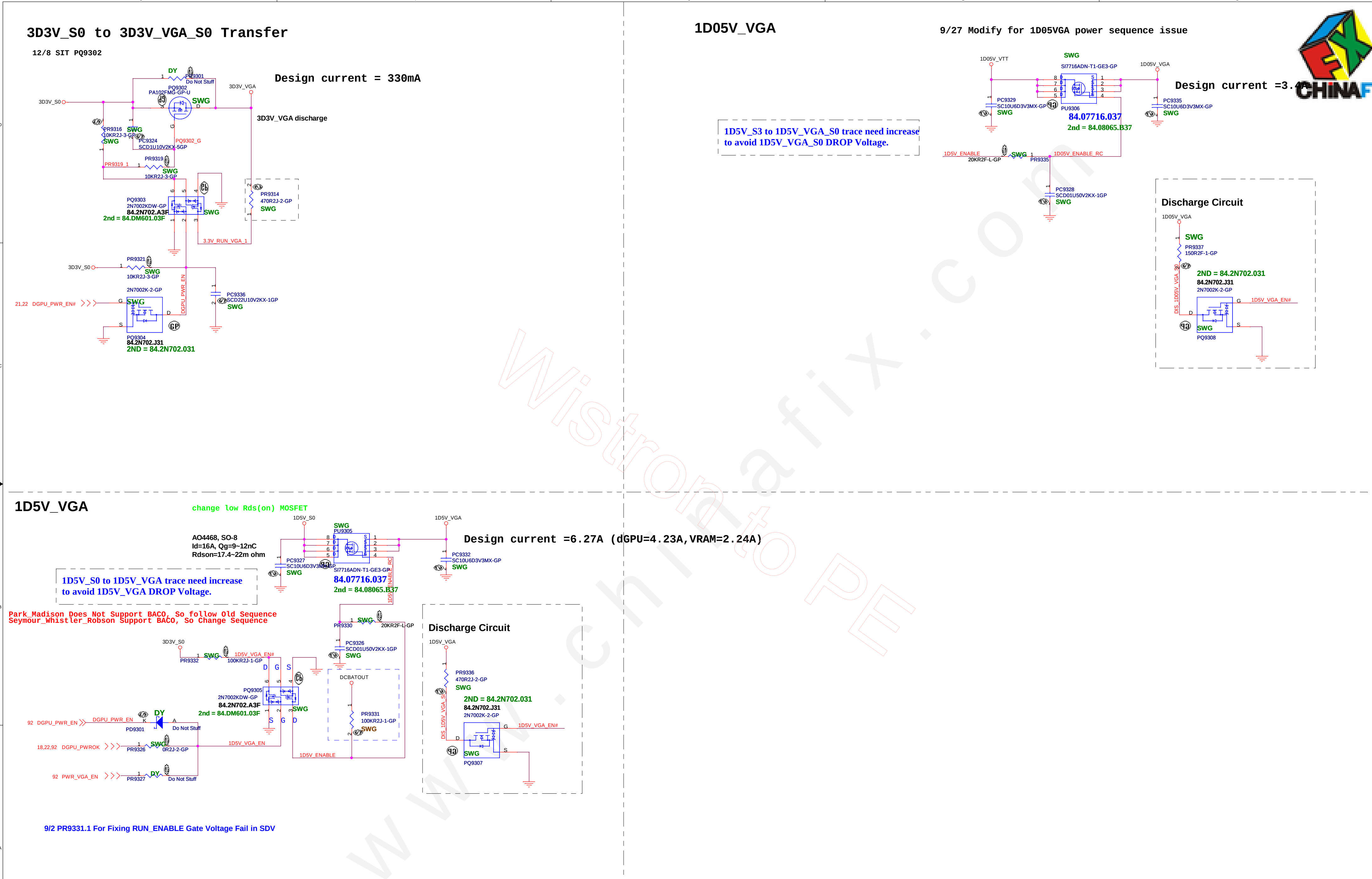
Title
ISL62882_+VGA_CORE
Size Custom
Document Number
CD1 DIS
Rev SC
Date: Tuesday, December 13, 2011
Sheet 92 of 102



BOM1

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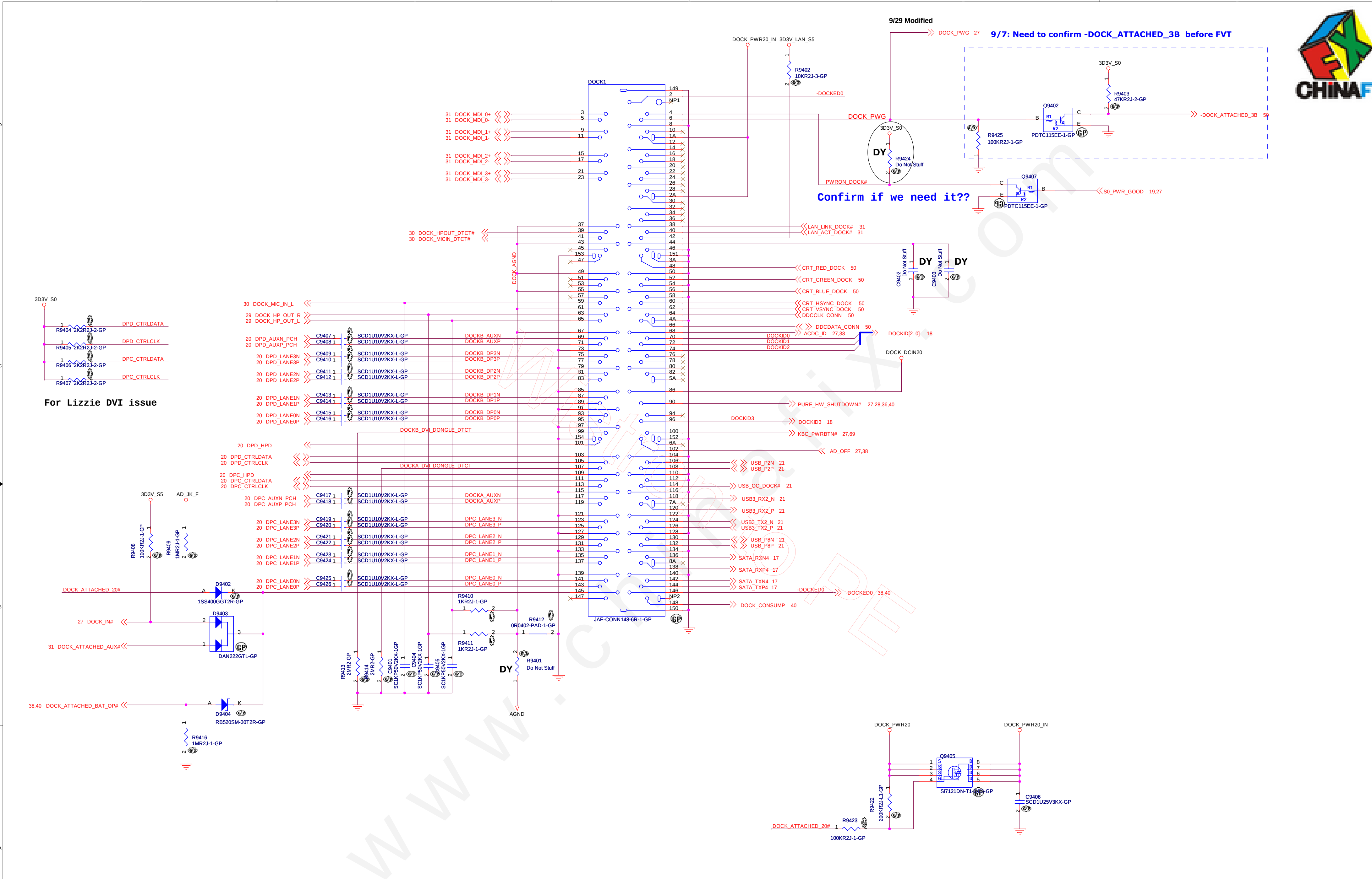
Title	DISCRETE VGA POWER	
Size: A2	Document Number: CD1 DIS	Rev: SC
Date: Tuesday, December 13, 2011	Sheet 93 of 102	



BOM1

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21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title	**Dock Connector**	
Size: A2	Document Number: CD1 DIS	Rev: SC
Date: Tuesday, December 13, 2011	Sheet 94 of 102	

BOM1

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21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title
GPIO Extender

Size	Document Number	Rev
A4	CD1 DIS	SC

Date: Tuesday, December 13, 2011 Sheet 95 of 102

5 4 3 2 1

D

C

B

A

(Blanking)

BOM1

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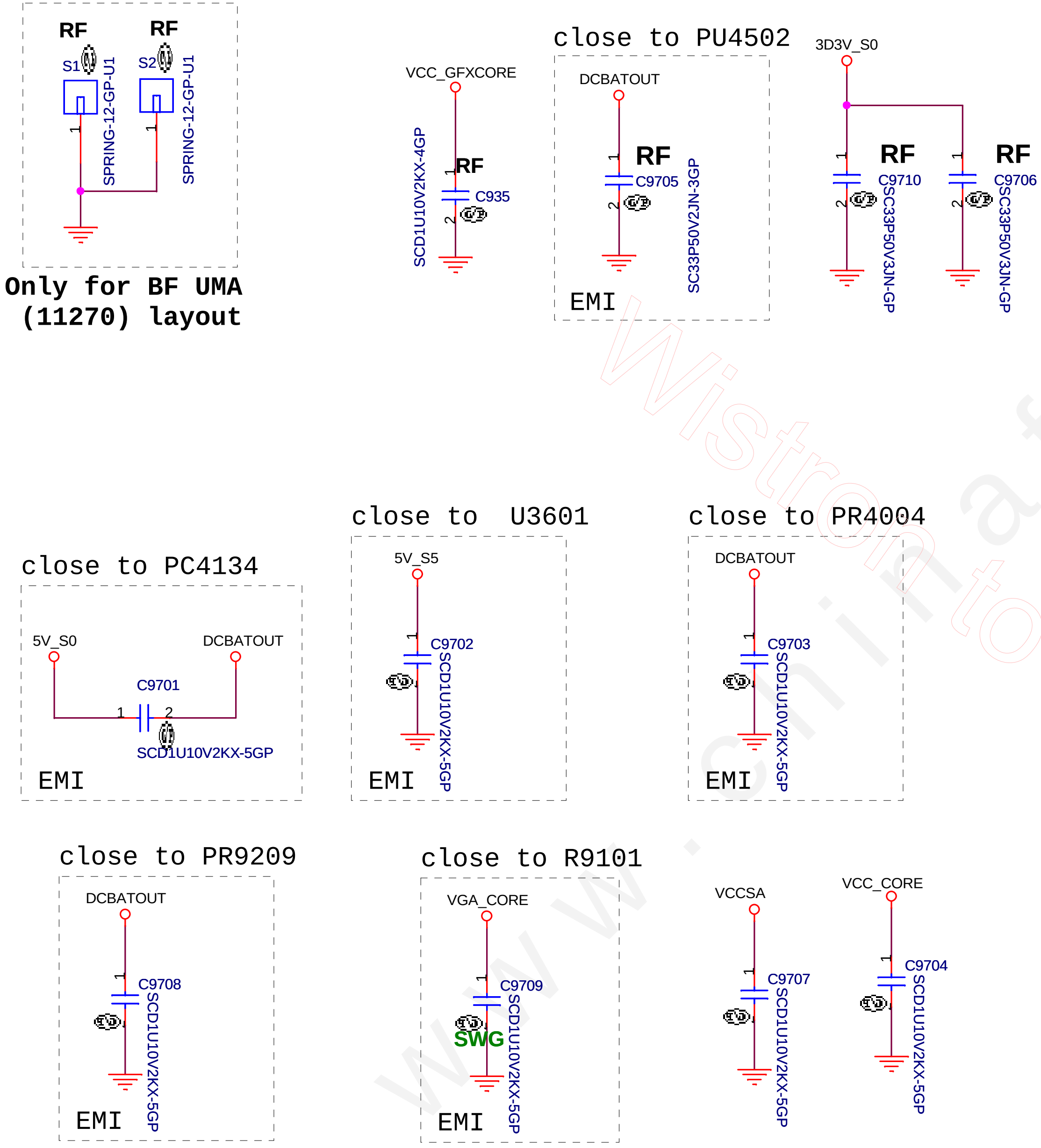
Title		
	Reserved	
Size A3	Document Number **CD1 DIS**	Rev **SC**
Date: Tuesday, December 13, 2011	Sheet 96 of 102	



12/6 RF Solution Table:

Location	CD DIS	CD UMA	BF UMA
C1427 33PF	No_ASM	No_ASM	ASM
EC4601 33PF	ASM	ASM	No_ASM
PC4126 33PF	ASM	ASM	No_ASM
PC4120 33PF	ASM	ASM	No_ASM
PC4613 33PF	ASM	ASM	No_ASM
C9705 33PF	ASM	ASM	No_ASM
PC4622 330PF	ASM	ASM	No_ASM
PR4612 2.2ohm	ASM	ASM	No_ASM
C9710 C9706 33PF	No_ASM	No_ASM	ASM
C935 0.1uF	No_ASM	No_ASM	ASM
S1 S2 Spring	No_ASM	No_ASM	ASM

BOM1

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Title: **EMI Capacitors**

Size	Document Number	Rev
Custom	CD1 DIS	SC

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Schematic Change List

Date	Page number	Description	Remark
2011/07/12	28	Add U2802,R28015~R2818,C2810~C2811,RT2801~RT2814 for Thermister protection function.	
	22	Change R2211 to ASM.	
	49	Change LVDS connector reference to LVDS1 and pin1,5,37 NC.	
	66	Add R6610~R6612,C6612,U6602,U6603 for SMbus alternative path (Intel Anti-sheft 3.0)	
	17	Add intersheet connection for LPC_AD[3:0] and LPC_FRAME#.	
	45,....	Change power rail name VCCP_CPU to 1D05V_VTT, Delete PR4514 (for iAMT),Change 1D05V_LAN to 1D05V_S0.	
	69	Change net TP4CLKPAD to TP4CLK, net TP4DATAPAD to TP4DATA.	
	42	Remove PR4227,PR4229 because of double pull-up.	
	41	Change net I244143961 to 5V_S5.	
	27	Change pull-up power of FAN_ID to 3D3V_S0.	
2011/07/13	27	Reserve test pads TP2709,TP2710 for net KCOL16,KCOL17.	
	2	Remove Media card reader block right side.	
	28,86	Delete Q2801,Q2802,C2805,R2813,R2814,wire themal sensor channel1 to dGPU inside	
	52	Change DP net name to DPC to DPB of connector side.	
	92	Remove U9204,U9207 and output cap because of space reason.	
	93	Remove 1D8V_VGA_S0 power rail.Change 1V_VGA_S0 to 1D05V_VGA_PLL Add power for 1D05V_VGA_S0 power rail.	
	86,92,93	Change 3D3V_VGA_S0 to 3D3V_VGA.	
	93	Change 1D5V_VGA_S0 to 1D5V_VGA.	
	40,94	Change net -PWRSHUTDOWN to PURE_HW_SHUTDOWN#.	
	61	Change U6102.5 connecting to 5V_S5.	
2011/07/14	93	Remove LDE RD3552 and the 1D05V_VGA_PLL. Change net 1D05V_VGA_PLL to 1D05V_VGA.	
	83~87	Connect GPU core power named as "VGA_CORE", Connect 1D05V power named as"1D05V_VGA".	
	8	Remove R801 and PTC803 because of layout space reason.	
	46	Add PTC4603 (330UF BULK CAP).	
	45	Remove PTC4502 because of layout space reason.	
	42~44	Change CPU core power controller to ISL95838 because of layout requirement.	
2011/07/15	21,31	Reserve R3120,R3121 to select LAN power control by EC or PCH,change PCH GPIO14 as LAN_PWERON_PCH.	
	29	Add 10uF hook and DY on signal MIC_GPI.	
	27	Change EC pin16 from CAPSLOCK_LED# to PCIE_WAKE#.	
	28	Change Thermistor comparator power from VCC5M to 5V_S5.	
	85	Removed power supply to IFPx modules and add R8507~R8514 to pull down.	
	86	Removed power supply to DAC modules and add R8646 to pull down.let DAC I/O pins NC.	
	93	Add PC9336 0.22uF for DGPU power sequence adjustment.	
	84	Connect pin FB_CLAMP to GND, let pin FB_VREF NC.	
2011/07/16	28,86	Add pull up resistor R2819 net on net THERM_ALERT#.	
	69	Change TPFP1 type, remove pins for POA function.	
	86	Remove R8623~R8625.	
2011/07/17	18	Remove R1825 because of double pull down.	
2011/07/18	49	Change U4901 from GMT part to TI22922.	
	27	Change U2702 from GMT part to TPS3809,Remove R2710,R2713,R2718,R2724,R2726,R2741.	
	28,86	Connect GPU GPIO9 signal VIDEO_THERM_ALERT# to thermal sensor ALERT pin.	
	69	Update TPFP1 pin assignment.	
2011/07/19	45	Remove PR4517, Cahnge net 1.05VTT_PWRGD to 1D05VTT_PWRGD.	
	66	Change U6602,U6603 pin1 to SMB_DATA_WWAN,SMB_CLK_WWAN.	
	21,63,82	Change BLUETOOH LED control from PCH to BLUETOOTH, Add R2108 as pull up of REQ2#.	
	5	Remove R511,R516 because of duplication.	
	26	Move R2602 from XDP_DBRESET# to PM_RSMRST#, and correct connection of PXDP1 pin 10,17,18.	
	18,22,27	Correct EC_SCI#,EC_SMI#,EC_SWI connection,leave only EC_SCI# from PCH to EC.	
	14,15	Add C1423,C1424,C1425,C1426,C1503,C1504,C1517,C1522 for VREF CA,VREF DQ power. Add 10uF caps C1427,C1523,C1524 on 0D75V_S0.	
	66	Add C6613,C6614,C6615,C6616 for mSATA.	
	18	Change RN1802 as pull up SMB_CLK_PCH/SMBDATA_PCH	
	56	Remove connection of SATA_ODD_DA# and RN2102.	
2011/07/19	9	C934 pin2 change to GND	
	42	PR4230 pin1 and PR4229 pin1 change to 1D05V_VTT	
	13	Del 3D3V_PCIE_AOAC PWR Plane	
	66	Change RN6601 pin1,2 net name to SMB_CLK_WWAN and SMB_DATA_WWAN	
	69	Del RN6901	
	77	Cut U7701 pin15	
	82	RJBD1 pin34 change to LED_BDC_ON	
	93	Change net name to 1D05V_VGA_EN	
2011/07/20	17	PCH_JTAG_TDI keep PU to follow LKN4	
	93	PR9335 pin2 change to 1D5V_ENABLE,R9326 change to SWG,R9327 change to DY to adjust GPU power up sequence.	
	43,44	Add PC4312, PC4313, PC4314 for EMI	

BOM1

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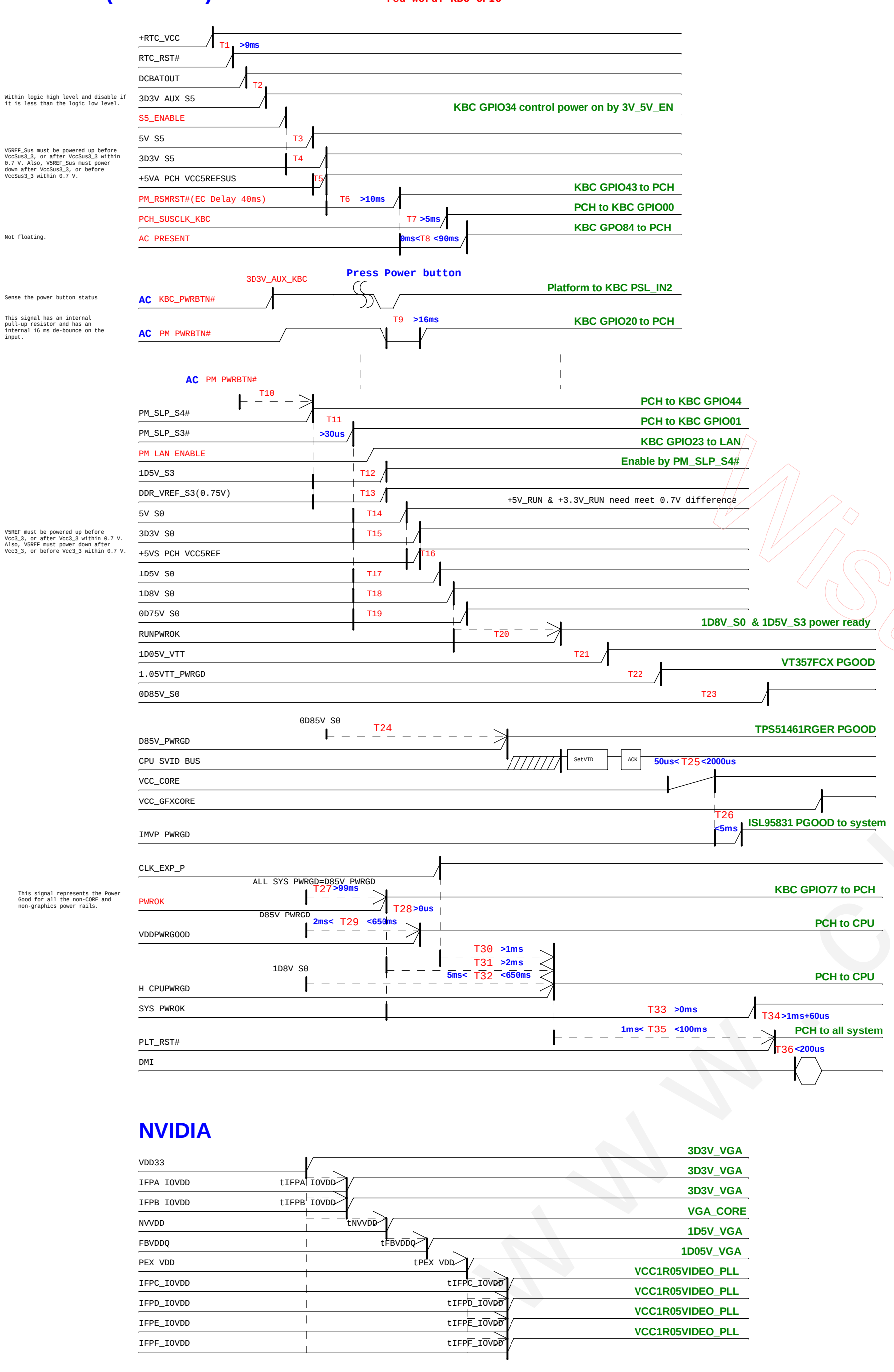
Title: Change History

Size: A2 | Document Number: CD1 DIS | Rev: SC

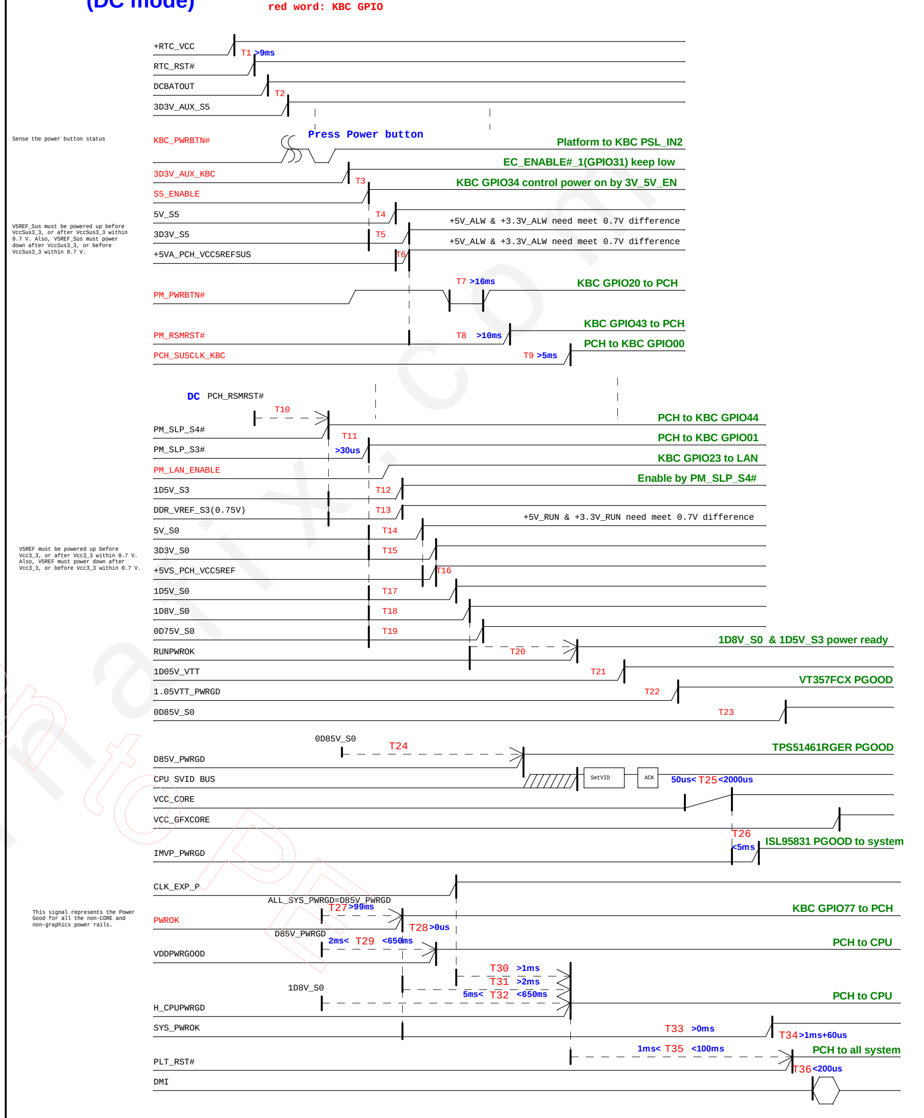
Date: Tuesday, December 13, 2011 | Sheet 98 of 102

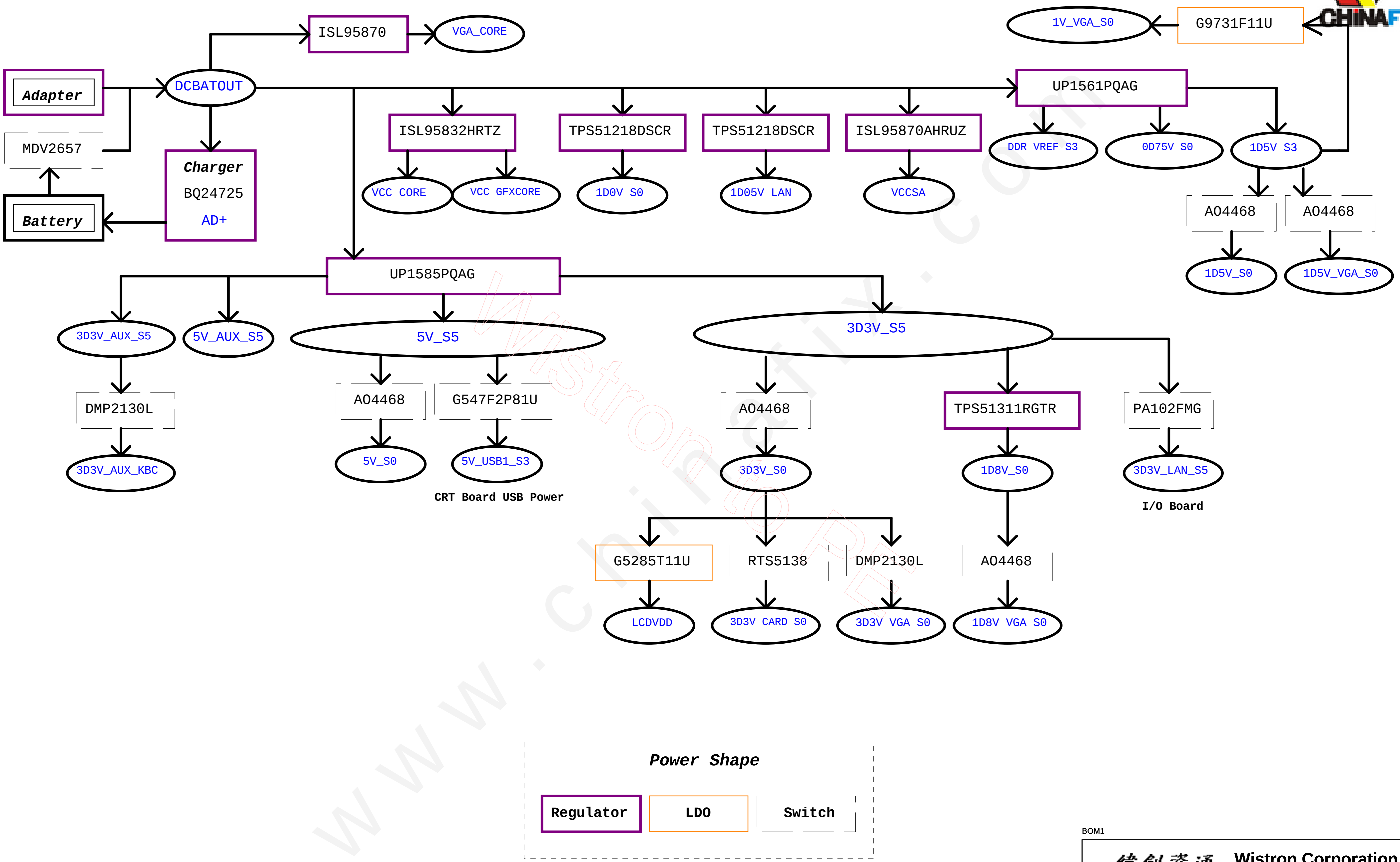
Chief River Platform Power Sequence

(AC mode)



(DC mode)





BOM1

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Title: **Power Block Diagram**

Size	Document Number	Rev
A3	CD1 DIS	SC

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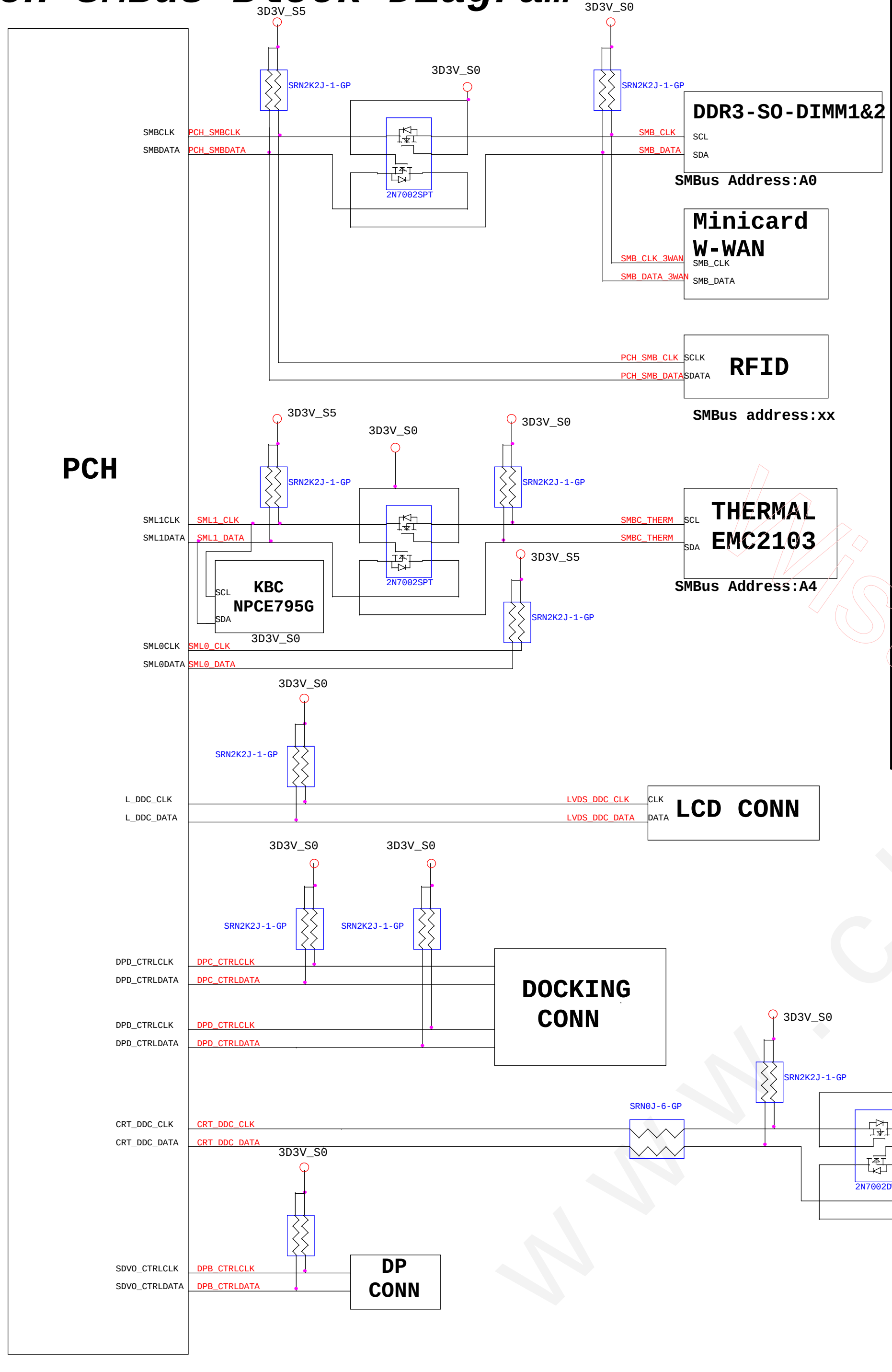
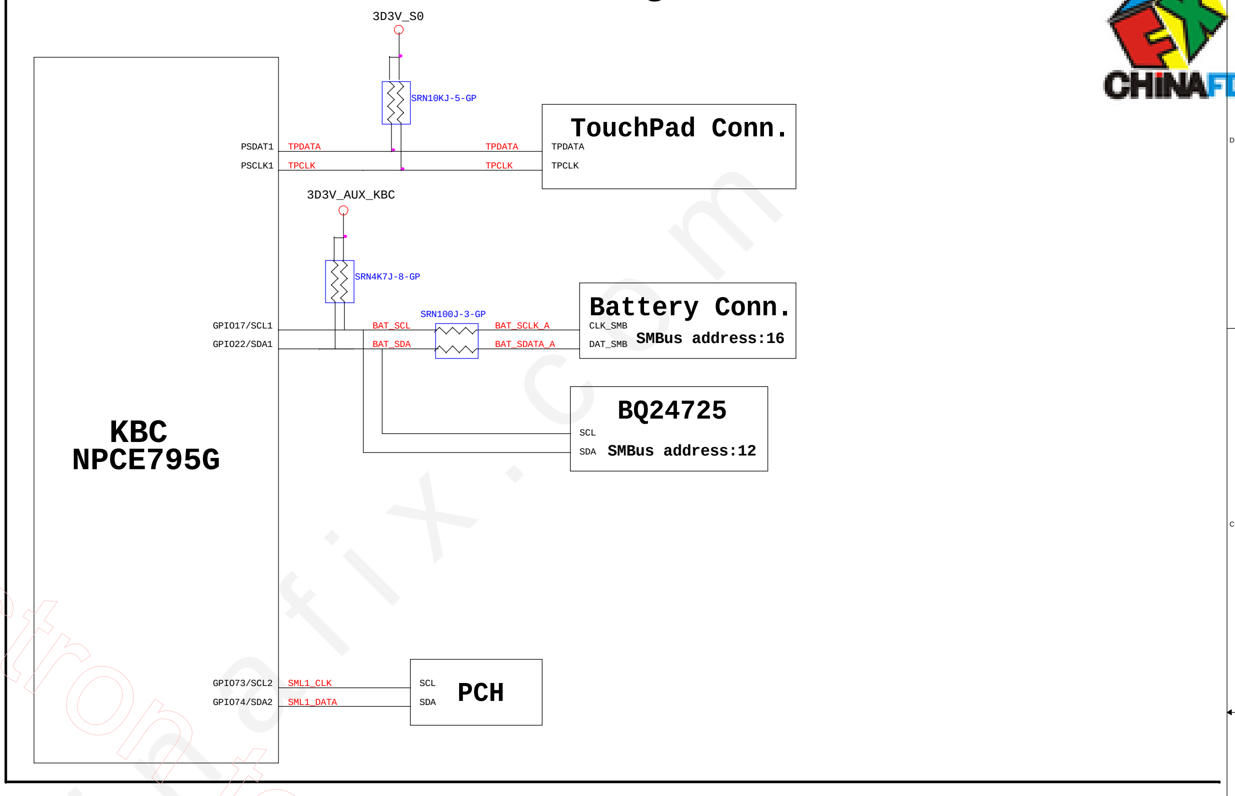
PCH SMBus Block Diagram



KBC SMBus Block Diagram



BOM1

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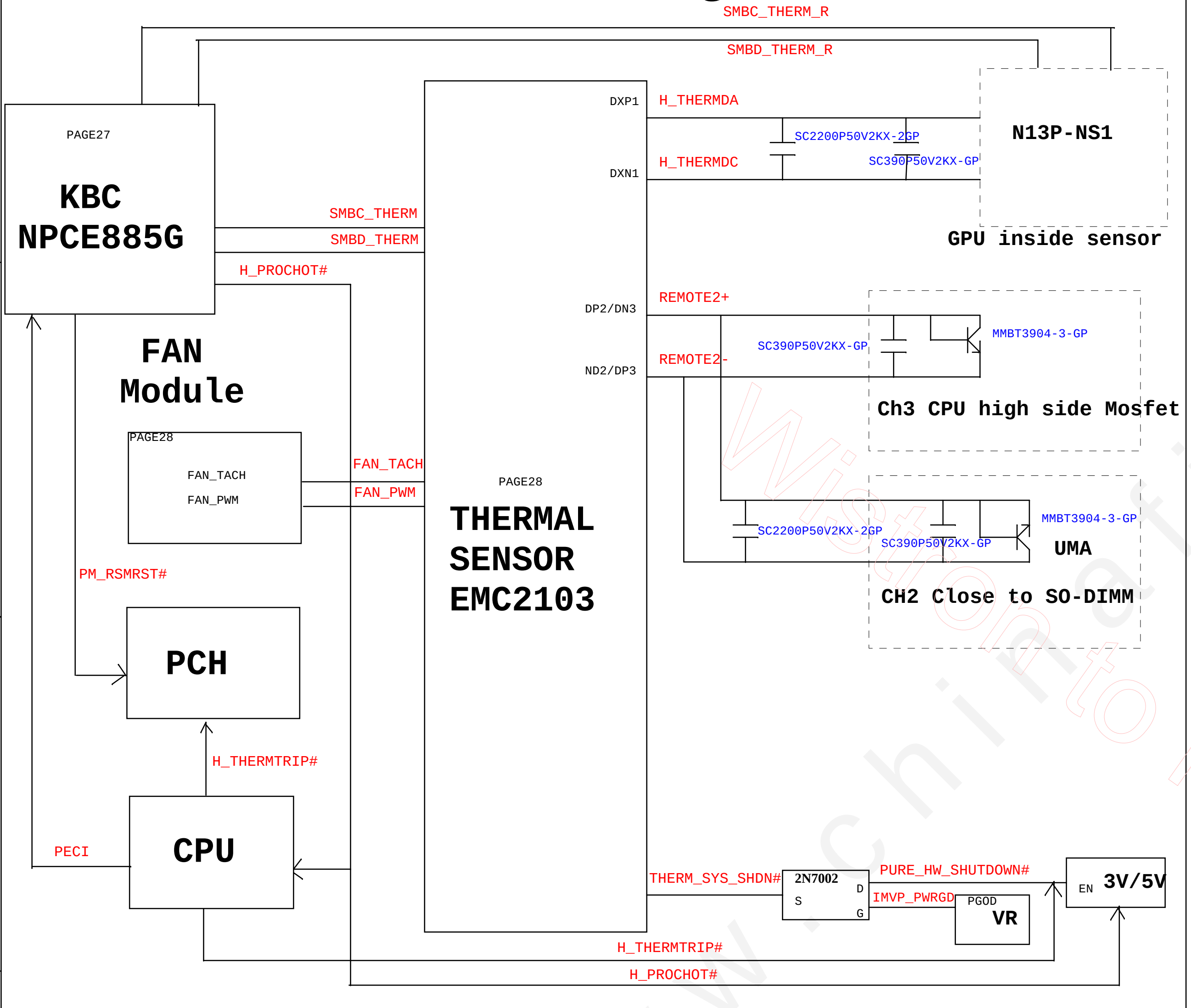
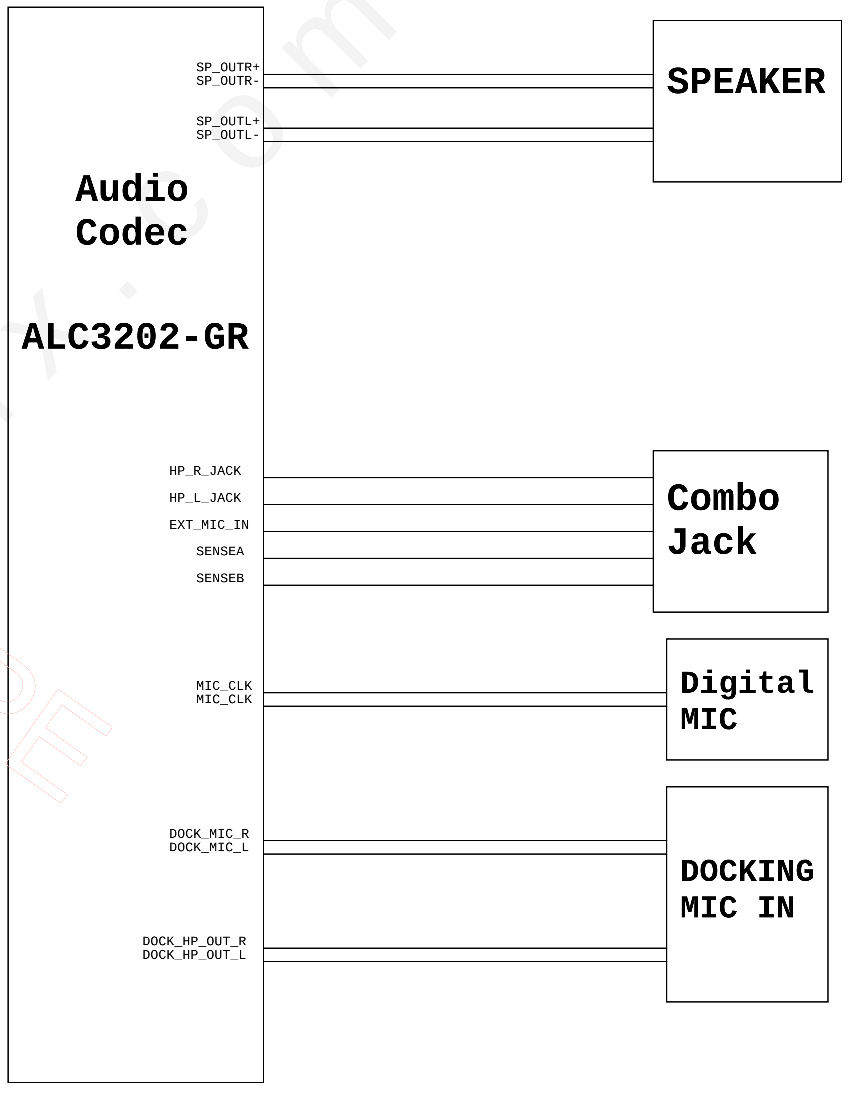
Title	SMBUS Block Diagram	
Size A2	Document Number CD1 DIS	Rev SC
Date: Tuesday, December 13, 2011	Sheet 101 of 102	

Thermal Block Diagram



Audio Block Diagram



BOM1

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21F, 88, Sec.1, Hsin Tai Wu Rd., Hsichih,
Taipei Hsien 221, Taiwan, R.O.C.

Title		
Thermal/Audio Block Diagram		
Size A3	Document Number **CD1 DIS**	Rev **SC**
Date: Tuesday, December 13, 2011	Sheet 102 of 102	